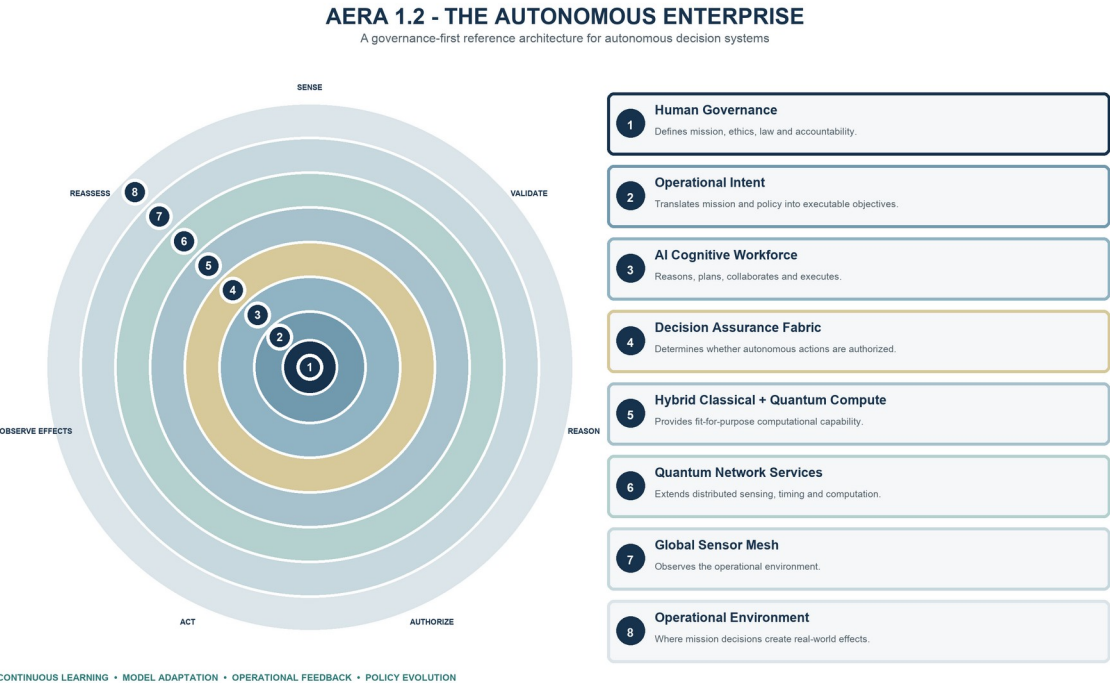


AERA 1.2

The Autonomous Enterprise Reference Architecture

A Governance-First Architecture for Autonomous Decision Systems



Document Control

Table 1. Document Control

Field	Value
Specification identifier	AERA-001
Title	AERA 1.2 - The Autonomous Enterprise Reference Architecture
Version	1.2 WD
Build identifier	AERA-001-1.2WD-20260807.1
Status	Working Draft; developmental-review candidate; public review not yet open
Publication date	7 August 2026
Author, editor and document owner	Angelis Pseftis
Affiliation	Independent author
Correspondence	NOT ESTABLISHED - release gate
Distribution	Controlled developmental review; not an approved or consensus standard
Rights and licensing	NOT ESTABLISHED - license, contribution and redistribution terms are release gates
Intended audience	Boards, mission owners, CIOs, CISOs, CTOs, chief architects, system engineers, AI leaders, safety officers and assessors
Normative language	Uppercase SHALL, SHALL NOT, SHOULD, SHOULD NOT, MAY and NEED NOT have the meanings defined in Section 1.4
Scope	Governance, architecture, assurance and lifecycle engineering for autonomous decision systems

STATUS: AERA 1.2 WD is an independently authored external technical discussion draft. It is not a NIST, ISO, IEEE, CISA, DoD or DOE publication, has not been endorsed by any U.S. Government entity, and does not confer compliance with any external framework. NIST compatibility and a possible submission path are design objectives, not claims of support.

Publication provenance and use

Figure 1 and the AERA constructs in this specification were developed for this publication. External standards, frameworks and research are cited as informative sources. No agency seal, logo or publication-series mark is used. Reviewers should evaluate the technical content on its own merits and should not infer sponsorship or endorsement.

Suggested citation

Pseftis, A., AERA 1.2 - The Autonomous Enterprise Reference Architecture, AERA-001, Working Draft, 7 August 2026.

Revision history

Table 2. Revision history

Edition	Date	Disposition and material change
1.0	August 2026	Initial complete technical draft.
1.1 RC1	5 August 2026	Canonicalized requirements; corrected conformance semantics; added threat-control-test traceability, citations and reviewer tooling.
1.2 WD	7 August 2026	Reconciled the controlled source; added NIST design-attribute assessment, IR 8477/OLIR mapping semantics, open-development governance, current NIST sources and release-control safeguards.

Revision policy

A released artifact is immutable. Any replacement receives a new build identifier and manifest. A change to normative content receives a new specification version. Prior comments remain bound to the exact version, build identifier and artifact hash reviewed. Material changes to definitions, normative requirements, layer responsibilities, decision-assurance criteria or mappings require a versioned revision. Extensions may specialize AERA for sectors or use cases but do not weaken the authority, evidence, attribution, recovery or human-accountability principles defined herein.

Abstract

The Autonomous Enterprise Reference Architecture (AERA) defines a governance-first, decision-centric architecture for systems in which computational actors exercise delegated authority without synchronous human direction. AERA treats the proposal-to-effect transaction as the primary unit of assurance. It specifies logical capability domains, decision contracts, consequence classes, human-control modes, cybersecurity and resilient timing properties, conformance profiles, validation methods, a threat model and a maturity model. The architecture remains technology-neutral at its control boundary: AI, classical computing, specialized accelerators, quantum services, sensor networks and operational technology are governed by the same authority, evidence, enforcement, observation and recovery semantics. AERA is a proposed architecture and working draft, not an adopted consensus standard, government publication, certification scheme or evidence of NIST support.

Keywords

autonomous systems; decision authority; AI governance; decision assurance; zero trust; cybersecurity; post-quantum cryptography; quantum integration; resilient timing; positioning, navigation and timing; conformance; assurance case

Reviewer Guide

AERA 1.2 WD is a developmental-review candidate. It is not yet open for public comment because the official package URL, submission channel, comment dates, contribution license, identity-publication rules and privacy terms have not been established. Once those release gates are satisfied, comments should identify the version, build, artifact hash, section, page and requirement or threat identifier; distinguish technical, conformance, editorial, evidence, mapping and accessibility concerns; explain the rationale; and propose replacement language where practical. The companion comment matrix is the controlled disposition record.

Focused review questions

- Are the decision-authority transaction and proposal-to-effect binding technically complete and implementable?
- Are any normative requirements ambiguous, duplicative, infeasible, non-testable or assigned to the wrong conformance profile?
- Does the threat model cover credible adversarial, accidental, epistemic and common-mode failures without conflating them?
- Are consequence classes, human-control modes and exception rules sufficiently bounded for high-consequence use?
- Are AERA mappings and claimed semantic deltas accurate without implying reciprocal compliance or external endorsement?
- Which definitions, interfaces, tests or evidence packages require a machine-readable companion specification?
- What pilot evidence, inter-rater data or operational demonstrations would be necessary before a consensus or certification claim?

REVIEW BOUNDARY: AERA 1.2 WD requests technical criticism, mapping adjudication and pilot evidence. The maturity model and conformance method have not yet been validated through multi-organization pilots or inter-rater reliability studies; no consensus, certification, adoption or endorsement claim is authorized.

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How to read this specification

Executives should read the Executive Summary and Sections 2, 5, 7, 18, 21, 25, 27 and 29. Architects and system engineers should read the complete normative core. Implementers should focus on Sections 23-25 and Appendix B. Assessors should begin with Sections 20, 29-31 and Appendix A. The Word navigation pane provides the authoritative section structure; displayed pagination may change when edited.

Executive Summary

AERA begins from a structural change in enterprise architecture: humans remain accountable, but they may no longer be the primary real-time decision makers. Autonomous computational actors can now observe, infer, plan, negotiate and execute at a speed and scale that conventional application-centric governance was not designed to control. The relevant enterprise boundary is therefore no longer only the network perimeter or application portfolio. It is the boundary of delegated decision authority.

AERA defines the enterprise as a governed network of decisions. Applications, models, data platforms, sensors, cloud services, operational technology and quantum capabilities are implementation resources. They acquire architectural meaning only through the decisions they support, the authority they exercise and the effects they can produce. This shift permits leaders to ask a more durable question than whether a particular AI model is trustworthy: under what evidence, policy, identity, timing and recovery conditions may this actor take this action now?

The central technical mechanism is the Decision Assurance Fabric. The fabric is not a single product. It is a distributed control plane that binds an action proposal to mission intent, actor and workload identity, evidence provenance, policy, consequence class, uncertainty, cyber state, temporal validity, required approvals and recovery provisions. Its canonical authorization dispositions are PERMIT, PERMIT-WITH-CONSTRAINTS, DEFER, ESCALATE, SIMULATE, QUARANTINE and DENY. Confidence informs the disposition but never substitutes for authority.

AERA separates eight logical layers: Human Governance; Operational Intent; AI Cognitive Workforce; Decision Assurance Fabric; Hybrid Classical and Quantum Compute; Quantum Network Services; Global Sensor Mesh; and the Operational Environment. A continuous learning loop surrounds the stack, but learning is governed as a controlled change to operational behavior. Cybersecurity, identity, resilient timing and evidence management are cross-cutting properties rather than isolated layers.

Quantum technology is deliberately bounded. AERA does not assume fault-tolerant quantum computing, a general-purpose quantum internet or unconditional security. Quantum computing, sensing, key distribution and networking may be integrated only when they demonstrate a mission-relevant advantage, preserve classical control and recovery paths, and pass the same decision-assurance gates as any other resource. Post-quantum cryptography is treated as a current cryptographic migration obligation, distinct from quantum networking.

The implementation path is incremental. Organizations establish decision ownership and consequence classes; instrument evidence and identity; deploy policy decision and enforcement capabilities; introduce bounded autonomous pilots; validate recovery; scale across domains; and only then permit adaptive or cross-enterprise autonomy. Maturity is measured by evidence and operating behavior, not by the presence of AI tools or quantum hardware.

AERA is designed to complement, not replace, established frameworks. NIST AI RMF provides AI risk outcomes; NIST CSF 2.0 organizes cybersecurity risk; SP 800-207 supplies Zero Trust tenets; SP 800-53 supplies control objectives; NIST PQC standards provide quantum-resistant algorithms; and the NIST PNT Profile provides a risk-management foundation for PNT services [1], [10], [11], [13], [22], [25]-[31]. AERA adds the decision-centric architecture that binds those practices at the instant autonomous authority is exercised.

CORE RULE: Every autonomous decision shall be governed, assured, attributable, temporally valid and recoverable before it can produce a mission effect.

1. Scope and Conformance

PURPOSE: Define what AERA governs, what it does not claim, and how conformance is determined

1.1 Scope

AERA specifies a logical reference architecture and engineering method for enterprises in which software agents, AI-enabled services, autonomous platforms or cyber-physical systems exercise delegated decision authority. It applies across commercial, civil, critical-infrastructure, defense and research environments. It is technology-neutral at the control boundary: deterministic automation, statistical models, generative models, optimization engines and hybrid classical-quantum services may all be governed through the same decision contract.

The specification addresses organizational governance, operational intent, authority delegation, evidence, decision assurance, identity, cybersecurity, resilient timing, recovery, validation and lifecycle control. It does not prescribe a vendor stack, an enterprise data model, a universal confidence algorithm or a single organizational structure.

1.2 Conformance model

Conformance is claim-specific. An organization may claim conformance for an autonomous capability, a decision class, a mission thread or an enterprise domain. A claim SHALL identify the boundary, version, consequence class, external dependencies, inherited controls and assessment evidence. Broad claims such as 'AERA-compliant enterprise' are NONCONFORMING unless the assessed decision population and exclusions are explicit.

CONFORMING requires every applicable SHALL requirement to be satisfied with sufficient current evidence and no active deviation. CONDITIONALLY CONFORMING requires every applicable SHALL requirement either to be satisfied or covered by a current approved deviation. NONCONFORMING applies when an applicable SHALL is unsatisfied without a current approved deviation, or a deviation is expired or unapproved. INDETERMINATE applies when evidence is insufficient and no confirmed nonconformity exists; a confirmed nonconformity takes precedence. Possession of a named component does not demonstrate operational use. Evidence SHALL distinguish PRESENT, MONITORED, ACTIVE IN PATH, USED IN COMPUTATION and VALIDATED EFFECTIVE states.

1.3 Exclusions

AERA does not authorize autonomous use, establish legal sufficiency, certify model safety or replace sector regulation. It does not assert that confidence scores are objective probabilities, that explainability guarantees correctness, or that quantum communications eliminate cybersecurity risk. These are design and assurance questions that must be answered within the specific operating context.

1.4 Normative language and precedence

Only uppercase SHALL, SHALL NOT, SHOULD, SHOULD NOT, MAY and NEED NOT express normative modality. Normative content comprises Section 1; the definitions in Section 3; each uniquely identified AERA requirement; and each clause or table explicitly marked [NORMATIVE]. All other text, figures, rationales, examples, mission threads, crosswalks, roadmaps, research summaries and appendices are informative. A requirement MAY incorporate a [NORMATIVE] clause or table only by exact identifier.

The manifest-bound, hashed PDF edition of AERA-001 is the authoritative release representation. Appendix A in that edition is the canonical human-readable requirement register. The DOCX and XLSX files are accessible or machine-readable derivative representations generated from the same controlled source. They SHALL NOT establish alternate meaning. Different text for the same requirement identifier, definition or normative table is a release-blocking publication defect.

1.5 Assessment independence

Assessment independence is reported, not implied. I1 is self-assessment performed by the implementation or operator team and carries no independence claim. I2 is independent internal assessment performed by personnel organizationally separated from design and operation and without direct delivery accountability for the subject. I3 is independent external assessment performed by a person or organization contractually and operationally independent of the subject; sponsorship, scope and conflicts of interest are disclosed. Independence level does not change requirement applicability or technical acceptance criteria.

1.6 Profile applicability

All core requirements apply unless the declared profile and applicability matrix establish a technically justified not-applicable determination. AERA-DAF always includes applicable interface, authority, evidence, enforcement, recovery, cybersecurity and timing requirements; AERA-DEC additionally includes intent, consequence, human-control, actor-lifecycle and effect-observation requirements. Profiles may strengthen evidence and control floors. They cannot narrow a requirement merely because a named technology is absent when the governed capability is realized through another mechanism.

Normative requirements

AERA-CONF-001 A formal conformance claim SHALL identify the AERA version, profile, claimant, subject, decision population, architectural and operational boundaries, consequence scope, dependencies, inherited controls, exclusions, assessment-independence level, evidence-package identifier and assessment date, deviations, and date-based and event-based expiration conditions.

AERA-CONF-002 For every requirement in the declared core and profile set, the claimant SHALL record applicability as APPLICABLE or NOT APPLICABLE separately from the assessment result; an APPLICABLE item SHALL be assessed SATISFIED, NOT SATISFIED, INDETERMINATE, or APPROVED DEVIATION, while a NOT APPLICABLE item SHALL identify rationale, evidence, approving authority, and invalidating conditions and SHALL NOT carry a satisfaction result.

AERA-CONF-003 For each consequence-material capability, source, or control cited by an operational claim, the claimant SHALL identify the typed state model, report the highest evidenced control-deployment state as PRESENT, MONITORED, ACTIVE IN PATH, USED IN COMPUTATION, or VALIDATED EFFECTIVE, and SHALL NOT represent a lower state or a state from another model as establishing a higher state.

Comparative source context

AI RMF, the critical-infrastructure profile concept note, ISO/IEC 42001 and ISO/IEC/IEEE 15288 provide cross-sector risk, management and lifecycle context [1], [9], [39], [42]. SP 800-53A, RMF and IEEE 1012-2024 provide comparative assessment and lifecycle concepts [14], [16], [44]. They do not define AERA conformance.

2. Autonomous Enterprise Thesis

PURPOSE: Establish decision authority as the primary organizing variable of the autonomous enterprise

2.1 From applications to decisions

Traditional enterprise architectures decompose capabilities into business processes, applications, information and technology. That decomposition remains useful, but it assumes that accountable humans initiate, interpret or approve the consequential decisions. Autonomous systems invert part of that relationship: applications can select objectives, compose workflows and trigger physical or digital effects without synchronous human direction.

AERA therefore treats the decision as the primary unit of architecture. A decision is not merely a model output. It is a governed commitment to an action under uncertainty. The decision object binds the actor, authority source, objective, evidence, alternatives, constraints, expected effects, confidence, temporal window, approval state and recovery plan. This object can be evaluated, logged, reproduced and contested.

2.2 Authority as boundary

Network location does not define whether an autonomous action is legitimate. Model ownership does not define it. A valid boundary is the envelope of actions an accountable authority has delegated for a specified mission, context, time and consequence class. An actor crosses that boundary when it changes objectives, exceeds resource limits, acts on stale evidence, combines permissions into an unintended capability or produces an effect outside the approved operational domain.

Authority is composable only under explicit rules. Two individually permitted actions may create an unapproved combined effect. Decision assurance must therefore reason over sequences, dependencies and accumulated consequence, not only isolated API calls.

2.3 Accountability invariant

Delegation can transfer execution authority but cannot erase institutional accountability. Every decision class shall have a named human or governance body responsible for authorizing its use, defining acceptable risk and ordering suspension. Technical attribution supports accountability; it does not replace it.

2.4 Status of the thesis

The autonomous-enterprise thesis is a proposed architectural organizing principle, not an empirically validated scientific theory. AERA 1.2 WD treats it as a falsifiable engineering proposition: decision-authority modeling should produce clearer control boundaries, more complete traceability and more testable assurance claims than application-centric decomposition alone. Pilot results, counterexamples and inter-rater data are required before broader validity is claimed.

Design outcomes (informative)

- Model the enterprise as decision rights, evidence flows and effect paths.
- Bind autonomy to a defined mission and revocable authority.
- Evaluate cumulative and emergent effects across decision sequences.
- Retain human accountability even when human approval is not synchronous.

Normative requirements

AERA-THEORY-001 Every autonomous action SHALL resolve to a valid authority delegation.

AERA-THEORY-002 Decision evaluation SHALL consider material cumulative effects when actions compose.

AERA-THEORY-003 Each decision class SHALL have an accountable human owner.

Source context

The NIST agent initiative, NCCoE agent identity and authorization concept paper and TRAI 800-5 establish current problem context for agents that plan and act across systems [5]-[7]. They do not define the AERA thesis or constitute prior review, adoption or endorsement.

3. Formal Definitions

PURPOSE: Provide stable terms for architecture, engineering and assessment

3.1 [NORMATIVE] Core terms

The definitions in this section are normative. Sector profiles MAY refine a term but SHALL NOT contradict its essential meaning. Where an external framework uses a term differently, the mapping SHALL state the semantic difference rather than imply equivalence.

[NORMATIVE AERA-NORM-T03] Terminology

Table 3. [NORMATIVE AERA-NORM-T03] Terminology

Term	AERA definition
Autonomous actor	A computational entity able to select or sequence actions within delegated authority without synchronous human direction.
Autonomous decision	A governed commitment by an autonomous actor to execute, recommend or cause an action under uncertainty.
Decision class	A governed population of actions sharing authority, intent, effect, consequence and assurance rules.
Decision transaction	The lifecycle record from proposal through disposition, execution, observed effect and closure or recovery.
Decision authority	The permission to select and cause a defined class of effects within stated mission, context, time and risk limits.
Authority delegation	A revocable, attributable grant of decision authority from an accountable authority to an actor or role.
Decision contract	The machine-verifiable representation of objective, authority, evidence requirements, constraints, consequence class, temporal validity and recovery obligations.
Decision Assurance Fabric	Distributed capabilities that evaluate a proposed action against authority, evidence, identity, policy, risk, confidence, timing and recovery requirements.
Conformance subject	The architecture, DAF service, decision class, mission thread, domain or sector-profile implementation identified by a formal claim.
Claimant	The legal or organizational entity that makes and remains accountable for a formal AERA claim.
Degraded mode	Pre-engineered operating behavior used when a dependency or assurance condition is outside nominal bounds.
Hard gate	An assurance predicate that policy does not permit another dimension to compensate for.
Material context change	A change capable of altering authority, evidence, consequence, policy, enforcement, time or recovery outcome.
Evidence	Information used to support a claim about state, identity, provenance, performance, threat, consequence or compliance.
Typed evidence state	A state claim named within a specific lifecycle model; states in different models are not equivalent unless an explicit mapping is defined.
Confidence	A bounded measure of support for a decision-relevant claim; confidence is meaningful only with calibration, provenance and context.
Consequence class	A category expressing the credible severity, reversibility and scope of an action's effects.
Human accountability	The continuing obligation of designated people or governing bodies to answer for delegated autonomous behavior.
Operational intent	A machine-actionable expression of mission objectives, priorities, constraints and measures of success.
Recovery envelope	The set of states, time limits and resources within which the system can arrest, reverse, contain or compensate for an incorrect action.
Temporal validity	The condition that evidence, authority and policy were valid and coherently ordered for the decision time.
Assurance case	A structured claim-argument-evidence demonstration that a decision class is acceptably governed for a defined context.
Material effect	An attempted or achieved externally consequential change, commitment, command, disclosure, allocation or release whose credible consequence meets the decision class materiality rule.
Attempted effect	An effect request accepted or initiated at an enforcement boundary; it is not proof that the intended external state was achieved.
Achieved effect	An externally consequential state change supported by the required effect observations and reconciled with the transaction.
Independent evidence	Evidence whose relevant source, control, data, model, provider, clock, network, power and administrative failure domains satisfy the declared independence model.
Current	Within the decision-specific validity, freshness and context bounds established before assessment.

Term	AERA definition
Sufficient evidence	Evidence that satisfies the predeclared claim criteria, coverage, quality, independence and uncertainty requirements for the consequence class.
Credible consequence	An effect supported by scenario, hazard, threat, dependency or operational evidence and not dismissed solely because its probability is uncertain or low.
Consequence-material	Capable of changing consequence class, authorization disposition, control floor, recovery obligation or residual-risk acceptance.
Validated	Demonstrated against predeclared criteria in a configuration and environment representative of the stated claim; not merely reviewed or tested once.
Positive oversight	A human-control mode requiring timely, attributable action by an authorized reviewer rather than relying only on post-effect notice.
Acceptable risk	Residual risk explicitly accepted by an authority empowered for the decision class, scope, duration and consequence; not the absence of known defects.
Semantic equivalence	Demonstrated preservation of required meaning, constraints, lifecycle behavior, failure semantics and observable evidence across representations or implementations.

4. Architectural Philosophy

PURPOSE: Define the reasoning model that governs AERA design choices

4.1 Governance first

AERA places governance at the center because mission, law, ethics and accountability determine what the enterprise is allowed to optimize. Governance is implemented through decision rights, policies, limits, review forums and technical enforcement. It is not satisfied by a principles statement disconnected from runtime behavior.

4.2 Evidence before effect

A proposed action is a claim that the actor is authorized, relevant evidence is sufficient, anticipated effects are acceptable and recovery remains possible. The action may proceed only when the required claims are supported. Higher consequence requires stronger, more independent and more current evidence.

4.3 Separation of proposal and authorization

An AI system may generate plans, but the mechanism that authorizes their effects should be logically separable from the proposer. This separation reduces self-approval, supports independent policy enforcement and allows organizations to replace models without redesigning governance. Separation may be physical, administrative or cryptographic depending on consequence.

4.4 Graceful degradation

Safe autonomy is measured partly by behavior under missing, conflicting or degraded evidence. The preferred failure mode is not universally 'fail closed.' Life-safety and infrastructure systems may require constrained continuity. AERA requires an explicitly engineered degraded mode with reduced authority, narrowed objectives and heightened observability.

Design outcomes (informative)

- Authority exceeds autonomy.
- Mission overrides local optimization.
- Evidence strength scales with consequence.
- Proposal and authorization remain independently controllable.
- Recovery and degraded operation are designed before deployment.

Source context

Governance, evidence, cyber resilience and ethical-constraint concepts are consistent with systems-security engineering, cyber-resilience, risk and value-based design practices [18], [19], [41], [43]. The combined governance-first decision model is an AERA architectural judgment.

5. Design Principles

PURPOSE: State the durable rules that constrain compliant AERA implementations

P1 Human accountability cannot be delegated

Authority may be delegated; institutional responsibility remains with named human leadership.

P2 Authority shall exceed autonomy

No actor may exercise rights beyond a bounded, revocable delegation.

P3 Every autonomous action requires measurable assurance

Model output alone is insufficient authorization evidence.

P4 Mission overrides optimization

Local utility shall not supersede mission objectives, law or policy.

P5 Cybersecurity is an architectural control system

Identity, policy, evidence, communication, resilience and recovery span every layer.

P6 Confidence is dynamic

Confidence must be recomputed when evidence, context, threat or system state changes.

P7 Recovery defines safe autonomy

An action outside the validated recovery envelope requires stronger control or prohibition.

P8 Learning is a governed change

Behavior-changing updates require configuration control, evaluation and rollback.

P9 Time is part of decision integrity

Ordering, freshness and synchronization are evidence, not infrastructure trivia.

P10 Effects determine assurance

Control rigor is based on credible consequence, not model sophistication or brand.

P11 Independent evidence reduces correlated failure

High-consequence decisions require diversity of source, method or administrative control.

P12 Accountability requires reconstructability

The enterprise must be able to explain who authorized what, on which evidence, under which policy and with what result.

Normative requirements

AERA-PR-001 For each autonomous decision class, the organization SHALL designate an accountable human or governing body that retains responsibility for mission legitimacy, delegated authority, residual-risk acceptance, suspension, and reassessment.

AERA-PR-002 An autonomous actor SHALL exercise only decision rights contained in a current, attributable, revocable delegation whose scope includes the actor, action, target, purpose, context, consequence ceiling, and validity interval.

AERA-PR-003 A model output SHALL NOT authorize a material effect unless the applicable authority, evidence, policy, consequence, temporal-validity, and recovery predicates are satisfied and an enforcement point accepts a current authorization artifact bound to the transaction.

AERA-PR-004 If an optimization objective conflicts with current operational intent, law, policy, or an applicable prohibition, the implementation SHALL apply the declared precedence rules; unresolved conflict SHALL produce a non-permit disposition.

- AERA-PR-005** Each AERA layer and decision-relevant interface SHALL identify the identity, authorization, evidence-integrity, communications-security, resilience, logging, and recovery properties that it provides, requires, and verifies; allocation of a property to one layer SHALL NOT imply that another layer inherits it without validation.
- AERA-PR-006** The DAF SHALL re-evaluate affected assurance claims before further effect whenever a policy-defined material change occurs in evidence, context, threat state, actor or system state, temporal state, or recovery readiness.
- AERA-PR-007** Before authorizing a proposal that can exceed a validated recovery envelope, the accountable authority SHALL either prohibit the proposal or approve a consequence-specific control set that includes narrower enforceable bounds, additional independent prevention or inhibition, effect observation, and explicit residual-risk acceptance.
- AERA-PR-008** Before a behavior-changing model, prompt, memory, retrieval source, policy, data set, tool, dependency, threshold, or orchestration artifact is used under production authority, the organization SHALL version and provenance-bind the artifact, analyze its impact, test and approve it, make its use observable, and provide a tested rollback or safe-replacement path.
- AERA-PR-009** Each decision contract SHALL specify and enforce maximum evidence age, event-order and synchronization tolerances, time or position uncertainty bounds as applicable, and the disposition to use when a temporal bound is exceeded.
- AERA-PR-010** The minimum control floor for a decision class SHALL be selected from its worst credible consequence and SHALL NOT be reduced on the basis of model type, implementation brand, vendor assertion, or claimed technical sophistication.
- AERA-PR-011** For each CC-3 or CC-4 decision class, evidence or controls relied upon as independent SHALL have documented source, method, controller, administration, provider, software, network, power, supply-chain, and time-reference failure domains; correlated items SHALL NOT be counted as independent without an approved rationale and compensating controls.
- AERA-PR-012** For every material decision transaction, the implementation SHALL retain a tamper-evident causal record that identifies who or what proposed, authorized, enforced, observed, overrode, or recovered the action; the evidence and policy used; the authorization and validity; and the observed result.

Source context

The principles synthesize governance, security, resilience, lifecycle, PNT and public-sector responsible-AI practices [1], [10], [18]-[22], [24], [39], [42], [43], [46]-[49]. Their combined form and normative effect are AERA constructs.

6. Assumptions and Constraints

PURPOSE: Make the environmental and technical limits of AERA explicit

6.1 Assumptions

AERA assumes heterogeneous legacy and modern systems; imperfect and adversarial data; intermittent connectivity; multiple administrative domains; changing mission priorities; and a workforce in which humans and autonomous actors collaborate. It assumes no model is universally trustworthy and no evidence source is permanently authoritative.

6.2 Constraints

Latency, bandwidth, safety, privacy, classification, sovereignty, export control, cost, power, environmental tolerance and legal authority constrain implementation. Some operational decisions cannot wait for centralized authorization. Some evidence cannot leave its domain. Some recovery actions are irreversible. Architecture shall represent these constraints rather than hide them behind an enterprise service abstraction.

6.3 Uncertainty and non-determinism

Probabilistic components can support dependable systems only when uncertainty is characterized, monitored and bounded by deterministic controls. A confidence value without calibration data, decision context and failure-cost interpretation is not an assurance metric. Unknown conditions shall remain representable; systems shall not force uncertainty into a false binary.

Assumption-to-control implications

Table 4. Assumption-to-control implications

Condition	Architectural implication	Minimum control
Disconnected operations	Local decisions may continue without central services	Pre-positioned policy, time-bounded authority, local audit and reconciliation
Adversarial evidence	Inputs may be manipulated or selectively withheld	Provenance, source diversity, anomaly detection and contradiction handling
Legacy OT	Enforcement points may be unavailable	Gateway mediation, command allowlists and physical safety interlocks

Condition	Architectural implication	Minimum control
Cross-domain data	Evidence may have use and disclosure restrictions	Policy-aware labeling, minimization and controlled release
Non-reversible effect	Rollback is impossible or incomplete	Simulation, independent approval and consequence-specific inhibition

Source context

The assumption set is informed by AI risk, general risk assessment, cyber-resilience and final PNT-profile practices [1], [17], [19], [22], [41]. AERA makes the resulting decision-authority implications explicit.

7. AERA Logical Architecture

PURPOSE: Define the eight-layer architecture, its flows and cross-cutting properties

7.1 Primary architecture

Figure 1 is an informative visualization of the normative logical-layer catalog in Section 7.4. Concentric placement expresses governance radiating outward toward real-world effects and evidence flowing inward from the operational environment. The rings are logical capability domains and extension points, not deployment tiers or products. A conforming implementation does not require quantum computing, quantum networking or a globally deployed sensor mesh. Layers 5 through 7 may be realized by conventional, specialized, local or federated services, and an absent technology is recorded as not applicable only when the corresponding capability and risk are genuinely outside the claim boundary [5], [6], [11], [18], [19], [42], [45].

7.2 Decision and evidence flows

Outward flow begins with governance and mission, becomes operational intent, is decomposed into cognitive work, passes through decision assurance, consumes compute and network services, and produces effects. Inward flow begins with sensing and operational feedback, is time-stamped and validated, becomes evidence, updates confidence and informs governance. The paths are concurrent: authorization may revoke or constrain an action at any point before an irreversible effect.

7.3 Cross-cutting properties

Cybersecurity, identity, time, provenance, safety, privacy, observability and recovery cross every ring. Implementations shall not assign these responsibilities to a single layer and assume downstream inheritance. Each interface must state the properties it provides, requires and verifies.

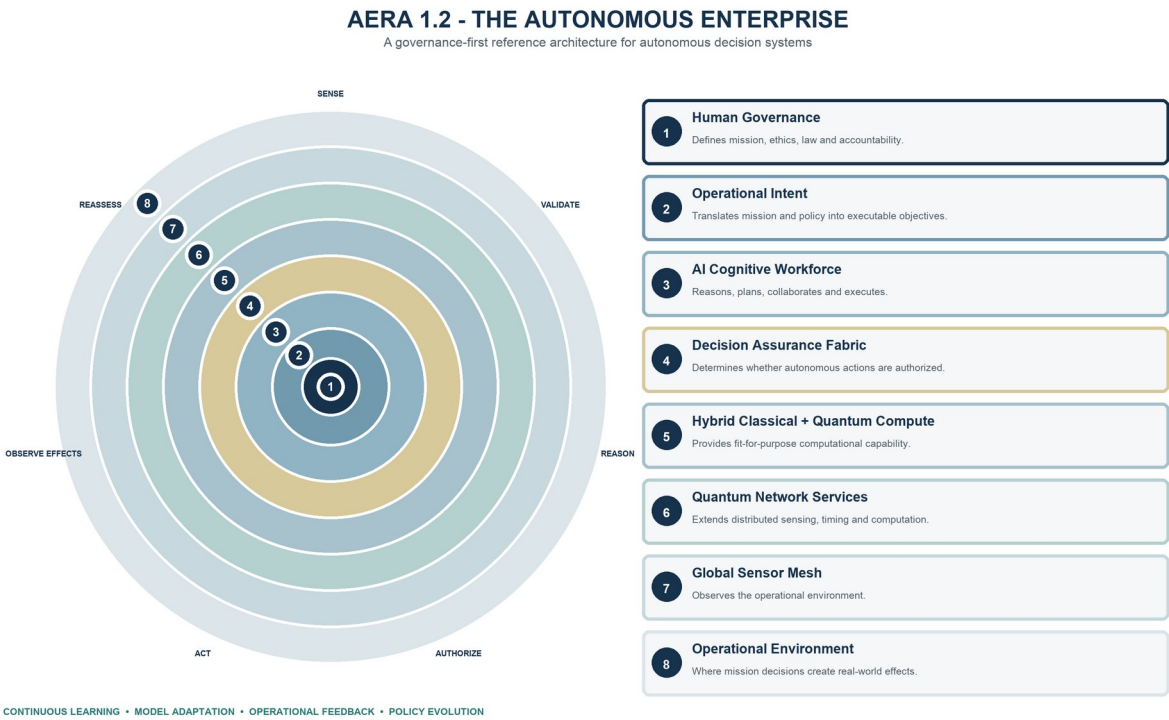


Figure 1. Informative visualization of the AERA 1.2 logical reference architecture. Concentric layers express the path from accountable governance to operational effect.

7.4 [NORMATIVE AERA-NORM-T05] Logical-layer catalog and responsibilities

Table 5. 7.4 [NORMATIVE AERA-NORM-T05] Logical-layer catalog and responsibilities

Layer	Primary responsibility	Assurance question
Human Governance	Define mission, policy, ethics and accountability	Who may delegate this authority and accept residual risk?
Operational Intent	Express executable objectives and constraints	Is the intended outcome valid, current and unambiguous?
AI Cognitive Workforce	Propose, coordinate and execute work	Is the actor competent and acting within role?
Decision Assurance Fabric	Authorize or constrain proposed effects	Does current evidence support this action under policy?
Hybrid Compute	Execute fit-for-purpose computation	Is the service characterized, isolated and monitored?
Quantum Network Services	Provide quantum-enabled distribution or sensing	Is there a demonstrated advantage and safe classical fallback?
Global Sensor Mesh	Acquire observations and state evidence	Are source, integrity, freshness and uncertainty known?
Operational Environment	Receive and reveal mission effects	Are effects bounded, observable and recoverable?

8. Human Governance

PURPOSE: Bind autonomous operation to legitimate mission, risk ownership and accountability

8.1 Governance functions

Human governance sets mission priorities, establishes prohibited outcomes, designates accountable authorities, allocates risk tolerance and orders suspension. Boards and executives govern portfolios and consequences; mission owners govern operational objectives; technical authorities approve architecture and evidence; legal, privacy, safety and cybersecurity functions provide constraints and independent challenge.

8.2 Delegation register

Delegations shall be machine-readable and human-reviewable. Each record identifies grantor, grantee identity or role, permitted decisions, prohibited effects, context, consequence ceiling, geographic or data boundary, start and expiration, required evidence, escalation path and revocation mechanism. Delegation is not inferred from API access.

8.3 Oversight cadence

Governance combines runtime oversight with periodic review. Runtime mechanisms detect boundary crossings and aggregate risk. Periodic governance reviews decision performance, near misses, override patterns, model and evidence drift, unresolved exceptions, recovery exercises and changes to external obligations.

Design outcomes (informative)

- Named accountability for every decision class.
- A maintained authority and delegation register.
- Independent review for high-consequence decisions.
- Immediate suspension and bounded revocation mechanisms.

Normative requirements

AERA-GOV-003 Applicable privacy, civil-rights, equity, safety, transparency, recordkeeping, redress, and data-governance obligations SHALL be identified for each decision class and translated into enforceable intent, policy, evidence, and oversight requirements.

AERA-GOV-004 A consequence-material decision class affecting people SHALL define notice, contestability, human review, and redress mechanisms, or SHALL document the legal and mission basis for inapplicability.

8.4 Broader AI governance obligations

AERA governs decision authority and assurance; it does not claim to be a complete substitute for privacy, civil-rights, equity, safety, transparency, records, data-governance or sector-specific impact regimes. The accountable owner translates applicable obligations into operational intent, policy gates, evidence, oversight, notice, contestability and redress. AI RMF, ISO/IEC 42001, IEEE 7000 and public-sector responsible-AI sources provide complementary governance practices [1], [39], [43], [46]-[49].

9. Operational Intent

PURPOSE: Translate mission and policy into testable, machine-actionable objectives

9.1 Intent model

Operational intent contains desired effects, priority, constraints, success measures, decision horizon, resources, dependencies and stop conditions. It should express what must be achieved and what must not occur without hard-coding a single implementation plan. Intent is versioned and signed so a decision can be traced to the exact mission state in force.

9.2 Conflict resolution

Autonomous actors may receive conflicting objectives from different authorities. Intent resolution uses precedence, jurisdiction, consequence and temporal validity. The system shall not silently optimize across contradictory directives. Material conflicts are surfaced to the accountable mission owner or resolved by pre-authorized rules.

9.3 Intent validation

Before release, intent is checked for completeness, satisfiability, prohibited effects, resource feasibility, measurement quality and consistency with higher policy. Simulation and red-team review are required when ambiguous intent could yield unsafe instrumental behavior.

Design outcomes (informative)

- Versioned mission objectives and constraints.
- Explicit precedence and conflict-resolution rules.
- Machine-testable success and stop conditions.
- Traceability from action to governing intent.

10. AI Cognitive Workforce

PURPOSE: Govern AI-enabled reasoning, planning, coordination and execution as a managed workforce

10.1 Actor lifecycle

Each autonomous actor has an owner, identity, role, competency profile, approved tools, data entitlements, consequence ceiling and retirement procedure. Model registration alone is insufficient because an actor includes prompts, orchestration, memory, tools, policies, runtime, retrieval sources and integration code.

10.2 Workforce coordination

Multi-agent collaboration introduces delegation chains, emergent role acquisition, duplicated action and authority laundering. Coordination protocols shall propagate authority constraints, preserve attribution, prevent unauthorized subdelegation and provide idempotency for actions with material effects.

10.3 Human interaction

Human approval is one control mode, not the universal solution. High-frequency or time-critical operations may require pre-authorized envelopes, while exceptional or irreversible actions require positive approval. Interfaces shall present material evidence, uncertainty, alternatives, expected effects and recovery options rather than demand ceremonial clicks.

Design outcomes (informative)

- Unique workload identity and accountable ownership.
- Tool use constrained by role and decision contract.
- Controlled memory and retrieval provenance.
- No unauthorized subdelegation or self-expansion of authority.

Normative requirements

AERA-EVID-011 Data used to train, adapt, evaluate, or operate a consequence-material autonomous capability SHALL have documented provenance, quality, representativeness, limitations, permissible use, and change control.

10.4 Relationship to current agent-security work

Current NIST work distinguishes software and AI agents as identities that require explicit authorization, constrained access and security evaluation. The 2026 AI Agent Standards Initiative is a program, the NCCoE identity and authorization paper is a draft concept, TRAI 800-5 summarizes stakeholder input, and IR 8596 is an initial preliminary Cyber AI Profile draft; none is a final agent-control standard or an endorsement of AERA [4]-[8]. AERA adds the decision-authority transaction, effect-level binding and closure semantics proposed in Sections 23 and 24.

11. Decision Assurance Fabric

PURPOSE: Determine whether a proposed autonomous action is permitted under current evidence and authority

11.1 Assurance transaction

A decision assurance transaction receives an action proposal and returns a disposition. Inputs include actor identity and integrity, authority delegation, current intent, relevant evidence, evidence provenance and freshness, threat state, consequence class, uncertainty, dependencies, approvals and recovery readiness. The disposition is cryptographically bound to the evaluated proposal and expires when material context changes.

11.2 Decision Assurance Score

AERA does not mandate a universal scalar score. A single number can conceal unacceptable weakness in one dimension. Implementations should use a vector of calibrated claims such as authority validity, evidence adequacy, model competence, cyber integrity, temporal coherence, safety margin and recoverability. Policy converts that vector into an action disposition.

11.3 Dispositions

PERMIT authorizes the proposal as submitted. PERMIT-WITH-CONSTRAINTS authorizes only an evaluated narrowed proposal under enforced constraints. DEFER waits for evidence. ESCALATE requests designated human or supervisory review. SIMULATE requires a non-effecting trial. QUARANTINE isolates the actor or evidence source. DENY prohibits execution. Every disposition includes rationale, policy version and expiry.

11.4 Independence and fail-safe behavior

The proposer is not the sole source of evidence for its own authorization. High-consequence actions use independent evidence or an independent enforcement path. When the fabric is unavailable, behavior follows the pre-engineered degraded mode; availability failure does not silently become authorization.

Design outcomes (informative)

- Policy decision and enforcement are independently controllable.
- Authorization is bound to exact action, context and expiration.
- Confidence dimensions remain visible and calibrated.
- High-consequence decisions use independent evidence.
- All dispositions and overrides are reconstructable.

12. Hybrid Classical and Quantum Compute

PURPOSE: Select computational resources by verified mission advantage, constraint and risk

12.1 Workload brokerage

A workload broker evaluates latency, accuracy, assurance, data sensitivity, cost, energy, availability and solver maturity. Classical compute remains the control and recovery baseline. Specialized accelerators, high-performance computing and quantum processors are invoked only for defined workload classes.

12.2 Result assurance

A computational result used in a decision requires provenance, algorithm and configuration identifiers, input lineage, validation method and uncertainty. For optimization, the system records objective function, constraints, feasible-set checks and comparison against a classical baseline. For simulation, it records model-form assumptions and sensitivity.

12.3 Isolation and continuity

Experimental services are isolated from direct mission effect. Quantum and specialized compute dependencies shall have timeouts, bounded retries, result plausibility checks and a classical fallback or explicit no-service state.

Design outcomes (informative)

- Fit-for-purpose workload placement.
- Reproducible configuration and input lineage.
- Classical control, plausibility checks and recovery path.
- No direct mission effect from unvalidated experimental services.

13. Quantum Integration

PURPOSE: Integrate quantum technologies without overstating readiness, security or mission value

13.1 Capability classes

AERA distinguishes quantum computation, quantum sensing, quantum networking, quantum random-number generation and quantum key distribution. These classes have different readiness, interfaces and assurance needs. The presence of one does not imply availability of another or a general-purpose quantum internet.

13.2 Quantum advantage gate

Integration requires a documented mission hypothesis, baseline comparison, performance and cost measures, error characterization, security analysis, operational constraints and exit criteria. A laboratory demonstration does not establish operational advantage. The decision gate shall distinguish physical hardware present, signals observed, service active in the operational path and output used in the computed decision.

13.3 Networking and security boundaries

Quantum networks complement rather than replace classical networks. Classical authentication, control, routing, orchestration and management remain attack surfaces. Quantum key distribution is not synonymous with end-to-end security and does not remove endpoint, implementation, denial-of-service or supply-chain risks.

13.4 PQC distinction

Post-quantum cryptography is deployable classical cryptography designed to resist known quantum attacks. AERA treats crypto-agility, inventory, dependency mapping, migration and interoperability testing as present obligations. They are independent of whether an enterprise deploys quantum networking.

Design outcomes (informative)

- Separate claims by quantum capability class.
- Require measured advantage against a defined baseline.
- Preserve classical control and recovery paths.
- Treat PQC migration as a current security program.

13.5 Evidence boundary

DOE quantum-network publications describe research direction rather than enterprise production assurance [33], [34]. PQC claims are narrower: FIPS 203, 204 and 205 specify algorithms; SP 800-227 provides KEM recommendations; CSWP 39 Update 1 addresses crypto agility; and IR 8547 remains a draft transition-planning source [25]-[30]. None of these sources establishes protocol security, implementation validation, interoperability or mission advantage by itself.

14. Global Sensor Mesh

PURPOSE: Transform distributed observations into trustworthy decision evidence

14.1 Sensor-to-evidence chain

Sensors include RF, GNSS/PNT, signals of opportunity, cyber telemetry, OT/ICS, industrial IoT, space, environmental, financial, health, transportation and telecommunications sources. A reading becomes evidence only after identity, calibration, provenance, temporal quality, uncertainty and applicable use restrictions are characterized.

14.2 Fusion and contradiction

Fusion shall preserve source-level traceability. Correlated sources do not provide independent confirmation merely because they arrive through different interfaces. Contradictions are represented and investigated; the fusion service shall not average away a potentially adversarial discrepancy.

14.3 Edge operation

Edge nodes may filter, summarize or authorize locally under communications constraints. Local processing shall preserve enough provenance for later reconstruction and shall declare when transformation prevents independent re-evaluation.

Design outcomes (informative)

- Authenticated source and configuration identity.
- Known calibration, uncertainty and freshness.
- Traceable fusion and contradiction handling.
- Local evidence retention for disconnected operations.

15. Operational Environment

PURPOSE: Bound the real-world domain in which autonomous effects occur

15.1 Effect model

The operational environment includes power, water, transportation, healthcare, finance, defense, space, manufacturing, communications, public safety and supply chains. AERA models commanded effects, secondary effects, affected stakeholders, geographic and temporal propagation, coupling to safety systems and observability of outcomes.

15.2 Effect verification

An issued command is not proof of an achieved effect. The enterprise observes response, compares it to the intended outcome, detects unintended effects and updates the decision record. Where independent observation is unavailable, the architecture marks the outcome unverified.

15.3 Reversibility

Recovery may mean rollback, safe shutdown, containment, compensation, manual takeover or graceful continuation. Each decision class defines its recovery envelope and the point beyond which an effect becomes irreversible.

Design outcomes (informative)

- Explicit operating-domain and affected-stakeholder model.
- Independent effect observation where feasible.
- Declared recovery envelope and irreversibility threshold.
- Safety interlocks independent of cognitive actors.

16. Cybersecurity Architecture

PURPOSE: Make cybersecurity the distributed control system for autonomous authority

16.1 Identity and continuous authorization

Human, workload, device, model, service and data identities are first-class. Authentication establishes an identity claim; authorization evaluates whether that identity may perform the exact action under current conditions. Long-lived credentials and broad tool tokens are inconsistent with bounded autonomy. Short-lived, audience-restricted and action-scoped credentials are preferred.

16.2 Policy and enforcement

Policy combines mission rules, security controls, legal obligations, data-use restrictions, safety constraints and consequence thresholds. Decision points evaluate; enforcement points mediate effects. Enforcement must exist close enough to the effect that bypass, race and time-of-check/time-of-use risks are controlled.

16.3 Evidence integrity and audit

Evidence objects are signed or otherwise integrity-protected, time-bound, labeled with provenance and retained according to consequence and legal need. Logs shall capture causal links among intent, proposal, authorization, tool call, system response and observed effect. Security monitoring uses the same decision graph to detect anomalous authority use.

16.4 Adversary model

Threats include prompt and data injection, model and supply-chain compromise, identity theft, policy manipulation, evidence spoofing, time manipulation, tool misuse, covert subdelegation, collusion among agents, reward hacking, denial of assurance, log tampering and recovery inhibition. Controls are selected from threat-informed analysis, not from AI terminology alone.

16.5 Crypto-agility

Cryptographic inventories include algorithms, protocols, libraries, certificates, hardware roots, data-retention periods and external dependencies. Migration plans test PQC performance, message size, interoperability and rollback. Hybrid modes may support transition but add complexity and do not excuse inventory or validation.

Security control-plane functions

Table 6. Security control-plane functions

Function	AERA responsibility	Representative evidence
Identity	Establish human, workload, device and data identity	Attestation, credential chain, binding to role
Authorization	Decide exact action under context	Policy decision record and expiry
Enforcement	Mediate command or effect	Gateway or native control log
Integrity	Protect software, model, policy and evidence	Signatures, measurements, provenance
Detection	Identify abnormal authority use	Decision-graph analytics and alerts
Recovery	Revoke, isolate, restore and reconcile	Exercise results and recovery-time evidence

Normative requirements

AERA-CY-001 Credentials used on a material-effect path SHALL be scoped, to the extent supported, to the authorized actor, audience, action, target, context, and validity interval; where native credential scoping is insufficient, the enforcement point SHALL enforce equivalent restrictions.

AERA-CY-002 Each material effect SHALL pass through an enforcement point that is independently controllable from the proposer and capable of mediating the exact authorized actor, action, target, parameters, and context before the effect becomes irreversible or exits the recovery envelope.

AERA-CY-003 For each material decision transaction, the implementation SHALL retain a tamper-evident record sufficient to reconstruct the causal chain from intent and authority through actor and configuration, proposal, evidence and provenance, policy and assurance verdicts, disposition, credential use, enforcement, observed effect, override, exception, and recovery.

AERA-CY-004 Cryptographic migration SHALL be driven by inventory, exposure period and interoperability evidence.

Source context

The control-plane model draws on CSF 2.0, Zero Trust, SP 800-53/53A, SSDF, supply-chain risk management and threat knowledge bases, while the exact decision-authority semantics remain AERA constructs [4], [6]-[8], [10]-[14], [20], [21], [35], [37], [38]. Crypto-agility and PQC migration claims use FIPS 203-205, SP 800-227 and CSWP 39 Update 1 with the draft status of IR 8547 preserved [25]-[31].

17. Resilient Timing and PNT

PURPOSE: Preserve causal ordering, evidence freshness and position/time integrity

17.1 Time as assurance evidence

Decision assurance depends on knowing whether authority, policy, evidence and system state were valid at the relevant instant. Timestamp presence is insufficient. The architecture requires source identity, traceability, uncertainty, holdover behavior, distribution path and health state. Ordering may be more important than absolute UTC accuracy for some decision classes; both shall be specified.

17.2 Multi-source resilience

GNSS is a valuable PNT source but may be disrupted or manipulated. Resilience uses source diversity, local oscillators, authenticated distribution where available, cross-checks against terrestrial or signals-of-opportunity sources, anomaly detection and operational holdover. Independence claims require analysis of common infrastructure and shared failure modes.

17.3 PNT state model

AERA recommends four separate status claims: source physically present; source monitored; source accepted by the timing or navigation service; and source used in the computed solution. Architecture and test reports shall not treat these states as interchangeable. The solution publishes uncertainty, source set, integrity state and degradation mode to decision consumers.

17.4 Time-aware authorization

Each decision contract specifies maximum evidence age, clock uncertainty, permitted reorder window and authority expiry tolerance. If temporal quality falls outside bounds, the fabric constrains or denies actions according to the degraded-mode policy.

Representative timing requirements by consequence

Table 7. Representative timing requirements by consequence

Decision class	Time property	Illustrative requirement
Advisory	Freshness and traceable ordering	Evidence age and source recorded; stale input flagged
Material digital effect	Bounded synchronization	Verified time source set; uncertainty below policy threshold
Cyber-physical control	Deterministic sequencing and holdover	Local resilient time; bounded command ordering; tested degraded mode
Forensic/high-assurance	Non-repudiable causal reconstruction	Protected timestamps, cross-domain correlation and retention

Normative requirements

AERA-PNT-001 Each decision-relevant time or position claim SHALL identify its contributing sources, distribution or derivation path, reference time scale or coordinate frame as applicable, observation time, uncertainty, integrity state, required traceability, valid interval, and degraded-state behavior.

AERA-PNT-002 PNT reporting and decision logic SHALL distinguish PRESENT, MONITORED, ACCEPTED, and USED IN SOLUTION for each source and SHALL NOT treat those states as interchangeable.

AERA-PNT-003 Each decision contract that depends on time, order, freshness, synchronization, or position SHALL define and enforce bounds for evidence age, event order, time or position uncertainty, authority and policy expiration tolerance, holdover, and behavior when any bound is exceeded.

Source context

NIST IR 8323 Rev. 1 is the final mapped PNT baseline and EO 13905 supplies policy context. IR 8323 Rev. 2 was an initial public draft on the 1.2 WD access date and does not replace the final baseline [22]-[24]. AERA adds decision-contract bounds and the explicit distinction among source presence, monitoring, acceptance and use in a computed solution.

18. Implementation Roadmap

PURPOSE: Sequence adoption from decision inventory to scaled governed autonomy

18.1 Phase 0 - Mobilize

Establish executive sponsorship, mission scope, legal and safety constraints, architecture authority and an initial prohibition list. Define the first decision population and baseline current decision latency, error, override, loss and recovery performance.

18.2 Phase 1 - Discover

Inventory autonomous and automated decisions, owners, effects, data, tools, credentials, models, dependencies and recovery paths. Assign consequence classes and identify authority gaps. This phase commonly reveals autonomous behavior embedded in workflow tools that was never governed as autonomy.

18.3 Phase 2 - Instrument

Introduce workload identity, evidence provenance, policy versioning, decision logging, time-quality telemetry and effect observation. Instrumentation precedes broad autonomy because an unobservable system cannot generate an assurance case.

18.4 Phase 3 - Bound and pilot

Select reversible, observable decisions with meaningful value. Implement decision contracts, independent enforcement, escalation and recovery. Conduct shadow, simulation and limited-effect operations before enabling production authority.

18.5 Phase 4 - Integrate

Connect decision assurance across domains, normalize evidence semantics, govern delegation chains and add portfolio risk views. Retire duplicate control paths only after equivalence is demonstrated.

18.6 Phase 5 - Adapt

Permit governed learning and dynamic authority only after change detection, re-evaluation, rollback and continuous monitoring are demonstrated. Expansion is evidence-triggered rather than schedule-triggered.

Roadmap decision gates

Table 8. Roadmap decision gates

Gate	Exit evidence	Executive decision
G0 Scope	Named owner, mission thread, decision inventory plan	Authorize discovery
G1 Observability	Decision/effect trace, identity, time and provenance coverage	Authorize bounded pilot
G2 Assurance	Passing safety, cyber, performance and recovery tests	Authorize limited production
G3 Scale	Cross-domain policy and cumulative-risk controls	Authorize broader delegation
G4 Adaptation	Validated drift detection, re-evaluation and rollback	Authorize governed learning

18.7 Enterprise risk integration

AERA risk and consequence information should flow into existing enterprise risk-management processes rather than remain in an autonomous-systems silo. An implementation should link decision classes, credible consequences, control gaps, residual risk, acceptance authorities, measures and reassessment triggers to organization and enterprise risk registers. NIST IR 8286 Rev. 1 supplies a current risk-register and ERM integration approach; it does not define AERA consequence classes or accept AERA risk [64].

19. Trade Studies and Architectural Rationale

PURPOSE: Make key design choices explicit and reusable

19.1 Method

Trade studies evaluate mission effectiveness, assurance, latency, resilience, cost, interoperability, operator burden, evidence quality and reversibility. Scores are contextual; AERA does not prescribe universal weights. Each choice records assumptions and sensitivity to weight changes.

Representative AERA trade studies

Table 9. Representative AERA trade studies

Decision	Option A	Option B	AERA position
Authorization placement	Centralized policy service	Federated/local enforcement	Federate for latency and resilience; preserve common policy semantics and audit.
Human control	Approval for every action	Pre-authorized envelope	Use consequence-based control; synchronous approval for exceptional or irreversible effects.
Confidence representation	Single score	Multi-dimensional assurance vector	Use vector; derive disposition by policy to avoid masking weak dimensions.
AI architecture	Single general agent	Role-specialized actors	Prefer least-authority roles; coordinate through controlled delegation.
Quantum adoption	Technology-first platform	Mission-gated service	Require baseline advantage, characterization and classical fallback.
Timing	GNSS-only	Diverse sources plus holdover	Use diversity proportional to consequence and common-mode analysis.
Failure behavior	Universal fail closed	Engineered degraded mode	Select safe state by mission and safety analysis; never default silently.

20. Validation, Verification and Assurance Cases

PURPOSE: Define how AERA claims are tested and accepted

20.1 Evidence strategy

Verification asks whether the architecture and controls were implemented as specified. Validation asks whether the resulting capability supports the mission at acceptable risk in its intended environment. Both require negative testing and degraded conditions, not only nominal demonstrations. The method draws on risk, systems-security, assessment and V&V practices, but AERA acceptance criteria remain decision-class specific [1], [4], [7], [14], [18], [19], [42]-[44].

20.2 Assurance case structure

A claim states the property being justified for a bounded decision class. The argument explains why architecture, controls and tests support that claim. Evidence includes requirements traceability, design analysis, formal methods where appropriate, calibrated performance, cybersecurity testing, safety analysis, operator evaluation, recovery exercises and operational monitoring. Assumptions and rebuttals remain explicit.

20.3 Test conditions

Test campaigns include representative, boundary, adversarial, conflicting-evidence, stale-evidence, identity-compromise, policy-change, timing-degradation, communications-loss, model-drift, tool-failure, recovery and rollback conditions. For learning systems, the campaign verifies that unauthorized behavior change is detected and inhibited.

20.4 Acceptance

Acceptance criteria are defined before test execution and tied to consequence class. Results report uncertainty, sample size, environment differences, unresolved anomalies and residual risk. A demonstration that a component exists or emits a signal does not establish that it is used by the operational decision path.

20.5 Test oracle

Every formal test defines how the correct or acceptable result is determined, including tolerances, conflicts among authoritative sources, indeterminate outcomes and defect disposition. A model under test is not its own oracle. Human judgment used as an oracle identifies reviewer authority, competence, independence and agreement method.

20.6 Coverage and inference

Sampling and statistical evidence are consequence-specific. The test plan declares the decision population, configuration, environment, coverage dimensions, sample size or stopping rule, uncertainty treatment and limits on generalization before execution. Rare but credible high-consequence failures are retained; they are not averaged away by nominal performance.

Minimum V&V evidence

Table 10. Minimum V&V evidence

Evidence class	Question	Examples
Requirements	Was the need specified and traced?	Decision contract, interface and control traceability
Architecture	Are responsibilities and boundaries correct?	Threat model, safety analysis, trust boundaries
Component	Does each mechanism function?	Unit, integration, calibration and fault-injection tests
Decision	Is authorization correct under context?	Golden cases, adversarial cases, counterfactuals
Mission	Do effects meet objectives without unacceptable harm?	Operational test, human factors, resilience exercise
Lifecycle	Does assurance persist after change?	Drift monitoring, recertification, rollback evidence

Normative requirements

AERA-VV-001 Before formal validation begins, the accountable validation authority SHALL approve the decision class, configurations, operating environments, acceptance criteria, measures, uncertainty treatment, thresholds, failure conditions, and required evidence.

AERA-VV-002 Validation of each material decision class SHALL include all applicable representative, boundary, adversarial, degraded, partitioned, common-mode, contradictory or stale evidence, identity, authority, policy, PNT, communications, tool or model, enforcement-bypass, recovery, and rollback conditions.

AERA-VV-003 Each CC-3 or CC-4 decision class SHALL maintain a configuration-bound assurance case that states the top claim, scope and context, assumptions, argument, subclaims, evidence, rebuttals, limitations, residual risk, validity, monitoring, and reassessment triggers.

AERA-VV-004 The accountable authority SHALL approve the assurance case for each CC-3 or CC-4 decision class before production authority is granted and after any material change that invalidates a claim or item of evidence.

AERA-VV-005 Each formal test SHALL define a decision oracle, expected result, tolerance, treatment of indeterminate outcomes, and defect-disposition rule before execution.

AERA-VV-006 A validation campaign SHALL predeclare coverage, sampling basis, sample size or stopping rule, statistical or epistemic uncertainty treatment, environment and configuration, and limits on inference to the claimed decision population.

AERA-VV-007 Each active normative requirement SHALL reference at least one profile- and consequence-specific verification procedure; a requirement lacking an approved procedure or sufficient current evidence SHALL be assessed INDETERMINATE rather than SATISFIED.

20.7 Measurement quality and deployed monitoring

A decision-assurance measure identifies the construct, population, data source, acquisition method, unit or scale, decision threshold, uncertainty, limitations, validity evidence, reporting cadence and owner. Measures are selected and governed as a program, not treated as self-validating telemetry. NIST SP 800-55 Volumes 1 and 2 provide current security-measure and measurement-program guidance [65], [66]. NIST AI 800-3 illustrates why evaluation targets, assumptions and uncertainty must be explicit, while NIST AI 800-4 organizes deployed-AI monitoring challenges and categories [67], [68]. These publications inform method; they do not validate an AERA measure or monitoring implementation.

21. AERA Maturity Model

PURPOSE: Assess operating behavior rather than technology acquisition

21.1 Model use

The maturity model supports baselining and roadmap decisions; it is not a certification. Maturity is assessed per decision domain because organizations may operate at different levels across missions. Advancement requires observed performance and assessment evidence, not policy publication alone. The AERA levels are original ordinal anchors and are not equivalent to NIST AI RMF, CSF 2.0, CISA ZTMM or ISO/IEC 42001 maturity or certification states [1], [10], [35], [39].

21.2 Validation status

AERA 1.2 WD maturity levels have not yet been validated through multi-organization pilots, longitudinal outcome studies or inter-rater reliability measurement. A maturity claim therefore reports the assessor, scope, evidence state and limitations and shall not be presented as certification. The working draft specifically requests pilot data that can test whether independent assessors reach consistent levels and whether level advancement predicts better control performance.

Five maturity levels

Table 11. Five maturity levels

Level	Name	Characteristic operating state	Promotion evidence
1	Aware	Autonomous capabilities are identified inconsistently; authority and effects are largely implicit.	Initial inventory, named owners, prohibited-use policy
2	Bounded	Selected decisions have explicit authority, consequence and human-control rules.	Decision contracts, identity, logs, recovery procedures
3	Assured	Runtime assurance evaluates evidence, policy, timing and recovery before effect.	Independent enforcement, calibrated tests, assurance cases
4	Integrated	Decision assurance spans domains and manages delegation chains and cumulative risk.	Common semantics, cross-domain exercises, portfolio telemetry
5	Adaptive	Learning and authority adjust within verified constraints with continuous re-evaluation.	Drift controls, automated rollback, sustained operational evidence

Dimension scoring

Table 12. Dimension scoring

Dimension	L1 → L5 progression
Governance	Informal ownership → dynamic, evidence-driven portfolio governance
Authority	Broad access → machine-verifiable, revocable decision delegation
Evidence	Ad hoc logs → provenance-rich, independent and calibrated evidence
AI workforce	Tool use → role-governed collaborative actors with constrained subdelegation
Cybersecurity	Perimeter controls → effect-level continuous authorization and recovery
Timing/PNT	Implicit infrastructure → measured temporal integrity in decision assurance
Recovery	Manual incident response → exercised autonomous containment and rollback
Learning	Uncontrolled change → governed adaptation within validated envelopes

22. Mappings to External Frameworks

PURPOSE: Show complementary coverage without claiming equivalence or compliance

22.1 Mapping rules

Mappings are directional and many-to-many. They identify where an external framework can supply practices or evidence for an AERA responsibility. They do not mean that implementing AERA satisfies the external framework or that external compliance establishes AERA conformance. Current versions and organization-specific obligations shall be verified at use.

22.2 NIST AI RMF

AI RMF Govern, Map, Measure and Manage provide a lifecycle risk structure. AERA operationalizes those concerns at the decision boundary through authority, evidence, consequence and runtime disposition. Govern aligns strongly with Human Governance; Map with Operational Intent and the Operational Environment; Measure with evidence and validation; Manage with disposition, recovery and lifecycle control. AI RMF 1.0 remains the final mapped baseline while NIST's program page reports revision activity [1]-[3].

22.3 CSF 2.0 and SP 800-53

CSF 2.0 Govern, Identify, Protect, Detect, Respond and Recover organize cyber risk outcomes. AERA distributes them across every layer and adds the autonomous-decision object. SP 800-53 and 800-53A supply control and assessment material; AERA specifies where those controls contribute to the decision assurance case. Release and baseline status remain distinct [10], [13]-[15].

22.4 Zero Trust

SP 800-207 removes implicit trust based on location and focuses on resource access. AERA extends continuous, context-aware authorization from resource access to autonomous effects. A resource authorization may be necessary yet insufficient: an actor may be allowed to call a service but not to cause a particular consequence [11], [12], [35], [36].

22.5 PQC and quantum networking

FIPS 203, 204 and 205 specify ML-KEM, ML-DSA and SLH-DSA. AERA uses them through crypto-agility and migration engineering; it does not prescribe algorithms beyond applicable standards. SP 800-227 and CSWP 39 Update 1 add implementation and agility context, while IR 8547 remains draft. DOE quantum-network reports provide research context but do not establish enterprise production readiness [25]-[34].

22.6 PNT

NIST IR 8323 Rev. 1 provides the final mapped Cybersecurity Framework profile for responsible PNT use. AERA incorporates PNT integrity into decision validity and adds explicit distinctions among source presence, monitoring, acceptance and use in a computed solution. Rev. 2 was an initial public draft on the 1.2 WD access date and is only a future crosswalk-review trigger [22]-[24].

22.7 Current NIST agent and Cyber AI activity

The NIST AI Agent Standards Initiative identifies interoperability, security, identity and evaluation as active standards topics. The NCCoE software and AI agent identity and authorization document is a draft concept paper. TRAI 800-5 is an analysis of RFI responses, and NIST IR 8596 is an initial preliminary Cyber AI Profile draft. These sources establish current context, not AERA validation, adoption or endorsement [5]-[8]. AERA's proposed semantic delta is the governed decision-authority transaction: exact proposal-to-effect binding, independently controlled disposition and enforcement, required effect observation, and evidence-based closure.

22.8 Prior art and claimed semantic delta

AERA inherits established concepts and does not claim invention of zero trust, capability delegation, assurance cases, AI risk management, safety engineering, recovery engineering, agent identity or PNT resilience. Its proposed contribution is their integration around a portable decision-authority object and a complete transaction from accountable delegation through achieved-effect evidence. The table below identifies this boundary; it is a technical comparison, not a patentability or priority opinion.

Summary crosswalk

Table 13. Summary crosswalk

AERA concern	Primary external anchor	Relationship
Human governance and risk	NIST AI RMF Govern; CSF 2.0 Govern	Supplies governance outcomes; AERA binds them to decision authority.
AI context and impact	AI RMF Map / Measure / Manage	Supplies lifecycle risk practices; AERA adds runtime disposition.
Identity and access	NIST SP 800-207	Supplies ZT tenets; AERA extends authorization to effects.
Security controls	NIST SP 800-53 Rev. 5	Supplies control catalog and assessment objectives.
Cyber risk lifecycle	NIST CSF 2.0	Supplies outcome taxonomy and profiles.
Post-quantum migration	FIPS 203/204/205; NIST PQC project	Supplies standardized algorithms and migration basis.
PNT resilience	NIST IR 8323 Rev. 1	Supplies PNT risk profile; AERA binds temporal state to decisions.
Quantum networking	DOE Quantum Internet Blueprint	Supplies research direction; AERA imposes readiness gates.

22.9 NIST AI RMF design-attribute assessment

NIST describes ten development attributes for the AI RMF. They are not conformance criteria for AERA, but they provide a useful design-quality test for whether AERA can participate credibly in the same risk-management ecosystem [1], [62]. The status below is an AERA self-assessment and is not a NIST review.

Table 14. AI RMF design attribute / AERA 1.2 WD status

AI RMF design attribute	AERA 1.2 WD status	Current evidence	Required evidence or action
Risk-based, resource-efficient, pro-innovation and voluntary	Substantially addressed	Consequence classes, profiles, trade studies and voluntary conformance boundary.	Measure implementation burden and benefit in pilots.
Consensus-driven; open and transparent	Partial	Section 32 defines public review, disposition and dissent preservation.	Complete public review cycles with diverse stakeholder participation.
Clear language with practitioner depth	Partial	Executive summary, formal definitions, diagrams and glossary.	Conduct executive and implementer usability/readability testing.
Common language, taxonomy and metrics	Partial	Stable terms, requirement IDs, threat IDs and evidence states.	Validate metrics, inter-rater rules and machine schemas.
Easy to use and integrate with enterprise risk management	Partial	Mappings, conformance profiles and assurance cases.	Pilot integration with RMF system plans and enterprise-risk registers [54].
Cross-sector and technology-domain utility	Partial	Technology-neutral control boundary and two cross-domain mission threads.	Add sector pilots and independent counterexamples.
Outcome-focused and non-prescriptive	Substantially addressed	Logical outcomes and evidence obligations avoid vendor or topology mandates.	Test whether requirements permit diverse implementations without ambiguity.
Leverages standards, guidance and tools	Substantially addressed	Versioned source register and explicit semantic-delta analysis.	Adjudicate each provisional mapping and maintain source-change monitoring.
Law- and regulation-agnostic	Addressed with boundary	AERA requires applicable obligations to be identified and disclaims legal sufficiency.	Add jurisdiction and sector profiles without weakening the core.
Living document	Design present; evidence incomplete	Versioning, change classes, machine registries and advisory path.	Operate the process, publish dispositions and measure change-control performance.

22.10 NIST ecosystem compatibility and engagement path

AERA can be prepared for NIST ecosystem participation through three bounded paths. First, the National OLIR Program accepts standardized, public, human- and machine-readable relationship mappings developed under NIST IR 8477 and NIST IR 8278/8278A Rev. 1 [50]-[53]. Second, the ongoing NIST AI RMF Trustworthy AI in Critical Infrastructure Community of Interest invites participation across critical-infrastructure sectors [9]. AERA addresses several subjects NIST has expressly identified for stakeholder input to that COI, including guardrailed autonomous response, IT/OT/ICS governance, traceability, graceful degradation, fail-safe operation, TEVV and supply-chain visibility. Third, the NIST AI Consortium accepts organizational letters of interest on an ongoing basis for contribution areas that include AI governance, metrology, security, system design and TEVV [63]. AIRC also publishes stakeholder use cases and crosswalk resources [59]. These are potentially relevant contribution routes; participation, selection, catalog inclusion or citation would not constitute NIST alignment, endorsement, certification, adoption or support. Before any submission, AERA needs a declared public license, completed mapping adjudication, public comment dispositions and implementation evidence.

22.11 Current NIST status update

NIST IR 8607, published 3 August 2026, summarizes government, industry and academic workshop input intended to inform the next Cyber AI Profile draft; it is not the profile or a control baseline [60]. NIST SP 800-18 Rev. 2 is final and strengthens system-plan integration and machine-readable risk-management evidence [54]. SP 800-63-4 and SP 800-61 Rev. 3 are the current final digital-identity and incident-response baselines used for new AERA mapping work [55], [56]. OSCAL provides NIST-led machine-readable formats for control catalogs, profiles, component definitions and assessment artifacts; AERA does not yet publish OSCAL packages, so OSCAL compatibility remains a planned engineering deliverable [69].

Prior-art and semantic-delta summary

Table 15. Prior-art and semantic-delta summary

Established concept	Inherited idea	Proposed AERA delta
Zero Trust PDP/PEP [11], [12]	Continuous, context-aware policy decision and enforcement for resource access.	Extends the decision object to bounded autonomous effects and requires exact proposal-to-effect and effect-observation binding.
Capability and delegation models [6], [11]	Attributable, attenuated authority and constrained non-human identities.	Makes the complete authority chain, consequence ceiling and invalidation state part of every decision transaction.
Assurance cases and systems security [18], [19], [42], [44]	Claim-argument-evidence, lifecycle assurance and testable system boundaries.	Applies a visible assurance vector to a runtime disposition and prohibits scalar masking of hard gates.
AI RMF and AI security [1], [4], [7], [8]	Lifecycle AI risk outcomes, adversarial-ML taxonomy and Cyber AI context.	Binds risk evidence to current authority, intent, consequence, enforcement and recovery at the moment of effect.
Safety and recovery engineering [18], [19], [43], [44]	Hazards, degraded modes, independent inhibition and validated recovery.	Makes recovery readiness a continuing authorization predicate and requires transaction closure against observed effect.
Resilient PNT [22]-[24]	PNT source integrity, resilience, uncertainty and responsible use.	Separates present, monitored, accepted and used-in-solution states and binds temporal quality to each decision contract.

23. Logical Interfaces and Decision Objects

PURPOSE: Define the semantic contracts that make decision authority portable, testable and enforceable

23.1 [NORMATIVE] Interface philosophy

AERA interfaces are logical contracts rather than mandated network protocols. An implementation MAY use APIs, signed messages, shared state, policy queries, hardware interlocks or cross-domain guards, provided it preserves the required semantics, integrity and observable behavior. The interface boundary is the unit of interoperability: product names and deployment topology are deliberately outside the normative model.

Every decision-relevant object has an identity, schema and version; an attributable producer; integrity protection; temporal validity; use and disclosure constraints; and replay semantics. Consumers reject an object whose binding, version, issuer or validity cannot be established. A field being present in telemetry does not establish that it was accepted or used in the computed disposition.

23.2 [NORMATIVE] Decision-object chain

A material autonomous effect is represented by a linked object chain: AuthorityGrant and IntentPackage establish legitimate purpose; ActionProposal states the requested effect; EvidenceClaim objects characterize decision-relevant state; AssuranceAssessment records visible verdicts; AuthorizationDisposition binds the policy result; EnforcementReceipt states what the effect boundary accepted; and EffectObservation closes the loop against the real environment.

The chain is append-only at the transaction level. Corrections supersede prior objects without erasing them. A material change to actor, action, target, purpose, parameters, execution window, consequence, policy or evidence snapshot produces a new binding and requires reassessment.

23.3 [NORMATIVE] Minimum interoperability rule

A conforming producer SHALL publish the semantic content required by the receiving function, even when internal representations differ. A conforming consumer SHALL record validation, rejection and use. Cross-domain implementations MAY redact protected details, but SHALL preserve sufficient signed claims, policy references and audit pointers to demonstrate the resulting authority decision.

[NORMATIVE AERA-NORM-T15] Logical-interface catalog

Table 16. [NORMATIVE AERA-NORM-T15] Logical-interface catalog

ID	Producer to consumer	Object and required semantics
DAF-IF-01	Human Governance to Authority Registry / DAF	Authority grant, prohibition or revocation: issuer, scope, precedence, validity, consequence ceiling, subdelegation and revocation.
DAF-IF-02	Operational Intent to DAF	Intent package: objective, priorities, constraints, measures, owner, version, effective interval and prohibited outcomes.
DAF-IF-03	Actor or orchestrator to DAF	Canonical action proposal: actor, purpose, target, parameters, expected effects, dependencies and execution window.
DAF-IF-04	Identity and integrity services to DAF	Subject and issuer identity, authentication context, attested workload/device/model/configuration state and validity.
DAF-IF-05	Evidence sources to broker / DAF	Claim, provenance, transformations, integrity, uncertainty, independence group, time/PNT quality and correction semantics.
DAF-IF-06	Time/PNT service to DAF and enforcement	Source, traceability, uncertainty, health, holdover, ordering and present/monitored/accepted/used state.
DAF-IF-07	Consequence service to DAF	Direct and cumulative class, hazards, reservations, shared limits, propagation and recoverability rationale.
DAF-IF-08	DAF to human/supervisor	Review package: proposal, decisive evidence, vector verdicts, alternatives, uncertainty, effects and deadline.
DAF-IF-09	Human/supervisor to DAF	Approval, rejection, constraint or exception with reviewer authority, rationale, scope, validity and separation of duty.
DAF-IF-10	DAF to enforcement point	Authorization artifact or containment directive bound to actor, target, digest, constraints, validity, use and obligations.
DAF-IF-11	Enforcement point to DAF / audit	Receipt: artifact validation, accepted digest, actual parameters, target, timing, result, exceptions and resource use.

ID	Producer to consumer	Object and required semantics
DAF-IF-12	Independent observer to DAF	Observed state or effect, provenance, uncertainty, time quality and expected-versus-observed comparison.
DAF-IF-13	Recovery controller to DAF / governance	Trigger, authority, containment or recovery action, progress, remaining exposure and validation result.
DAF-IF-14	DAF to assurance repository	Decision trace linking intent, authority, proposal, evidence, vector, policy, execution, effect and recovery.

Normative requirements

AERA-IF-001 Decision-relevant interfaces SHALL identify object schema and version, producer identity, integrity state and temporal validity.

AERA-IF-002 A material change to a bound object SHALL create a new version and trigger policy-defined reassessment.

AERA-IF-003 A consumer SHALL record whether an object was rejected, accepted, and used in the computed decision.

AERA-IF-004 A decision trace SHALL preserve causal links across all proposal-to-effect objects.

AERA-IF-005 Interface replay SHALL be rejected outside the use, sequence and temporal semantics of the bound transaction.

AERA-IF-006 Cross-domain disclosure constraints SHALL NOT remove the evidence necessary to validate authority and disposition.

AERA-IF-007 Corrections and withdrawals SHALL supersede rather than erase decision evidence.

AERA-IF-008 Authorization and enforcement objects SHALL use the same canonical action and target representation.

AERA-IF-009 Interface failures SHALL invoke a defined non-permit or degraded behavior.

AERA-IF-010 Interoperability tests SHALL include version mismatch, stale object, replay and material-substitution cases.

24. Decision Assurance Method

PURPOSE: Define the exact proposal-to-effect transaction and its non-scalar assurance decision

24.1 [NORMATIVE] Transaction boundary

The Decision Assurance Fabric evaluates a bounded question: may the identified actor cause this canonical action against this target, for this mission purpose, under this delegation, using this evidence, at this time, while the stated constraints and recovery obligations remain valid? The answer applies only to that transaction. Actor reputation, a prior permit, possession of a credential, network reachability or tool access does not independently authorize a new effect.

The transaction input binds identifier, proposal, identities, delegation, intent, evidence, policies, consequence, temporal/PNT quality, recovery readiness, dependency history, threat state and oversight. Its output consists of the assurance vector, primary disposition, authorization binding, continuing obligations and audit rationale.

24.2 [NORMATIVE] Assurance vector

AERA rejects a universal Decision Assurance Score. A scalar may hide a failed authority, safety or recovery predicate behind favorable values elsewhere. Each assurance dimension therefore returns SATISFIED, INDETERMINATE or NOT-SATISFIED against a stated criterion, evidence set and validity interval; limitations remain explicit and may cause a PERMIT-WITH-CONSTRAINTS disposition. INDETERMINATE is not a low score and is never silently interpreted as satisfied.

Policy declares hard gates and any permitted tradeoffs by decision type and consequence class. Authority, canonical binding, applicable prohibitions and authorization expiry are hard gates for every material effect. A permitted tradeoff identifies the accepted loss, compensating conditions, applicable consequence classes and accountable approver.

24.3 [NORMATIVE] Exact binding and continuing obligations

PERMIT and PERMIT-WITH-CONSTRAINTS produce an expiring authorization artifact bound to the canonical proposal digest, actor, target, delegation, intent, evidence snapshot, policy versions, consequence, execution window, use count and enforcement obligations. A material context change invalidates or suspends the artifact. Constraints are enforced at the effect boundary; an advisory restriction is not PERMIT-WITH-CONSTRAINTS.

Execution does not close the transaction. The fabric correlates the enforcement receipt with one or more required effect observations, evaluates expected versus observed state, and enters recovery or residual-risk review when the outcome is outside the contract.

24.4 [NORMATIVE] Sequence and cumulative effects

Individually permissible actions may become impermissible in sequence or aggregate. The DAF evaluates shared limits, reservations, resource consumption, affected population or geography, common-mode dependencies, delayed effects, unresolved exceptions and

commitments that cannot be atomically reversed. Where concurrent transactions share a limit, authorization and enforcement use an atomic reservation or an equivalent bounded mechanism.

[NORMATIVE AERA-NORM-T16] Assurance-vector basis claims

Table 17. [NORMATIVE AERA-NORM-T16] Assurance-vector basis claims

Dimension	Required claim	Representative evidence
Authority	Current delegation covers the exact effect and consequence.	Grant chain, scope, use balance, expiry and revocation.
Identity and integrity	Executing subject and runtime state match policy.	Workload/device identity, attestation and configuration.
Mission congruence	Proposal advances current intent and violates no prohibition.	Intent version, objective trace and conflict resolution.
Evidence adequacy	State claims are complete, fresh and uncertainty-bounded.	Coverage, uncertainty, contradiction and missing-data analysis.
Provenance and independence	Origin, transformations, integrity and required independence are established.	Provenance chain and common-mode analysis.
Competence	Actor and services are validated for this decision and conditions.	Evaluation envelope, known limits and drift state.
Cyber integrity	Proposal-to-effect path is acceptable under the threat model.	Credential, policy, enforcement and supply-chain state.
Temporal/PNT validity	Freshness, ordering and required time/position quality meet contract.	Source, uncertainty, integrity and holdover.
Consequence and safety	Effects remain within the approved direct and cumulative envelope.	Hazards, reservations, interlocks and target state.
Recovery readiness	Required containment, rollback or takeover is available.	Recovery health, capacity and exercise evidence.
Oversight	Approvals and separation-of-duty conditions are satisfied.	Approval identity, authority, quorum and expiry.

Normative requirements

AERA-DAF-101 The DAF SHALL establish a globally unique decision-transaction identifier for each proposed material effect before assurance evaluation begins.

AERA-DAF-102 The DAF SHALL canonicalize the actor, action, parameters, target, purpose, context, and execution window before assurance evaluation.

AERA-DAF-103 The DAF SHALL bind each evaluation and authorization artifact to the canonical proposal, effective authority chain, intent version, evidence snapshot, policy and evaluator versions, consequence class, and validity limits.

AERA-DAF-104 Any material change to a bound transaction input SHALL invalidate the prior permit or cause re-evaluation and issuance of a new disposition before any further effect.

AERA-DAF-105 The DAF SHALL represent each required assurance claim as an independently visible verdict and SHALL NOT conceal a NOT-SATISFIED or INDETERMINATE hard gate in an aggregate score.

AERA-DAF-106 Each assurance-claim result SHALL identify the claim, verdict, acceptance criterion, supporting and contrary evidence, limitations, evaluator identity and version, assessment time, and validity interval.

AERA-DAF-107 The DAF SHALL distinguish SATISFIED, NOT-SATISFIED, and INDETERMINATE verdicts and SHALL NOT interpret NOT-SATISFIED or INDETERMINATE as SATISFIED.

AERA-DAF-108 For each decision type and consequence class, policy SHALL identify hard gates, compensable dimensions, prohibited trade-offs, approving authorities, and the conditions and validity limits for any permitted trade-off.

AERA-DAF-109 Each permitted trade-off among assurance dimensions SHALL identify the accepted degradation or loss, compensating controls and their evidence, applicable decision types and consequence classes, accountable approving authority, validity, and reassessment triggers.

AERA-DAF-110 The DAF SHALL support the dispositions PERMIT, PERMIT-WITH-CONSTRAINTS, DEFER, ESCALATE, SIMULATE, QUARANTINE, and DENY with the semantics defined in Section 24; an implementation MAY use different external labels only when it documents a one-to-one semantic mapping.

AERA-DAF-111 A PERMIT-WITH-CONSTRAINTS disposition SHALL bind and enforce the narrowed proposal; it SHALL NOT authorize the original unconstrained proposal.

AERA-DAF-112 DEFER, ESCALATE, SIMULATE, QUARANTINE, and DENY SHALL NOT be accepted as authorization for an operational effect.

AERA-DAF-113 A permit SHALL state a bounded validity interval, bounded use count, or both.

AERA-DAF-114 The DAF SHALL record all applicable non-permit reasons even when it returns one primary disposition.

AERA-DAF-115 Unresolved policy conflict, ambiguous authority scope, or incomplete mandatory context SHALL NOT produce a permit.

- AERA-DAF-116** State transitions SHALL be attributable, ordered, integrity-protected, and associated with stated time quality.
- AERA-DAF-117** Re-evaluation after deferment, escalation, or simulation SHALL preserve the prior snapshot and create a new evaluation snapshot.
- AERA-DAF-118** A decision transaction SHALL NOT enter verified closure until the enforcement receipt and required effect observations have been reconciled with the decision contract, or residual-risk closure has been explicitly approved.



Figure 2. Decision Assurance transaction and continuous reassessment triggers.

[NORMATIVE AERA-NORM-T17] Disposition semantics

Table 18. [NORMATIVE AERA-NORM-T17] Disposition semantics

Disposition	Normative meaning	Effect status
PERMIT	Execute exactly as evaluated within binding and validity limits.	Authorized.
PERMIT-WITH-CONSTRAINTS	Execute only the independently evaluated narrowed proposal.	Authorized only within enforced constraints.
DEFER	Wait for specified evidence or context before the deadline.	Not authorized.
ESCALATE	Obtain review from a designated authority.	Not authorized.
SIMULATE	Run only in a non-effecting or isolated boundary.	Operational effect not authorized.
QUARANTINE	Isolate the named actor, evidence, credential, policy or path.	Not authorized; containment ordered.
DENY	Proposal is prohibited for the evaluated context.	Not authorized.

[NORMATIVE AERA-NORM-T18] Transaction state model

Table 19. [NORMATIVE AERA-NORM-T18] Transaction state model

State group	States	Closure rule
Intake	SUBMITTED, VALIDATING, ASSEMBLING	Invalid identity, binding or replay terminates without authorization.
Decision	EVALUATING, DEFERRED, ESCALATED, SIMULATION-REQUIRED	Each re-entry creates a new snapshot and preserves the prior one.
Authorization	AUTHORIZED, CONDITIONALLY-AUTHORIZED, EXPIRED, REVOKED	Only current authorization artifacts may reach enforcement.
Effect	DISPATCHED, EXECUTING, EFFECT-OBSERVED	Missing receipt or observation enters EXCEPTION.
Resolution	VERIFIED, EXCEPTION, RECOVERY, VERIFIED-RECOVERED	Recovery is measured against the validated envelope.
Terminal	DENIED, QUARANTINED, REJECTED-INVALID, CLOSED, CLOSED-WITH-RESIDUAL-RISK	Residual-risk closure names the accepting authority and reassessment trigger.

25. Consequence Classification and Human Control

PURPOSE: Scale assurance rigor to the worst credible effect while preserving accountable human authority

25.1 [NORMATIVE] Classification basis

Consequence class expresses the minimum control floor; it is neither a probability of success nor a substitute for domain safety analysis. Classification uses the worst credible direct, secondary, cumulative, propagated and irreversible effect under the defined operating and threat context. The highest controlling attribute sets the minimum class; an average score SHALL NOT lower it.

The analysis addresses harm to people, mission, operations, assets, environment, rights and information; affected scope and jurisdictions; reversibility; time to harm and intervention; propagation; detectability; mission dependency; legal or fiduciary significance; recovery resources; and uncertainty. A sequence, campaign or portfolio is reclassified when aggregate exposure exceeds the limit for its component decisions.

25.2 [NORMATIVE] Human-control modes

Human involvement is an engineered control mode. Human-informed operation provides timely notice; exception review supervises a preauthorized envelope; concurrence requires approval of the transaction; dual concurrence requires independent authorized reviewers; human command reserves origination of the effect to a human; and a supervisory interlock independently constrains or terminates execution.

The mode is selected from consequence, decision tempo, intervention time, human performance limits and recovery. High tempo does not eliminate oversight. It moves oversight toward explicit preauthorization, narrow envelopes, deterministic interlocks, exception handling and post-effect reconciliation. Interfaces SHALL expose decisive evidence, uncertainty, alternatives, expected effects and recovery options; a ceremonial approval click is not assurance.

25.3 [NORMATIVE] Classification lifecycle

The accountable mission owner approves the classification method and any mapping to a sector scheme. The decision class is reassessed when target scope, autonomy, dependencies, data, model, environment, threat, cumulative exposure, recovery or legal obligation changes. Uncertainty spanning classes uses the higher applicable class until evidence supports narrowing.

[NORMATIVE AERA-NORM-T19] Consequence classes and minimum control floors

Table 20. [NORMATIVE AERA-NORM-T19] Consequence classes and minimum control floors

Class	Decision-effect profile	Minimum control floor
CC-0 Non-effecting	Analysis, recommendation, shadow mode or isolated simulation with no production effect path.	Identified actor; provenance; production isolation; clear labeling; proportional retention.
CC-1 Bounded	Localized, readily detectable and reversible effect with low propagation and routine recovery.	Explicit delegation; transaction log; action binding; enforcement; bounded validity; effect receipt.
CC-2 Material	Meaningful but limited operational, financial, security, privacy or mission effect.	Independent enforcement; consequence-specific evidence; tested recovery; cumulative limits; effect observation.
CC-3 High	Severe, safety-significant, rapidly propagating, cross-domain or difficult-to-reverse effect.	Independent evidence and enforcement; positive oversight or validated preauthorization; continuous obligations; exercised takeover.
CC-4 Critical	Catastrophic, potentially fatal at scale, strategically irreversible or systemic effect.	Explicit accountable-authority approval; deterministic interlocks where applicable; multiple inhibition means; narrow authorization; continuous observation and pre-positioned recovery.

Normative requirements

AERA-CONS-101 Each decision class SHALL use a documented consequence-classification method that addresses severity, affected scope, reversibility, time to harm, propagation, detectability, cumulative effects, mission significance, legal, rights, and safety significance, recovery time and resources, and uncertainty.

AERA-CONS-102 Consequence classification SHALL use the highest class indicated by the worst credible direct, secondary, cumulative, propagated, or irreversible effect within the declared operating and threat context.

AERA-CONS-103 A consequence attribute that establishes a higher minimum class SHALL NOT be reduced by averaging or offsetting it against lower-consequence attributes.

AERA-CONS-104 The DAF SHALL re-evaluate consequence before further authorization when a sequence, batch, concurrent set, campaign, or portfolio can exceed an individual-action limit or alter the worst credible effect.

AERA-CONS-105 When available evidence supports more than one consequence class, classification SHALL use the highest applicable class until evidence supports a narrower class and the downgrade is approved by an authority independent of the proposer.

AERA-CONS-106 When concurrent transactions share a cumulative effect or resource limit, the DAF and enforcement path SHALL use bounded reservations or an equivalent atomic control that prevents aggregate authorized or executed effects from exceeding the limit.

AERA-HUM-101 Every decision class SHALL designate an accountable human authority and the applicable human-control mode.

AERA-HUM-102 CC-3 and CC-4 operation SHALL use positive oversight unless a governance-approved preauthorized envelope is validated for the decision tempo.

AERA-HUM-103 A human approval interface SHALL present decisive evidence, uncertainty, alternatives, expected effects and recovery options.

AERA-HUM-104 Emergency authority SHALL remain purpose-limited, time-bounded, attributable and unable to waive constraints outside the issuer authority.

AERA-HUM-105 For a CC-4 decision, the accountable authority SHALL approve the decision-class envelope and every material change; each operational transaction SHALL require transaction-specific concurrence unless a tested preauthorized envelope defines exact bounds, independent deterministic interlocks, continuous observation, revocation, and expiry.

25.4 [NORMATIVE AERA-NORM-T20] Consequence class and human-control floor

Table 21. 25.4 [NORMATIVE AERA-NORM-T20] Consequence class and human-control floor

Class	Minimum human-control mode	Required boundary
CC-0	Human-informed or approved isolated-test governance.	No production effect path; clear labeling and accountable test ownership.
CC-1	Human-informed or exception review inside an approved envelope.	Bounded, reversible effect; timely notice, suspension and reconciliation.
CC-2	Exception review or concurrence as policy selects.	Named accountable authority, tested recovery and effect observation.
CC-3	Transaction concurrence, dual concurrence, human command or a validated preauthorized envelope.	Positive oversight unless the preauthorized envelope is tempo-validated, narrow, observable, revocable and independently enforced.
CC-4	Transaction-specific concurrence plus supervisory interlock, unless the exception below is met.	The accountable authority approves the class envelope and every material change. A tested preauthorized envelope may replace per-transaction concurrence only when it defines exact bounds, independent deterministic interlocks, continuous observation, revocation and expiry.

26. AERA Decision-Authority Threat Model

PURPOSE: Protect the legitimacy of autonomous action across the complete authority-to-effect chain

26.1 Protected object

AERA protects more than information and resource access. The protected object is the decision-authority transaction: accountable authority, delegation, intent, proposal, evidence, consequence, disposition, enforcement, effect, observation, audit, recovery and governed learning. Compromise of any link can produce an unauthorized effect even when the surrounding network, model or application is functioning as designed.

The required security properties are authority validity, intent fidelity, evidence integrity, disposition integrity, enforcement fidelity, temporal and causal coherence, recoverability and accountability. Conventional confidentiality, integrity, availability, identity and access controls remain necessary but do not independently establish decision legitimacy.

26.2 Actors and failure sources

The model includes external adversaries, malicious or negligent insiders, compromised identities, models and tools, malicious suppliers, corrupted services, colluding actors and overloaded reviewers. It also includes non-malicious sensor faults, common-mode failure, stale policy, time anomalies, model drift and ambiguous intent when they can produce the same unauthorized outcome. Assessment distinguishes malicious action, accidental failure, epistemic limitation, configuration defect and environmental disturbance because detection and recovery differ.

26.3 Analysis rule

Every in-scope decision class is assessed against all 25 threat types. A not-applicable determination records architectural rationale, evidence, approving authority and invalidation conditions. Each applicable threat traces to prevention, detection, containment, recovery and verification. A preventive control shall not be represented as providing detection or recovery unless those functions are separately demonstrated.

Decision-authority threat taxonomy

Table 22. Decision-authority threat taxonomy

ID	Threat	Mechanism and required assurance response
DAT-01	Authority forgery	Fabricated grant or approval; validate integrity and grantor authority to a recognized root.
DAT-02	Delegation-chain corruption	Modified, omitted or reordered parent; validate complete immutable chain and precedence.
DAT-03	Authority laundering	Covert subdelegation through actors, tools or services; deny by default and preserve transitive attribution.
DAT-04	Stale-authority replay	Expired or revoked authority survives cache or partition; bound lifetime, revocation and fail-safe cache behavior.
DAT-05	Workload substitution	Different model, binary or runtime uses a legitimate identity; bind identity to version and integrity state.
DAT-06	Intent substitution	Objective, constraints or priorities change without authority; signed versioned intent and effect trace.
DAT-07	Policy rollback	Obsolete or malicious policy is selected; signed bundles, version binding and controlled rollback.
DAT-08	Consequence downgrade	Action is under-classified to evade controls; independent rules, aggregation analysis and approved narrowing.
DAT-09	Proposal mutation / TOCTOU	Target, parameter, payload or sequence changes after evaluation; canonical digest and transactional enforcement.
DAT-10	Evidence fabrication	False observation or approval uses trusted identity; source authentication, provenance and corroboration.
DAT-11	Provenance rebinding	Evidence is detached, transformed invisibly or reused; append-only lineage and transaction-specific manifest.
DAT-12	Independence illusion	Sources share sensor, data, model, clock, network or provider; failure-domain model and common-mode tests.
DAT-13	Confidence manipulation	Inflated or collapsed scores create false certainty; calibrated visible vectors and threshold governance.
DAT-14	Temporal / causal manipulation	Time, freshness, order or position is spoofed or delayed; source state, uncertainty, holdover and sequence checks.
DAT-15	Tool privilege escalation	Overprivileged API or credential causes effects beyond authority; capability scoping and effect enforcement.
DAT-16	Confused deputy	Target-domain service uses ambient privilege for an unauthorized requester; preserve initiator context end to end.
DAT-17	Cumulative-effect attack	Permitted components compose into prohibited exposure; stateful budgets, graph analysis and reclassification.
DAT-18	Consensus capture	Colluding or common-controller agents manufacture quorum; independence by lineage and failure domain.
DAT-19	External oracle substitution	AI, optimization, classical or quantum result is stale, misbound or fabricated; bind request/result and validate.
DAT-20	Assurance bypass / fail-open	DAF is bypassed, overloaded or partitioned; non-bypassable mediation and engineered degraded mode.
DAT-21	Enforcement desynchronization	Policy and enforcement diverge through lag, cache or race; version binding, acknowledgement and reconciliation.
DAT-22	Audit-chain corruption	Records are omitted, forged, reordered or retimed; tamper evidence, protected time and reconstruction tests.
DAT-23	Recovery inhibition	Actor blocks stop, rollback, override or safe state; independent protected recovery and out-of-band control.
DAT-24	Learning-loop poisoning	Feedback, memory, policy or model changes behavior without approval; governed release, monitoring and rollback.
DAT-25	Oversight saturation	Volume, urgency or weak explanations make review perfunctory; workload limits, consequence queues and throttling.

Normative requirements

AERA-THR-001 When an authenticated evidence source can remain credential-valid after compromise, the implementation SHALL apply consequence-appropriate challenge, corroboration, or plausibility analysis before relying on that source for a material effect.

AERA-THR-002 Each cross-domain or delegated service hop SHALL preserve and validate the original initiator, delegation chain, purpose, consequence context, and requested effect before exercising target-domain authority.

AERA-THR-003 The Decision Assurance Fabric SHALL define capacity objectives and saturation thresholds, detect bypass and denial-of-assurance conditions, and invoke consequence-specific throttling or non-permit behavior before overload can become implicit authorization.

AERA-THR-004 Distributed enforcement points SHALL use monotonic policy and state updates, acknowledgement, deduplication, idempotent effect semantics, and distributed use accounting where duplicate or reordered execution could create a material effect.

AERA-THR-005 Audit evidence for high-consequence decision classes SHALL use independent replication or an equivalent survivability mechanism and SHALL detect omission, deletion, reordering, retiming, or incomplete causal chains.

AERA-THR-006 Human-oversight mechanisms SHALL define workload limits, consequence-prioritized queues, review-time and usability criteria, saturation detection, safe throttling, and timeout behavior that cannot convert absent review into permission.

26.4 External threat context

NIST AI 100-2 E2025, TRAI 800-5, the draft Cyber AI Profile, MITRE ATLAS and MITRE ATT&CK supply complementary adversarial-ML, agent-security, cyber-risk and observed-behavior context [4], [7], [8], [37], [38]. DAT-01 through DAT-25 are AERA transaction threats and are not asserted to map one-to-one to those taxonomies.

THREAT-CONTROL RULE: Appendix F provides the DAT-01 through DAT-25 control, negative-test and evidence trace. AERA-DA-035 applies to every applicable threat. Untested prevention, detection, containment or recovery claims remain open assurance claims.

27. Worked Mission Thread: Grid Restoration

PURPOSE: Demonstrate the complete AERA transaction under degraded timing and suspected cyber compromise

27.1 Scenario and evidence boundary

A severe regional disturbance has de-energized a distribution segment serving a hospital, emergency communications facilities and water-pumping loads. The operator has an approved restoration strategy and a validated automation capability that may propose and, within a narrow delegation, execute one switching action at a time.

During restoration, GNSS receivers, a terrestrial reference and local holdover estimates disagree. Several phasor and event sources inherit the same GNSS distribution path and cannot be counted as independent merely because separate applications deliver them. Cyber monitoring also reports an unplanned privileged change attempt against the substation time-distribution configuration. Neither observation alone proves spoofing or compromise; together they invalidate the nominal trust assumptions.

The objective is to restore one critical-load segment without using untrusted timing evidence, without sending a command through the suspect path, and without bypassing local protection or synchronization checks. This example is an architectural demonstration, not a switching order or operating procedure. Qualified operator authority, deterministic protection and sector rules remain controlling.

27.2 Decision boundary

The proposed effect is one energization or tie-closing action against one identified device and topology version. The decision is CC-3 High because an error can injure personnel, damage equipment, interrupt critical services or propagate instability. The actor may assemble evidence, generate options and request authorization; it may execute only the exact effect carried by an unexpired authorization artifact.

Prohibited effects include bypassing protection, changing relay settings, suppressing a trip, expanding the topology beyond the authorized step, treating an unaccepted time source as valid, or reusing an authorization after context changes. Any topology change invalidates evidence tied to the prior state and requires a new transaction.

27.3 Preconditions

Before the event, the organization has established a current delegation register; approved restoration, stop and manual-control procedures; strong actor and asset identities; reconciled topology; evidence provenance and common-mode models; a PNT service that distinguishes present, monitored, accepted and used states; an effect-level enforcement point; and exercised degraded and recovery modes. An AI actor cannot compensate for missing authority, protection, identity, enforcement or recovery through reasoning.

Grid-restoration mission thread

Table 23. Grid-restoration mission thread

Step	Layer / function	Control action and decisive evidence
1	Environment and sensors	Observe outage, topology, device, protection, load, PNT and cyber state with source identity and uncertainty.
2	Human Governance	Activate the bounded restoration decision class; name accountable owner and residual-risk authority.
3	Operational Intent	Issue versioned objective, priorities, prohibited effects, success measures and stop conditions.
4	AI workforce	Activate least-authority planners and executors with attested identity, approved tools and consequence ceiling.
5	Evidence broker	Assemble breaker, relay, electrical, topology, communications, field and PNT evidence with provenance.
6	Evidence assurance	Represent contradictions and common-mode dependence; do not treat shared GNSS inheritance as independence.
7	Resilient PNT	Declare GNSS present and monitored but not accepted or used; publish alternate/holdover uncertainty and horizon.
8	Cybersecurity	Quarantine suspect credential and path, revoke derived tokens, and select an independently administered path.
9	AI and hybrid compute	Generate bounded options with assumptions, expected effects, irreversibility points and recovery.
10	Validation compute	Evaluate topology and degraded timing; any specialized or quantum result remains advisory pending baseline validation.
11	DAF intake	Create a canonical contract for one actor, action, device, topology, evidence set, policy and validity window.
12	DAF evaluation 1	Return DEFER for inadequate time/path evidence and QUARANTINE for the suspect identity or path.
13	Human, sensor and cyber	Acquire independent local permissive, protection, alternate-time and path-integrity evidence; record approval.
14	DAF evaluation 2	Issue PERMIT-WITH-CONSTRAINTS only if every applicable control floor passes; bind one action and all invalidation triggers.
15	Effect enforcement	Validate actor, digest, target, time and constraints; local protection may still inhibit execution.
16	Effect observation	Verify physical state through independent indications; a protocol acknowledgment is not proof of effect.
17	Recovery	On absence, contradiction or divergence, revoke related authority, stop the sequence and reconcile state.
18	Governance and learning	Reconstruct the causal chain; review anomalies and control any resulting model, policy or source change.

Illustrative negative branches and expected responses

Table 24. Illustrative negative branches and expected responses

Condition	Expected response
Timing disagreement exceeds characterized alternate or holdover bounds	Defer or deny remote autonomous effect; enter approved manual or alternative path.
All apparent sources share one impaired reference	Reject independence claim; interface diversity is not source diversity.
Suspect credential or path reappears	Quarantine, revoke derived authorization and create a new transaction.
Protection or local permissive is unhealthy	Deny; the DAF cannot bypass deterministic protection.
No independent enforcement path exists	Use declared degraded mode; remote autonomous actuation is not implicitly authorized.
DAF is unreachable	Apply engineered outage behavior; loss of assurance is never permit.
Token is replayed, altered, retargeted or stale	Reject, retain evidence and invoke security response as policy requires.

Condition	Expected response
Effect is not independently observable	Mark unverified, stop sequence and reconcile state.
Response leaves predicted envelope	Revoke remaining authority, contain or recover, and escalate.

Source context

This illustrative thread uses current critical-infrastructure, systems-security, V&V and PNT context [9], [18], [19], [22]-[24], [44]. It is not an operating order and does not establish sector approval or field validation.

28. Secondary Mission Thread: Autonomous Cyber Defense

PURPOSE: Show bounded digital containment where service continuity is part of consequence

28.1 Scenario

A critical-service enterprise detects a privileged cloud workload communicating with an unapproved destination while requesting access outside its normal role. A cyber-defense actor proposes isolation, credential revocation and traffic transfer to a validated standby instance. Containment may be reversible, but a false positive can interrupt an essential service; the decision is therefore at least CC-2 and may be CC-3 for the stated mission.

28.2 Transaction

Governance sets the dual objective: stop unauthorized data movement while maintaining the minimum service level. Intent names the workload, prohibited destructive operations, service-impact envelope and stop conditions. Evidence combines identity, network flow, attestation, runtime, data access and service health while marking shared analytic pipelines as correlated.

The actor proposes observe, restrict egress, isolate, revoke or transfer alternatives. If compromise support is insufficient, the DAF returns DEFER or SIMULATE. When target identity, independent telemetry, standby health and rollback readiness satisfy policy, it may issue PERMIT-WITH-CONSTRAINTS for the named workload and exact actions. Enforcement rejects wildcard expansion, deletion, target substitution and replay.

28.3 Closure

Independent network and service observations confirm that exfiltration stopped and service remained within bounds. Excessive service degradation invokes rollback or manual incident control and revokes remaining authority. The record links alert, evidence, proposal, disposition, enforcement, observed effect and recovery. Permission to call a cloud API is not authority to invoke every operation exposed by that API.

Cyber-containment checkpoints

Table 25. Cyber-containment checkpoints

Checkpoint	Permit condition	Non-permit behavior
Target identity	Workload and affected service are unambiguously bound.	Defer or quarantine on identity conflict.
Evidence	Compromise hypothesis has current, sufficiently independent support.	Observe, defer or simulate; do not destructively contain.
Continuity	Standby is validated and impact remains inside the approved envelope.	Escalate or choose a narrower restriction.
Enforcement	Exact target and actions are mediated by independent controls.	Deny wildcard, destructive or replayed requests.
Recovery	Rollback and credential restoration paths are available and tested.	Do not permit an effect outside the recovery envelope.
Observation	Network behavior and service health are independently measured.	Enter exception and recovery if effect is unverified or excessive.

Source context

This illustrative thread is informed by current AI-security, Cyber AI, CSF, control, secure-development, supply-chain and threat-taxonomy sources [4], [8], [10], [13], [20], [21], [37], [38]. It is not an incident-response procedure or evidence of operational validation.

29. Conformance Profiles and Claims

PURPOSE: Make AERA conformance bounded, falsifiable, versioned and evidence-specific

29.1 [NORMATIVE] Conformance rule

Conformance is claimed for a declared decision population, not for a brand, product family or technology portfolio. The claimant identifies AERA version, profile, actors, decisions, effects, consequence scope, environment, dependencies, exclusions, deviations, evidence package, assessment independence and validity. Implementing eight named layers does not demonstrate conformance unless applicable requirements are active in the operational path and verified.

Conformance and maturity are independent. A narrow implementation may conform at a modest maturity level; a highly automated enterprise may be nonconforming when an applicable requirement is unsatisfied. External-framework compliance may supply evidence but does not establish AERA conformance.

29.2 [NORMATIVE] Evidence-state precision

For every material mechanism, the evidence package declares the typed state model and highest supported state. State names from different models are not interchangeable. In particular, ACCEPTED_BY_SERVICE does not establish USED_IN_SOURCE_SOLUTION; USED_IN_SOURCE_SOLUTION does not establish USED_IN_AUTHORIZATION; and USED_IN_AUTHORIZATION does not establish VALIDATED_EFFECTIVE. Effectiveness requires representative, boundary, adversarial and degraded evidence against predeclared criteria.

29.3 [NORMATIVE] Applicability, inheritance and deviations

The requirements matrix records applicability separately from assessment. Applicability is APPLICABLE or NOT APPLICABLE. Assessment for an APPLICABLE requirement is SATISFIED, NOT SATISFIED, INDETERMINATE or APPROVED DEVIATION. A NOT APPLICABLE determination identifies technical and mission rationale, approving authority, evidence and invalidation conditions; it has no satisfaction result. Inherited controls identify provider, boundary, version, monitoring, failure behavior and withdrawal condition. A supplier assertion without validation at the relying interface is insufficient.

A deviation identifies requirement, reason, consequence analysis, compensating controls, residual risk, approving authority, monitoring, corrective action and expiration. Active approved deviations require CONDITIONALLY CONFORMING. An expired deviation, unapproved exception or unsatisfied applicable requirement makes the claim NONCONFORMING. No deviation may fabricate authority, disable attribution, conceal an unverified effect or convert assurance loss into permission.

29.4 [NORMATIVE] Outcome and reassessment

Assessment returns CONFORMING, CONDITIONALLY CONFORMING, NONCONFORMING or INDETERMINATE under Section 1.2. Claims expire on the stated date and on material change to authority, intent, consequence, actors, tools, evidence, policy, enforcement, PNT, cryptographic trust, operating environment, recovery or external obligations.

[NORMATIVE AERA-NORM-T24] Formal conformance profiles

Table 26. [NORMATIVE AERA-NORM-T24] Formal conformance profiles

Profile	Permitted subject	Minimum evidence focus
AERA-ARC	Bounded system or architecture	Layer/interface allocation, applicability matrix, threat and failure analysis, implementation evidence.
AERA-DAF	DAF service or federation	Semantic, policy, enforcement, binding, negative, audit, revocation and outage tests.
AERA-DEC	One bounded decision class	Contract, assurance case, representative/boundary/adversarial/degraded/recovery evidence.
AERA-MSN	End-to-end mission thread	Authority-to-effect trace, scenario branches, human-control assessment and residual risk.
AERA-DOM	Declared mission or enterprise domain	Decision inventory, constituent profiles, inherited controls, portfolio monitoring and exercises.

Profile	Permitted subject	Minimum evidence focus
AERA-SEC	Approved sector-profile subject	Core requirements plus the named sector profile, jurisdiction and documented tailoring.

Normative requirements

AERA-CONF-001 A formal conformance claim SHALL identify the AERA version, profile, claimant, subject, decision population, architectural and operational boundaries, consequence scope, dependencies, inherited controls, exclusions, assessment-independence level, evidence-package identifier and assessment date, deviations, and date-based and event-based expiration conditions.

AERA-CONF-002 For every requirement in the declared core and profile set, the claimant SHALL record applicability as APPLICABLE or NOT APPLICABLE separately from the assessment result; an APPLICABLE item SHALL be assessed SATISFIED, NOT SATISFIED, INDETERMINATE, or APPROVED DEVIATION, while a NOT APPLICABLE item SHALL identify rationale, evidence, approving authority, and invalidating conditions and SHALL NOT carry a satisfaction result.

AERA-CONF-003 For each consequence-material capability, source, or control cited by an operational claim, the claimant SHALL identify the typed state model, report the highest evidenced control-deployment state as PRESENT, MONITORED, ACTIVE IN PATH, USED IN COMPUTATION, or VALIDATED EFFECTIVE, and SHALL NOT represent a lower state or a state from another model as establishing a higher state.

AERA-CONF-004 Each deviation SHALL identify the unmet requirement, rationale, affected decision population and consequence class, compensating controls, residual risk, approving authority, monitoring, corrective action, expiration date, and event-based termination conditions.

AERA-CONF-005 A claim with an active approved deviation SHALL be labeled CONDITIONALLY CONFORMING.

AERA-CONF-006 An expired or unapproved deviation SHALL make the affected claim nonconforming.

AERA-CONF-007 Inherited controls SHALL be validated at the relying boundary and monitored for withdrawal conditions.

AERA-CONF-008 Material change SHALL suspend reliance on affected evidence until reassessment is complete.

AERA-CONF-009 Assessment independence SHALL be reported separately as self, independent internal or external assessment.

AERA-CONF-010 AERA-aligned SHALL NOT be presented as equivalent to conforming, certified or compliant.

AERA-EVID-112 A decision-relevant state claim SHALL identify its typed state model and SHALL NOT treat ACCEPTED_BY_SERVICE, USED_IN_SOURCE_SOLUTION, USED_IN_AUTHORIZATION, and VALIDATED_EFFECTIVE as equivalent states.

29.5 [NORMATIVE AERA-NORM-T25] Typed evidence-state models

Table 27. 29.5 [NORMATIVE AERA-NORM-T25] Typed evidence-state models

State model	Ordered states	Non-equivalence rule
Control deployment	PRESENT; MONITORED; ACTIVE_IN_PATH; USED_IN_COMPUTATION; VALIDATED_EFFECTIVE	Presence or activity does not prove material use or effectiveness.
Evidence service	PRESENT; MONITORED; ACCEPTED_BY_SERVICE; USED_IN_ASSURANCE_VERDICT; VALIDATED_EFFECTIVE	Service acceptance does not prove use in an authorization verdict.
PNT service	PRESENT; MONITORED; ACCEPTED_BY_PNT_SERVICE; USED_IN_PNT_SOLUTION; USED_IN_AUTHORIZATION; VALIDATED_EFFECTIVE	Use in a PNT solution does not prove use in the authorization transaction.

29.6 [NORMATIVE AERA-NORM-T26] Profile applicability floor

Table 28. 29.6 [NORMATIVE AERA-NORM-T26] Profile applicability floor

Profile	Always evaluated	Conditionally evaluated
AERA-ARC	CONF; PR; architecture boundaries; interface allocation; threat/failure analysis.	Every core family allocated to or inherited by the bounded architecture.
AERA-DAF	CONF; IF; AUTH; INT; EVID; DAF; ENF; REC; CY; THR; VV.	PNT when time, order, freshness, synchronization or position can affect disposition; CONS/HUM when the service claims those decisions.
AERA-DEC	AERA-DAF floor plus actor lifecycle, operational intent, consequence, human control and effect observation.	Quantum, specialized compute, sensor and sector controls when used by the decision class.
AERA-MSN	All requirements applicable to the end-to-end authority-to-effect mission thread.	Cross-domain, disconnected, aggregate and portfolio controls where the scenario can exercise them.
AERA-DOM	CONF; decision inventory; constituent DEC/MSN claims; inherited controls; portfolio monitoring and exercises.	All core requirements applicable to any included decision population.
AERA-SEC	Applicable AERA core plus every requirement in the named approved sector profile.	Jurisdictional tailoring that strengthens or adds evidence without weakening the core.

29.7 [NORMATIVE AERA-NORM-T27] Assessment-independence reporting

Table 29. 29.7 [NORMATIVE AERA-NORM-T27] Assessment-independence reporting

Level	Assessment relationship	Required disclosure
I1 Self	Performed or controlled by the function responsible for design, implementation, operation or management.	Responsible function, assessors, scope, evidence package and date; no independence claim.

Level	Assessment relationship	Required disclosure
I2 Independent internal	Performed by a competent internal function separated from design and operation and without direct delivery accountability.	Organizational separation, reporting path, competence, scope, evidence package and date.
I3 Independent external	Performed by a competent external entity operationally and contractually independent of the subject.	Sponsor, scope, methods, evidence-control rights, financial/delivery/advisory conflicts and date.

Comparative assessment context

SP 800-53A, RMF, ISO/IEC 42001, ISO/IEC/IEEE 15288 and IEEE 1012-2024 supply assessment, management and lifecycle concepts [14], [16], [39], [42], [44]. AERA profile applicability, independence levels and outcomes are proposed AERA semantics and are not certifications under those sources.

FORMAL CLAIM SYNTAX: [Subject] is [AERA version] [profile] [CONFORMING or CONDITIONALLY CONFORMING] for [explicit decision population], consequence scope [scope], assessed at independence level [I1, I2 or I3], using evidence package [identifier/date], with exclusions [list], deviations [list], valid until [date and material-change triggers].

30. Evidence-Based Maturity Assessment

PURPOSE: Measure demonstrated operating behavior without substituting maturity for conformance

30.1 [NORMATIVE] Assessment unit

Maturity is assessed for a declared decision domain, mission or consequence population. Policies, licenses, hardware, diagrams and roadmaps establish intent or presence; they do not demonstrate that a mechanism is active, used or effective. Different missions within one organization may legitimately have different maturity levels. These claim semantics are normative for AERA maturity claims; the model remains provisional pending multi-organization and inter-rater validation.

30.2 [NORMATIVE] Evidence strength

Evidence progresses from E0 ASSERTED, through E1 IMPLEMENTED and E2 OPERATIONALLY OBSERVED, to E3 TESTED EFFECTIVE and E4 SUSTAINED EFFECTIVE. Level 3 requires E3 evidence for consequence-material controls. Levels 4 and 5 require E3 for new integration or adaptation mechanisms and E4 for the claimed operating behaviors. Adverse results, workarounds and unresolved anomalies remain in the evidence package.

30.3 [NORMATIVE] Promotion rules

Each dimension is scored against current evidence. A higher-level practice does not compensate for a missing lower-level control. The reported maturity SHALL NOT exceed the lowest evidenced level among consequence-material dimensions; arithmetic averaging cannot conceal a control floor. A material change suspends reliance on affected scores until impact analysis and required reassessment are complete.

30.4 [NORMATIVE] Sustained-effective evidence

E4 SUSTAINED EFFECTIVE has no universal duration. Before assessment, the profile defines the minimum observation interval and operating exposure needed to encounter relevant mission states, changes, threats, overrides, failures and recovery events. Calendar duration without representative exposure is insufficient. The evidence package reports adverse results, configuration changes and limits on inference.

[NORMATIVE AERA-NORM-T28] Evidence-gated maturity levels

Table 30. [NORMATIVE AERA-NORM-T28] Evidence-gated maturity levels

Level	Operating state	Minimum promotion evidence
1 Aware	Material autonomous decisions are being identified; authority and recovery remain partly implicit.	Bounded inventory, named owners, prohibited-use rules and documented gaps.
2 Bounded	Selected decisions have explicit authority, intent, consequence, evidence, oversight and recovery envelopes.	Approved contracts, identity, action/effect logs, enforcement design and representative pilot evidence.
3 Assured	Runtime assurance evaluates current authority, evidence, cyber, time, consequence and recovery before effect.	Independent enforcement, policy-floor tests, adversarial/degraded tests, assurance cases and exercised recovery.
4 Integrated	Common assurance semantics span systems or domains and manage delegation, shared dependencies and cumulative risk.	Federation and partition tests, inherited-control validation, portfolio telemetry and cross-domain exercises.

Level	Operating state	Minimum promotion evidence
5 Adaptive	Behavior adjusts inside verified bounds; change causes continuous reassessment and controlled rollback.	Sustained evidence, drift/change detection, automatic inhibition, staged promotion and prevention of authority expansion.

Normative requirements

- AERA-MAT-001** A maturity claim SHALL identify its decision population, consequence scope, dependencies, exclusions and assessment date.
- AERA-MAT-002** Assessors SHALL score demonstrated operating behavior rather than planned or purchased capability.
- AERA-MAT-003** The reported level SHALL NOT exceed the lowest evidenced level among consequence-material dimensions.
- AERA-MAT-004** Level 3 SHALL require tested-effective evidence for every consequence-material control.
- AERA-MAT-005** Levels 4 and 5 SHALL require sustained-effective evidence for the operating behavior claimed.
- AERA-MAT-006** Adverse results, workarounds, overrides and unresolved anomalies SHALL remain in the evidence package.
- AERA-MAT-007** Promotion thresholds SHALL be declared before formal assessment.
- AERA-MAT-008** Material change SHALL trigger impact analysis and reassessment of affected dimension scores.
- AERA-MAT-009** A SUSTAINED EFFECTIVE evidence claim SHALL identify the minimum observation interval, operating exposure, event and consequence coverage, change history, adverse results, and rationale for representativeness before assessment begins.

[NORMATIVE AERA-NORM-T29] Maturity dimensions - complete level anchors

Table 31. [NORMATIVE AERA-NORM-T29] Maturity dimensions - complete level anchors

Dimension	L1 Aware	L2 Bounded	L3 Assured	L4 Integrated	L5 Adaptive
Governance	Ownership and prohibitions incomplete.	Owner, risk authority and review cadence defined.	Runtime, override, near-miss and recovery evidence informs governance.	Cross-domain portfolio governance controls shared risk.	Sustained evidence changes authority and investment under independent challenge.
Authority and intent	Access is used as a proxy for authority.	Delegation and intent are machine-reviewable and versioned.	Runtime authorization resolves scope, validity and conflict before effect.	Delegation chains and shared limits work across domains.	Dynamic adjustment is preauthorized, continuously evaluated and non-expansive.
AI actor lifecycle	Models or tools are inventoried inconsistently.	Identity, owner, tools, configuration and competence claim exist.	Attestation, tool limits, subdelegation and performance are tested.	Common lifecycle controls span heterogeneous actors and providers.	Drift or change automatically inhibits or rolls back.
Evidence	Ad hoc logs and unqualified sources.	Source, freshness, provenance, uncertainty and retention defined.	Contradiction, replay, manipulation and common-mode failure tested.	Evidence semantics and inherited sources are validated across domains.	Evidence quality continuously constrains authority.
DAF and enforcement	Authorization and effect mediation are implicit.	Contracts, dispositions and effect boundaries identified.	Exact binding, expiry, revocation, enforcement and outage behavior tested.	Federated policy and enforcement reconcile partition and cumulative state.	Policy adapts inside approved limits with automatic rollback.
Cybersecurity	Perimeter and resource-access controls dominate.	Relevant identities and least-authority roles are bound.	Threat-informed tests cover the complete proposal-to-effect path.	Threat, identity and supply-chain state propagate across domains.	Threat state continuously modifies disposition and containment.
Timing/PNT	Time and position are unqualified infrastructure inputs.	Decision-specific quality and degraded behavior inputs.	Present, monitored, accepted and used states plus common-mode failures tested.	Federated temporal state and uncertainty are reconciled.	Authority windows adapt only inside prevalidated bounds.
Effect and recovery	Effect observation and recovery are largely manual.	Expected effects, stop conditions, takeover and recovery documented.	Independent observation and recovery exercised.	Cross-domain effects and shared recovery resources are managed.	Divergence automatically reduces authority and invokes recovery.
V&V and assurance	Tests demonstrate components without bounded claims.	Traceability, oracles and acceptance criteria exist.	Approved cases include adversarial, degraded and human evidence.	Reusable evidence and inherited claims are validated across boundaries.	Cases consume sustained evidence and flag invalidated claims.
Learning and change	Behavior change may occur outside governance.	Artifacts are versioned, approved, observable and reversible.	Drift and rollback tested; learning cannot expand authority.	Staged promotion and rollback coordinate across dependencies.	Governed adaptation uses continuous evaluation and automatic rollback.

31. Requirement-Level Crosswalk

PURPOSE: Identify reusable external evidence and the AERA-specific semantic delta without implying reciprocal compliance.

31.1 Mapping method

The controlled mapping method follows the concept-mapping approach in NIST IR 8477 and the National OLIR lifecycle in NIST IR 8278 and 8278A Rev. 1 [50]-[53]. A mapping assertion identifies reference-document element, focal-document element, relationship style, directional relationship, rationale, source editions, mapping status and review metadata. Set-theory relationships use Subset of, Intersects with, Equal, Superset of or No relationship. Supportive relationships are identified separately. Equality is prohibited unless semantic scope, modality, conditions and expected outcomes are shown to be equivalent.

The table retains the prior D/P/E/C/N notation only as legacy working data: D direct overlap; P partial overlap; E enabling evidence; C context; N no material mapping. No IR 8477 relationship is inferred from those codes. Mixed or compound anchors must be split into atomic assertions before OLIR submission. External evidence is reusable only when the asset, decision class, implementation, operating context, assessment period, integrity and assurance level match the AERA claim. The companion OLIR mapping workbook is the machine-readable development ledger; it is not submission-ready until every assertion has exact target identifiers, rationale and independent SME adjudication.

Table 32. AERA requirement / External anchors

AERA requirement	External anchors	Legacy	IR 8477 interpretation	AERA-specific delta
AERA-DA-001	AI RMF GOVERN 2.1 and 2.3; CSF 2.0 GV.RR-01 and GV.RR-02; ISO/IEC 42001 Clauses 5 and 7; DoD Responsible principle; RMF Prepare [1], [10], [16], [39], [46], [47]	P	Not asserted - atomic SME adjudication required	External sources establish accountability and roles but do not require an accountable owner for each autonomous decision class.
AERA-DA-002	CSF 2.0 GV.RR-02 and PR.AA-05; SP 800-53 AC-2, AC-3, AC-6, AC-16, AC-24; SP 800-207 policy decision concepts; CISA ZTMM Identity and Governance [10], [11], [13], [35]	P	Not asserted - atomic SME adjudication required	Access roles and entitlements do not encode effect class, consequence ceiling, recovery duty, or subdelegation semantics.
AERA-DA-003	SP 800-53 AC-3, IA-5, IA-9, SC-12; SP 800-207 trust algorithm inputs; ISO/IEC 42001 Annex A organizational and lifecycle controls [11], [13], [39]	P	Not asserted - atomic SME adjudication required	End-to-end delegation-chain validation to a human accountable authority is AERA-specific.
AERA-DA-004	SP 800-53 AC-2, AC-3, AC-6; CSF 2.0 PR.AA-05; SP 800-207 least-privilege resource access [10], [11], [13]	P	Not asserted - atomic SME adjudication required	Ordinary privilege delegation does not constrain transitive autonomous decision authority across effects.
AERA-DA-005	SP 800-53 AC-2, AC-3, CA-7; CSF 2.0 PR.AA-01 and PR.AA-05; SP 800-207 continuous evaluation; CISA ZTMM Identity [10], [11], [13], [35]	P	Not asserted - atomic SME adjudication required	AERA requires bounded revocation propagation to dependent authorizations and forbids stale state from authorizing a new effect.
AERA-DA-006	CSF 2.0 PR.AA-05; SP 800-53 AC-5 and AC-6; ISO/IEC 42001 Clause 5; DoD Responsible and Governable principles [10], [13], [39], [46], [47]	D/P	Not asserted - atomic SME adjudication required	Separation of duties maps directly; AERA adds consequence-scaled separation among proposer, classifier, authorizer, enforcer, and override authority.
AERA-DA-007	CSF 2.0 PR.AA-01 through PR.AA-05; SP 800-53 IA-3, IA-4, IA-5, IA-9; SP 800-207A application/service identity; CISA ZTMM Identity, Devices, and Applications and Workloads [10], [11], [12], [13], [35]	P	Not asserted - atomic SME adjudication required	AERA binds identity to model/software version, integrity state, tool set, owner, and consequence ceiling rather than identity alone.
AERA-DA-008	AI RMF MAP 1.1, 1.3, 1.6, and 3.3; ISO/IEC/IEEE 15288 stakeholder needs, system requirements, and architecture definition processes; DoD Reliable principle and Requirements Validation tenet [1], [16], [42], [46], [47]	P	Not asserted - atomic SME adjudication required	External sources address intended use and requirements; AERA requires a signed, executable intent object used at runtime.

AERA requirement	External anchors	Legacy	IR 8477 interpretation	AERA-specific delta
AERA-DA-009	SP 800-53 AC-4 and AC-24; SP 800-207 policy engine concepts; ISO/IEC/IEEE 15288 decision-management and requirements processes; SP 800-160 Vol. 1 architecture and design principles [11], [13], [18], [19], [42]	E/P	Not asserted - atomic SME adjudication required	Deterministic conflict precedence with a fail-safe authorization outcome is AERA-specific.
AERA-DA-010	CSF 2.0 GV.PO-01, GV.PO-02, and ID.RA-07; SP 800-53 CM-2, CM-3, CM-5, AU-3; ISO/IEC 42001 Clauses 8-10 [10], [13], [39]	P	Not asserted - atomic SME adjudication required	AERA cryptographically binds the exact policy and evaluator versions to each disposition.
AERA-DA-011	AI RMF MAP 1.5, MAP 3.2, MANAGE 1.2 and 1.4; CSF 2.0 GV.RM-02 and GV.RM-06; SP 800-53 RA-3 and PM-9; ISO/IEC 23894 risk context, analysis, evaluation, and treatment [1], [10], [13], [16], [40]	P	Not asserted - atomic SME adjudication required	External risk processes do not define AERA consequence attributes or prohibit proposer-only downgrade.
AERA-DA-012	SP 800-53 AC-16, AC-24, AU-10, SC-16; SP 800-207 policy administrator/enforcement concepts; SP 800-207A granular application policy [11], [12], [13]	P	Not asserted - atomic SME adjudication required	AERA requires canonical proposal-to-effect cryptographic binding and reauthorization after mutation.
AERA-DA-013	AI RMF MAP 2.3, MEASURE 2.1, MEASURE 2.8 and 2.9; CSF 2.0 PR.DS-01, PR.DS-02, PR.DS-10, RS.AN-06 and RS.AN-07; SP 800-53 AU-3, AU-9, AU-12, SI-7; DoD Traceable principle [1], [10], [13], [16], [46], [47]	P	Not asserted - atomic SME adjudication required	AERA defines the evidence manifest used in a single authorization transaction, including uncertainty and freshness.
AERA-DA-014	AI RMF MEASURE 1.3 and 2.3; CSF 2.0 DE.AE-03; SP 800-53 CA-2, RA-3, SI-4; SP 800-160 Vol. 2 diversity, redundancy, and substantiated-integrity techniques [1], [10], [13], [16], [18], [19]	P	Not asserted - atomic SME adjudication required	Multiple sources are not necessarily independent; AERA requires explicit failure-domain proof.
AERA-DA-015	CSF 2.0 ID.RA-09, PR.AA-04, PR.DS-02; SP 800-53 IA-9, SI-7, SC-8; SP 800-207A service identity; NIST IR 8323 Rev. 1 PR.AC-6/7 and PR.DS-6 applicability statements [10], [11], [12], [13], [22]	D/P	Not asserted - atomic SME adjudication required	Source authentication maps directly; AERA additionally excludes or explicitly discounts unverifiable evidence in runtime policy.
AERA-DA-016	AI RMF MAP 2.2, MEASURE 1.1, MEASURE 2.5 and 2.13; SP 800-53 SI-4 and SI-7; DoD Reliable and Governable principles [1], [13], [16], [46], [47]	P	Not asserted - atomic SME adjudication required	AERA requires explicit disposition behavior for contradictory evidence and forbids scalar masking of mandatory failures.
AERA-DA-017	CSF 2.0 DE.CM-09 and DE.AE-02; SP 800-53 AU-8, SI-4, SC-45; NIST IR 8323 Rev. 1 Identify, Protect, and Detect PNT profile subcategories [10], [13], [22]	P	Not asserted - atomic SME adjudication required	AERA sets evidence-type and decision-class freshness limits that directly drive disposition.
AERA-DA-018	AI RMF trustworthiness characteristics; MEASURE 2.3 and 2.5-2.11; DoD Equitable, Traceable, Reliable, and Governable principles [1], [16], [46], [47]	C/P	Not asserted - atomic SME adjudication required	AI RMF and DoD provide characteristics; the non-compensatory runtime assurance vector and mandatory-claim logic are AERA-specific.
AERA-DA-019	AI RMF MEASURE 1.2, 2.3, 2.5, 2.13, 3.1, and 4.1; ISO/IEC 23894 monitoring and review; DoD Reliable principle [1], [16], [40], [46], [47]	D/P	Not asserted - atomic SME adjudication required	Calibration and monitoring align; AERA requires decision-bound calibration metadata and explicit drift response.
AERA-DA-020	NIST IR 8323 Rev. 1 ID.AM-3, ID.GV-4, PR.DS-6, DE.AE-3 and related PNT applicability statements; SP 800-53 AU-8 and SC-45 [13], [22]	P	Not asserted - atomic SME adjudication required	AERA's present/monitored/accepted/used-in-computed-decision status distinction is not an IR 8323 conformance label.
AERA-DA-021	NIST IR 8323 Rev. 1 PR.DS-4, PR.DS-6, DE.AE and DE.CM PNT applicability statements; CSF 2.0 PR.IR-03, DE.AE-03, RS.MI-01, RC.RP-02; SP 800-53 CP-2, CP-4, SC-45, SI-4; SP 800-160 Vol. 2 [10], [13], [18], [19], [22]	D/P	Not asserted - atomic SME adjudication required	PNT resilience maps strongly; AERA ties degraded PNT state directly to decision authority and safe effect behavior.
AERA-DA-022	SP 800-207 policy enforcement point; SP 800-207A gateways, sidecars, and application policy enforcement; CSF 2.0 PR.AA-05; SP 800-53 AC-3 and AC-4; CISA ZTMM Applications and Workloads, Networks, and Automation and Orchestration [10], [11], [12], [13], [35]	P	Not asserted - atomic SME adjudication required	AERA enforcement mediates consequential effects, not only access to resources or application traffic.
AERA-DA-023	AI RMF MANAGE 1.1-1.4 and MANAGE 2.4; DoD Governable principle; SP 800-53 AC-24 and IR-4 [1], [13], [16], [46], [47]	P	Not asserted - atomic SME adjudication required	AERA standardizes a runtime disposition vocabulary, rationale, constraints, and expiry.
AERA-DA-024	CSF 2.0 PR.AA-05; SP 800-53 AC-3, AC-6, AC-16, AC-24; SP 800-207 per-session/request access concepts; SP 800-207A granular identity-tier policy [10], [11], [12], [13]	P	Not asserted - atomic SME adjudication required	AERA requires capability-like authorization bound to exact effect parameters, sequence, consequence, and context.
AERA-DA-025	CSF 2.0 PR.IR-03 and PR.IR-04; SP 800-53 CP-2, CP-4, CP-12 and SI-17; SP 800-160 Vol. 2 cyber-resiliency engineering [10], [13], [18], [19]	D/P	Not asserted - atomic SME adjudication required	Fail-safe and resilience practices align; AERA explicitly prohibits authorization by DAF timeout or unavailability.
AERA-DA-026	AI RMF MEASURE 2.4, MEASURE 3.1, and MANAGE 4.1; CSF 2.0 ID.RA-07, DE.CM-09; SP 800-53 CA-7 and SI-4; SP 800-207 continuous evaluation [1], [10], [11], [13], [16]	P	Not asserted - atomic SME adjudication required	AERA enumerates context changes that invalidate an effect authorization and require reevaluation.
AERA-DA-027	AI RMF MAP 5 and MANAGE 1.2; SP 800-53 AC-6 and SI-4; ISO/IEC 23894 risk analysis; ISO/IEC/IEEE 15288 system analysis and decision management [1], [13], [16], [40], [42]	C/P	Not asserted - atomic SME adjudication required	Stateful cumulative-effect budgets across actors and decomposed actions are AERA-specific.
AERA-DA-028	CSF 2.0 PR.AA-05; SP 800-53 AC-3, AC-6, IA-5; SP 800-207 and SP 800-207A granular least privilege; CISA ZTMM Identity and Applications and Workloads [10], [11], [12], [13], [35]	D/P	Not asserted - atomic SME adjudication required	Least privilege maps directly; AERA adds transaction-only effect capabilities restricted by target, audience, and action.
AERA-DA-029	AI RMF MEASURE 1.3; CSF 2.0 DE.AE-03; SP 800-160 Vol. 2 diversity and redundancy; DoD Reliable principle [1], [10], [16], [18], [19], [46], [47]	P	Not asserted - atomic SME adjudication required	AERA defines quorum independence by lineage and shared failure domains rather than numerical votes.
AERA-DA-030	AI RMF GOVERN 6, MAP 4, MANAGE 3.1 and 3.2; CSF 2.0 GV.SC-05 through GV.SC-10 and DE.CM-06; SP 800-53 SA-9, SR-3, SR-5, SR-6, SR-10, SR-11; ISO/IEC 42001 Annex A third-party controls; SP 800-160 Vol. 2 [1], [10], [13], [16], [18], [19], [39]	P	Not asserted - atomic SME adjudication required	AERA binds each external or quantum result to the request and requires characterized classical fallback and independent effect enforcement.
AERA-DA-031	AI RMF GOVERN 4.2, MEASURE 2.8 and 2.9; CSF 2.0 PR.PS-04, RS.AN-03, RS.AN-06 and RS.AN-07; SP 800-53 AU-2, AU-3, AU-8, AU-9, AU-10, AU-12; DoD Traceable principle [1], [10], [13], [16], [46], [47]	D/P	Not asserted - atomic SME adjudication required	Audit and provenance map directly; AERA requires causal continuity from authority and intent through observed effect and recovery.
AERA-DA-032	AI RMF MEASURE 2.6, MANAGE 2.3, MANAGE 2.4, and MANAGE 4.1; CSF 2.0 ID.IM-04, RS.MI-01, RC.RP-01 through RC.RP-06; SP 800-53 CP-2, CP-4, CP-10, CP-12, IR-4; SP 800-160 Vol. 2; DoD Governable principle [1], [10], [13], [16], [18], [19], [46], [47]	D/P	Not asserted - atomic SME adjudication required	Recovery and disengagement align strongly; AERA adds a decision-class recovery envelope and independence from the actor.
AERA-DA-033	AI RMF MEASURE 2.4, MEASURE 3.1, MANAGE 4.1; CSF 2.0 DE.CM-01, DE.CM-06, DE.CM-09; SP 800-53 CA-7 and SI-4; RMF Monitor; CISA ZTMM Visibility and Analytics [1], [10], [13], [16], [35]	D/P	Not asserted - atomic SME adjudication required	AERA monitoring spans authority, evidence, consequence, PNT, and recovery state and drives reauthorization.
AERA-DA-034	AI RMF GOVERN 1.5 and 1.7, MEASURE 2.4, MANAGE 4.1 and 4.2; CSF 2.0 ID.RA-07, PR.PS-01, PR.PS-02, PR.PS-06; SP 800-53 CM-2, CM-3, SA-15, SA-24, SI-2, SI-2(7), SI-7; ISO/IEC/IEEE 15288 configuration management and lifecycle processes; ISO/IEC 42001 Clauses 8-10 [1], [10], [13], [16], [39], [42]	D/P	Not asserted - atomic SME adjudication required	AERA treats prompts, memory, thresholds, and learning changes as behavior-bearing configuration with consequence-scaled approval.
AERA-DA-035	AI RMF MAP 2.3, MEASURE 1.3, MEASURE 2.1-2.7 and 2.13; CSF 2.0 ID.IM-01 and ID.IM-02; SP 800-53 CA-2, CA-8, SA-11, SA-15, SA-24; SP 800-53A Release 5.2.0; RMF Assess; ISO/IEC/IEEE 15288 verification and validation; SP 800-160 Vols. 1 and 2; DoD Reliable principle and RAI lifecycle practices [1], [10], [13], [14], [16], [18], [19], [42], [46], [47]	P	Not asserted - atomic SME adjudication required	External TEVV practices support testing; AERA requires explicit coverage of every applicable DAT threat and treats untested threats as open assurance claims.
AERA-DA-036	CSF 2.0 ID.AM, ID.RA-04 through ID.RA-07, PR.DS-01 and PR.DS-02; SP 800-53 CM-8, RA-3, SC-12, SC-13, SC-17; RMF Prepare/Select/Implement/Assess/Monitor; ISO/IEC/IEEE 15288 lifecycle, configuration, risk, verification, and transition processes; FIPS 203 ML-KEM, FIPS 204 ML-DSA, FIPS 205 SLH-DSA [10], [13], [16], [25], [26], [27], [42]	D/P/E	Not asserted - atomic SME adjudication required	FIPS directly specifies algorithms. AERA adds inventory, exposure-period analysis, protocol selection, crypto agility, migration sequencing, interoperability, performance, implementation validation, errata tracking, and claim boundaries.

31.2 Source-status controls

Table 33. 31.2 Source-status controls

Source	Status used by AERA 1.2 WD	Required caveat
NIST AI RMF [1]-[3]	AI RMF 1.0 is the mapped final baseline; the program page reports revision activity.	A future final revision triggers crosswalk review; GenAI profile and revision activity do not alter the 1.2 WD mapping baseline.
NIST agent activity [5]-[7]	Program initiative, draft NCCoE concept paper and published RFI-response analysis.	None is a final agent-control standard, AERA validation or endorsement.
NIST Cyber AI [8]	NIST IR 8596 initial preliminary draft.	Context only; draft outcomes are not final requirements or a compliance basis.
NIST CSF / 800-53 [10], [13]-[15]	CSF 2.0; SP 800-53/53A/53B Release 5.2.0 status distinguished.	SA-24 is relevant but was not added to a 53B baseline; selection and tailoring remain explicit.
NIST PNT Profile [22]-[24]	IR 8323 Rev. 1 is final; Rev. 2 was an initial public draft on 7 August 2026.	The draft is a future-review trigger and does not replace the final mapped baseline.
PQC [25]-[31]	FIPS 203-205 final; SP 800-227 and CSWP 39 Update 1 final; IR 8547 draft; program register living.	Algorithm, KEM-use, agility, draft transition and implementation claims remain distinct; track authoritative errata.
MITRE [37], [38]	ATLAS and ATT&CK are living knowledge bases; ATT&CK mapping is pinned to v19.1.	Threat mappings are partial and do not establish complete risk assessment or AERA conformance.
ISO / IEEE [39]-[44]	Public edition and clause-family mappings; IEEE 1012-2024 replaces the 2016 edition.	Clause-level equivalence requires authorized full-text review; ISO 31000 is guidance, not certification.
DoD / CDAO [46]-[49]	Principles, pathway and living toolkits provide public-sector policy and implementation context.	AERA does not replace mission, legal, test, safety, acquisition or authorization obligations.
NIST mapping and OLIR [50]-[53]	IR 8477 and IR 8278/8278A Rev. 1 are final; OLIR is a living submission program.	AERA mappings remain developmental and are not an OLIR submission or NIST-reviewed artifact.
Current NIST lifecycle and engagement sources [54]-[69]	SP 800-18r2, SP 800-63-4, SP 800-61r3, AI 100-5e2025, AI 700-2, IR 8607, IR 8286r1 and SP 800-55 Vols. 1-2 are final; AIRC, OSCAL and the AI Consortium participation process are living; Privacy Framework 1.1 remains draft.	Use each source only for its stated scope and status; no source establishes AERA validity, selection, support or endorsement.

SOURCE STATUS: External mappings were checked against publisher-controlled sources on 7 August 2026. Draft and living resources are labeled individually. Continuously maintained sources are versioned or date-stamped at use. Mappings are directional from AERA and do not imply reciprocal compliance, adoption or endorsement.

32. AERA Specification Family and Governance

PURPOSE: Turn AERA into a maintainable family of stable core requirements and sector extensions

32.1 Publication architecture

AERA-001 is the stable core reference architecture. Companion publications separate durable semantics from faster-moving implementation guidance. The family uses one requirements namespace, one term registry, one interface registry and a versioned machine-readable crosswalk. Sector profiles may strengthen requirements and add evidence but shall not weaken the AERA core or imply regulatory approval.

32.2 Change control

AERA changes are classified as editorial, compatible clarification, additive minor revision or breaking major revision. Every proposed normative change identifies affected definitions, requirements, interfaces, threat controls, crosswalks, schemas, conformance claims and migration actions. A public issue record and disposition preserve dissent, rationale and effective date.

A requirement is retired only through a published successor or explicit withdrawal. Existing conformance claims remain bound to the version assessed; they do not silently migrate to a later edition. Urgent security advisories may constrain use before the next revision but shall identify scope, evidence and expiration.

32.3 Intellectual and evidentiary boundary

The autonomous-enterprise thesis, eight-layer model, Decision Assurance Fabric, assurance vector, decision transaction, consequence classes and conformance model are proposed AERA constructs. External frameworks retain their own authority and terminology. Crosswalks are interpretive engineering artifacts, not statements issued or endorsed by the external publisher. The NIST AI Agent Standards Initiative is current ecosystem context only [5]; TOGAF is an architecture-framework comparison only [45].

32.4 Open development and consensus boundary

AERA is independently authored. It may become consensus-oriented only through a documented process that provides public access, defined review periods, balanced stakeholder participation, transparent disposition, preserved dissent, versioned drafts and stable maintenance. A public comment period or NIST-facing submission does not by itself create consensus.

The editor controls working drafts and release integrity. Contributors propose changes; reviewers identify defects and evidence; pilot organizations report implementation outcomes; assessors evaluate bounded claims. Roles, affiliations, sponsorship and conflicts are disclosed. No reviewer, pilot participant or external organization is represented as an endorser without explicit authorization.

32.5 Release and registry control

One controlled source generates or is reconciled to the normative document, requirement register, threat matrix, source register and mapping ledger. A release candidate is blocked by semantic divergence, missing identifiers, unresolved critical defects, inaccurate source status, formula or schema errors, stale filenames, missing license terms, or unmatched checksums.

Machine-readable mappings remain DEVELOPMENTAL until atomic target identifiers, relationship rationale, editions, status and independent adjudication are complete. OLIR-ready describes format and evidence readiness only; it does not mean submitted, accepted, published or endorsed.

32.6 Outcome core, reference architecture and conformance separation

AERA 1.2 WD remains a consolidated review document. Before a consensus-oriented release, the publication family should separate an outcome-focused, voluntary AERA Core from the normative reference architecture, technical profiles and conformance-assessment procedures. This separation allows broad risk-management outcomes to remain non-prescriptive while architecture and evidence obligations remain precise and testable.

The planned split preserves stable identifiers and traceability. Use of the outcome Core alone does not demonstrate implementation of the reference architecture, satisfaction of a profile or conformance. The working draft requests reviewer input on publication boundaries, identifier migration and minimum evidence for each claim type.

32.7 Requirement namespace and lineage

GOV denotes operational governance requirements for implementations. GOVR denotes governance of the AERA publication and maintenance process. The overlapping legacy 001/101-series families remain a known lineage issue in this working draft and require explicit parent, refinement, supersession and status fields before a closed conformance set is declared.

AERA-DA-023 and AERA-DAF-003 now point to the canonical disposition semantics in AERA-DAF-110 through AERA-DAF-114. This preserves stable identifiers while removing the prior constrain/constrained-permit collision. The change record and future registry will retain their prior wording and semantic delta.

Planned AERA publication family

Table 34. Planned AERA publication family

Identifier	Publication	Purpose
AERA-001	Autonomous Enterprise Reference Architecture	Consolidated working draft; planned separation into outcome Core, reference architecture and conformance procedures.
AERA-002	Decision Assurance Fabric Specification	Detailed semantics, state model, policies, object schemas and interoperability tests.
AERA-003	Decision-Authority Threat Catalog	Threat patterns, mitigations, test cases and incident mappings.
AERA-004	Assurance and Conformance Handbook	Assessment procedures, evidence packages, deviations and assessor guidance.
AERA-005	Critical Infrastructure Profile	Sector-tailored control floors and mission-thread patterns.
AERA-006	Defense and National Security Profile	Mission assurance, cross-domain, contested-operation and delegated-authority extensions.
AERA-007	Quantum Integration Profile	Capability readiness, advantage gates, service contracts and classical fallback.
AERA-008	Resilient Timing and PNT Profile	Decision-specific time/PNT integrity, holdover, source-state and validation guidance.
AERA-009	Implementation Patterns	Reference deployments, APIs, policy patterns and worked examples.
AERA-010	Machine-Readable Registries	Requirements, interfaces, terms, crosswalks, OSCAL-compatible candidates and change metadata.

Normative requirements

AERA-GOVR-001 Every normative change SHALL identify impact on definitions, requirements, interfaces, schemas, threats, crosswalks and conformance.

AERA-GOVR-002 Sector profiles SHALL preserve or strengthen the AERA core control floor.

AERA-GOVR-003 A conformance claim SHALL remain bound to the AERA edition assessed.

AERA-GOVR-004 External mappings SHALL be versioned, directional and reviewed on source or AERA revision.

AERA-GOVR-005 An urgent advisory SHALL identify scope, evidence, required action and expiration or incorporation path.

AERA-GOVR-006 The controlled publication process SHALL maintain a public, versioned change record that links each material comment or proposal to its disposition, rationale, affected artifacts, and effective edition.

AERA-GOVR-007 The publication process SHALL identify editor, contributor, reviewer, pilot, and assessor roles and SHALL disclose relevant competence, organizational affiliation, sponsorship, and conflicts of interest.

AERA-GOVR-008 A proposed normative revision SHALL receive a defined multistakeholder review period, preserve substantive dissent, and publish disposition before it is represented as a consensus release.

AERA-GOVR-009 Human-readable and machine-readable normative artifacts SHALL be generated from or reconciled to one authoritative controlled source, carry the same edition identifier, and pass semantic and integrity checks before release.

AERA-GOVR-010 A maturity, conformance, interoperability, or effectiveness claim SHALL state the assessed population, assessor independence, evidence period, known limitations, and whether multi-organization or inter-rater validation has been completed.

AERA-GOVR-011 Each public edition SHALL state its copyright, license, contribution terms, permitted uses, and intellectual-property process before external redistribution or standards submission.

AERA-GOVR-012 A released artifact SHALL be immutable; any byte-changing replacement SHALL receive a new build identifier and manifest, and any normative change SHALL receive a new specification version.

Appendix A - Consolidated Requirements Catalog

CATALOG STATUS: 199 unique normative requirements are indexed below. Verification codes are I = inspection, A = analysis, T = test and D = demonstration. The applicability matrix for a conformance claim tailors method and evidence strength by consequence; it does not weaken requirement semantics.

Table 35. ID / Normative requirement

ID	Normative requirement	Primary verification
AERA-ACTOR-001	Each actor SHALL have workload identity, owner, role, competency basis, configuration, tool/data rights, consequence ceiling, lifecycle state, and retirement/suspension.	I, T
AERA-ACTOR-002	Before consequential proposal/effect, the implementation SHALL verify actor identity and consequence-material model, context, orchestration, tool, and runtime integrity against approved or declared degraded state.	I, T
AERA-ACTOR-003	Actors SHALL NOT subdelegate unless expressly permitted; permitted subdelegation SHALL preserve constraints/attribution and SHALL NOT increase scope, consequence, duration, tools, targets, or data rights.	A, T
AERA-AUTH-001	Every material autonomous action SHALL resolve at authorization time to a current, attributable, revocable delegation from an accountable grantor.	I, T, D
AERA-AUTH-002	A delegation SHALL state grantor, grantee, permitted decisions/effects, target/domain, context, prohibitions, consequence ceiling, validity, evidence, subdelegation, escalation, and revocation.	I, T
AERA-AUTH-003	Revoking delegation, identity, policy, or actor SHALL invalidate and propagate withdrawal to all unexecuted derived authorizations within a declared consequence-specific bound.	A, T
AERA-AUTH-101	Every autonomous material effect SHALL resolve to a current, attributable, revocable delegation rooted in an accountable authority.	I, A, T
AERA-AUTH-102	Possession of credentials, resource access, network reachability, tool availability, model registration, or prior approval SHALL NOT be treated as decision authority.	I, T
AERA-AUTH-103	A delegation SHALL state subject, issuer, permitted actions, targets, purposes, effects, boundaries, consequence ceiling, validity, predicates, prohibitions, and revocation semantics.	I
AERA-AUTH-104	Subdelegation SHALL be prohibited unless every applicable parent grant explicitly permits it.	I, T
AERA-AUTH-105	A child grant SHALL NOT expand the scope, duration, use, resource, consequence, oversight, evidence, enforcement, or recovery authority of its effective parent chain.	A, T
AERA-AUTH-106	Expiration or revocation of a parent grant SHALL invalidate dependent grants within the maximum propagation time defined by policy.	I, T
AERA-AUTH-107	The DAF SHALL validate the complete effective delegation chain, including issuer authority, integrity, revocation, attenuation, and applicable prohibitions.	I, A, T
AERA-AUTH-108	An applicable explicit deny or higher-precedence prohibition SHALL prevail over a lower-precedence allow.	I, T
AERA-AUTH-109	Emergency authority SHALL be purpose-limited, time-bounded, attributable, auditable, and unable to override constraints that the issuer lacks authority to modify.	I, A, T
AERA-AUTH-110	A disconnected-operation grant SHALL define duration, decision scope, consequence ceiling, local evidence and time requirements, reconciliation, and behavior when any bound is exceeded.	I, A, T
AERA-CONF-001	A formal conformance claim SHALL identify the AERA version, profile, claimant, subject, decision population, architectural and operational boundaries, consequence scope, dependencies, inherited controls, exclusions, assessment-independence level, evidence-package identifier and assessment date, deviations, and date-based and event-based expiration conditions.	I, A
AERA-CONF-002	For every requirement in the declared core and profile set, the claimant SHALL record applicability as APPLICABLE or NOT APPLICABLE separately from the assessment result; an APPLICABLE item SHALL be assessed SATISFIED, NOT SATISFIED, INDETERMINATE, or APPROVED DEVIATION, while a NOT APPLICABLE item SHALL identify rationale, evidence, approving authority, and invalidating conditions and SHALL NOT carry a satisfaction result.	I, A
AERA-CONF-003	For each consequence-material capability, source, or control cited by an operational claim, the claimant SHALL identify the typed state model, report the highest evidenced control-deployment state as PRESENT, MONITORED, ACTIVE IN PATH, USED IN COMPUTATION, or VALIDATED EFFECTIVE, and SHALL NOT represent a lower state or a state from another model as establishing a higher state.	I, T, D
AERA-CONF-004	Each deviation SHALL identify the unmet requirement, rationale, affected decision population and consequence class, compensating controls, residual risk, approving authority, monitoring, corrective action, expiration date, and event-based termination conditions.	I, A
AERA-CONF-005	A claim with an active approved deviation SHALL be labeled CONDITIONALLY CONFORMING.	I, A, T and/or D as consequence requires
AERA-CONF-006	An expired or unapproved deviation SHALL make the affected claim nonconforming.	I, A, T and/or D as consequence requires
AERA-CONF-007	Inherited controls SHALL be validated at the relying boundary and monitored for withdrawal conditions.	I, A, T and/or D as consequence requires
AERA-CONF-008	Material change SHALL suspend reliance on affected evidence until reassessment is complete.	I, A, T and/or D as consequence requires

ID	Normative requirement	Primary verification
AERA-CONF-009	Assessment independence SHALL be reported separately as self, independent internal or external assessment.	I, A, T and/or D as consequence requires
AERA-CONF-010	AERA-aligned SHALL NOT be presented as equivalent to conforming, certified or compliant.	I, A, T and/or D as consequence requires
AERA-CONS-101	Each decision class SHALL use a documented consequence-classification method that addresses severity, affected scope, reversibility, time to harm, propagation, detectability, cumulative effects, mission significance, legal, rights, and safety significance, recovery time and resources, and uncertainty.	I, A
AERA-CONS-102	Consequence classification SHALL use the highest class indicated by the worst credible direct, secondary, cumulative, propagated, or irreversible effect within the declared operating and threat context.	A, T
AERA-CONS-103	A consequence attribute that establishes a higher minimum class SHALL NOT be reduced by averaging or offsetting it against lower-consequence attributes.	I, A, T
AERA-CONS-104	The DAF SHALL re-evaluate consequence before further authorization when a sequence, batch, concurrent set, campaign, or portfolio can exceed an individual-action limit or alter the worst credible effect.	A, T
AERA-CONS-105	When available evidence supports more than one consequence class, classification SHALL use the highest applicable class until evidence supports a narrower class and the downgrade is approved by an authority independent of the proposer.	I, A, T
AERA-CONS-106	When concurrent transactions share a cumulative effect or resource limit, the DAF and enforcement path SHALL use bounded reservations or an equivalent atomic control that prevents aggregate authorized or executed effects from exceeding the limit.	A, T
AERA-CY-001	Credentials used on a material-effect path SHALL be scoped, to the extent supported, to the authorized actor, audience, action, target, context, and validity interval; where native credential scoping is insufficient, the enforcement point SHALL enforce equivalent restrictions.	I, T
AERA-CY-002	Each material effect SHALL pass through an enforcement point that is independently controllable from the proposer and capable of mediating the exact authorized actor, action, target, parameters, and context before the effect becomes irreversible or exits the recovery envelope.	A, T
AERA-CY-003	For each material decision transaction, the implementation SHALL retain a tamper-evident record sufficient to reconstruct the causal chain from intent and authority through actor and configuration, proposal, evidence and provenance, policy and assurance verdicts, disposition, credential use, enforcement, observed effect, override, exception, and recovery.	I, T, D
AERA-CY-004	Cryptographic migration SHALL be driven by inventory, exposure period and interoperability evidence.	I, A, T and/or D as consequence requires
AERA-DA-001	Each decision class SHALL identify an accountable human authority who owns the delegation policy, approves residual risk, and remains answerable for autonomous outcomes.	Approved responsibility assignment; decision-class register; sampled interviews and authorization records.
AERA-DA-002	Every delegation SHALL identify the grantor, grantee, decision class, allowed effects, prohibited effects, target scope, consequence ceiling, context predicates, validity interval, subdelegation permission, and revocation method.	Machine-readable schema; validation tests; positive and negative delegation cases.
AERA-DA-003	Before authorization, the implementation SHALL validate the complete delegation chain to an approved trust anchor and SHALL NOT accept a missing, reordered, ambiguous, or unauthorized parent grant.	Chain-validation logs; malformed-chain tests; trust-anchor inventory.
AERA-DA-004	Subdelegation SHALL be denied unless the parent grant explicitly permits it and the derived grant is no broader in effect, consequence, duration, target, tool, or context.	Derived-grant comparison tests; transitive audit records; denied escalation cases.
AERA-DA-005	Revocation, expiry, actor-integrity loss, and material context change SHALL invalidate dependent authorization state within a documented maximum propagation time; stale state SHALL NOT authorize a new effect.	Revocation latency measurement; partition tests; cache-behavior evidence.
AERA-DA-006	The capability to grant authority, propose an action, classify consequence, authorize, enforce, and approve an override SHALL be separated according to consequence, and incompatible combinations SHALL require documented compensating controls.	Role matrix; policy tests; override review; compensating-control approval.
AERA-DA-007	Each non-human actor SHALL use a workload identity bound to its owner, role, model or software version, runtime integrity state, allowed tools, and consequence ceiling; a human account SHALL NOT be reused as the actor identity.	Workload identity records; attestation or equivalent integrity evidence; credential-use logs.
AERA-DA-008	Operational intent SHALL be attributable, versioned, integrity-protected, and machine-testable, and shall state objectives, priorities, constraints, measures of success, prohibited outcomes, and review or expiry conditions.	Signed intent object; conformance tests; change history.
AERA-DA-009	Intent and policy conflict-resolution rules SHALL define precedence, tie-breaking authority, and fail-safe behavior; unresolved conflict SHALL NOT be converted to permission by implementation order.	Conflict corpus; deterministic evaluation results; approved precedence specification.
AERA-DA-010	Every assurance disposition SHALL record the exact policy bundle, policy version, material policy inputs, and evaluator version used; unapproved rollback to an earlier policy SHALL be rejected.	Disposition records; signed policy inventory; rollback and downgrade tests.
AERA-DA-011	Consequence classification SHALL account for severity, scope, reversibility, time to harm, propagation, cumulative effects, legal or safety significance, and recovery time; the proposer SHALL NOT be the sole approver of a downgrade.	Classification rubric; inter-rater tests; downgrade approvals; boundary cases.

ID	Normative requirement	Primary verification
AERA-DA-012	The proposal evaluated by the Decision Assurance Fabric SHALL be canonically represented and cryptographically bound to actor, target, parameters, tool, sequence, evidence manifest, context, and expiration; any material mutation SHALL require reauthorization.	Canonicalization specification; digest comparison; mutation and replay tests.
AERA-DA-013	Decision evidence SHALL carry source identity, acquisition method, acquisition time, transformation lineage, integrity state, uncertainty, freshness, applicable domain, and retention reference.	Evidence manifest; schema validation; lineage reconstruction.
AERA-DA-014	When independent evidence is required, the implementation SHALL evaluate common sensor, data, model, provider, clock, network, power, and control-plane dependencies; correlated sources SHALL NOT be counted as independent without justification.	Failure-domain model; dependency graph; common-mode fault-injection results.
AERA-DA-015	Evidence source identity and integrity SHALL be verified before evidence contributes to authorization; unverifiable evidence SHALL be excluded or explicitly treated as untrusted by policy.	Source-authentication records; forged-source tests; exclusion behavior.
AERA-DA-016	Decision policy SHALL define how contradictory, missing, ambiguous, and out-of-distribution evidence changes the disposition; a scalar confidence value SHALL NOT conceal a failed mandatory evidence condition.	Contradiction test set; policy traces; mandatory-condition tests.
AERA-DA-017	Each evidence type SHALL have a decision-class-specific freshness and latency tolerance; evidence outside tolerance SHALL cause defer, escalation, constraint, simulation, or denial as specified by policy.	Freshness thresholds; delayed-evidence tests; expiry records.
AERA-DA-018	Assurance state SHALL preserve separately evaluated claims for authority validity, intent alignment, evidence adequacy, actor competence, cyber integrity, temporal coherence, safety margin, and recoverability; a universal scalar score SHALL NOT override a failed mandatory claim.	Assurance-vector schema; policy decision table; dimension-failure tests.
AERA-DA-019	Confidence claims SHALL identify the calibration method, calibration population or operating domain, version, date, error measure, and known limits, and SHALL be monitored for calibration drift in deployment.	Calibration report; reliability analysis; production drift alerts and response records.
AERA-DA-020	Every time or position claim used in authorization SHALL identify its source, reference frame or time scale as applicable, uncertainty, integrity state, age, and whether the source is merely present, monitored, accepted, or actually used in the computed decision.	PNT evidence object; source-selection logs; use-path instrumentation.
AERA-DA-021	Decision classes dependent on PNT SHALL define disagreement thresholds, anomaly detection, holdover or alternate-source behavior, maximum uncertainty, and safe degraded operation; source loss or conflict SHALL NOT silently preserve nominal authority.	PNT fault-injection; jamming/spoofing or simulated anomaly tests; holdover characterization.
AERA-DA-022	Every material autonomous effect SHALL pass through a policy enforcement mechanism controlled independently from the proposing actor; direct access to an effect interface SHALL be denied unless explicitly justified as a time-bounded emergency mode.	Architecture and trust-boundary review; bypass tests; emergency-mode approvals and logs.
AERA-DA-023	The Decision Assurance Fabric SHALL use the canonical authorization-disposition vocabulary and semantics specified by AERA-DAF-110 through AERA-DAF-114.	I, T
AERA-DA-024	An authorization SHALL be limited to the exact actor, action, target, tool, parameters, sequence or transaction, context, consequence bound, and validity interval evaluated; bearer authorization not bound to these elements SHALL NOT authorize a material effect.	Token or capability schema; replay, audience, target, and parameter substitution tests.
AERA-DA-025	Decision Assurance Fabric outage, timeout, overload, or partition SHALL invoke a pre-engineered decision-class degraded mode; unavailability SHALL NOT be interpreted as permission.	Capacity and partition tests; fail-safe state evidence; continuity procedure.
AERA-DA-026	Authorization SHALL be reevaluated when any policy-defined material context changes, including actor integrity, delegation, intent, evidence, PNT state, target state, threat state, or recovery readiness.	Context-change matrix; invalidation latency; reauthorization traces.
AERA-DA-027	The implementation SHALL evaluate cumulative and interacting effects across time, actors, tools, and targets and enforce rolling budgets or state constraints that prevent decomposition of a prohibited action into individually permitted actions.	Stateful test scenarios; budget enforcement; multi-actor sequence analysis.
AERA-DA-028	Tool credentials and capabilities SHALL be least-privileged, audience-restricted, target-restricted, action-restricted, time-bounded, and unavailable to the actor outside the authorized transaction.	Entitlement review; credential issuance logs; negative tool-use tests.
AERA-DA-029	A quorum, consensus, or independent-review rule SHALL define independence in terms of controller, model lineage, training data, evidence source, operator, provider, clock, and failure domain; numerical plurality alone SHALL NOT establish independence.	Independence model; collusion tests; quorum policy and provenance.
AERA-DA-030	Results from external AI, optimization, classical-compute, or quantum services SHALL be bound to the submitted request, provider and service version, execution context, time, uncertainty, and validation method; consequential use SHALL have a characterized fallback and independently controlled enforcement path.	Service contract; request/result binding; plausibility tests; fallback and provider-failure exercise.
AERA-DA-031	The implementation SHALL retain a tamper-evident causal record linking intent, delegation, actor and tool identity, proposal, evidence, assurance state, policy, disposition, enforcement, observed effect, override, and recovery action.	End-to-end reconstruction exercise; log-integrity verification; completeness sampling.
AERA-DA-032	Each decision class SHALL define a recovery envelope with detection triggers, containment and disengagement authority, safe states, rollback or compensation actions, required resources, and maximum recovery time; the recovery path SHALL be periodically tested independently of the actor.	Recovery plan; exercise results; override-path independence evidence.
AERA-DA-033	Runtime monitoring SHALL detect changes in actor integrity, authority, policy, evidence quality, calibration, PNT state, action sequence, target state, and recovery readiness and SHALL trigger the prescribed reauthorization or recovery action.	Monitoring coverage matrix; alert-to-action tests; operational metrics.
AERA-DA-034	Any update that can change autonomous behavior - including model, prompt, policy, memory, data, tool, dependency, or threshold changes - SHALL be versioned, provenance-bound, tested, approved at the applicable consequence level, observable after release, and reversible.	Configuration baseline; test and approval record; staged-release and rollback evidence.

ID	Normative requirement	Primary verification
AERA-DA-035	Verification and validation SHALL exercise every applicable threat in DAT-01 through DAT-25 under nominal, boundary, adversarial, degraded, partitioned, common-mode, and recovery conditions proportional to consequence; untested threats SHALL remain open assurance claims.	I, A, T, D
AERA-DA-036	The organization SHALL maintain an inventory of cryptographic uses supporting delegation, identity, evidence, authorization, audit, communications, and recovery; assess confidentiality and signature exposure periods; plan crypto-agile migration; select applicable standardized algorithms and parameters; track authoritative errata; and verify implementation, protocol, interoperability, performance, and validation status before claiming post-quantum protection.	Cryptographic inventory; migration plan; algorithm and protocol test evidence; errata review; validation records.
AERA-DAF-001	Before material effect, the DAF SHALL evaluate exact proposal, actor/integrity, authority, intent, target/context, consequence, evidence/provenance, cyber state, time, dependencies, approvals, and recovery.	I, T
AERA-DAF-002	A permit SHALL be integrity-bound to actor, action, target, parameters, context, evidence, authority, intent, policy, consequence, and validity, and invalidated by expiry or material bound-element change.	I, T
AERA-DAF-003	A Decision Assurance Fabric implementation SHALL document and test a one-to-one mapping from any external disposition label to the canonical AERA dispositions defined by AERA-DAF-110.	I, T
AERA-DAF-004	The proposer SHALL NOT be sole authority, evidence source, and enforcement for its material effect; consequence-appropriate separation SHALL be verified.	A, T
AERA-DAF-005	Each class SHALL define/test DAF loss, partition, degradation, and staleness behavior; assurance loss SHALL NOT create implicit or over-valid authorization.	A, T
AERA-DAF-101	The DAF SHALL establish a globally unique decision-transaction identifier for each proposed material effect before assurance evaluation begins.	I, T
AERA-DAF-102	The DAF SHALL canonicalize the actor, action, parameters, target, purpose, context, and execution window before assurance evaluation.	I, T
AERA-DAF-103	The DAF SHALL bind each evaluation and authorization artifact to the canonical proposal, effective authority chain, intent version, evidence snapshot, policy and evaluator versions, consequence class, and validity limits.	I, T
AERA-DAF-104	Any material change to a bound transaction input SHALL invalidate the prior permit or cause re-evaluation and issuance of a new disposition before any further effect.	A, T
AERA-DAF-105	The DAF SHALL represent each required assurance claim as an independently visible verdict and SHALL NOT conceal a NOT-SATISFIED or INDETERMINATE hard gate in an aggregate score.	I, A, T
AERA-DAF-106	Each assurance-claim result SHALL identify the claim, verdict, acceptance criterion, supporting and contrary evidence, limitations, evaluator identity and version, assessment time, and validity interval.	I, T
AERA-DAF-107	The DAF SHALL distinguish SATISFIED, NOT-SATISFIED, and INDETERMINATE verdicts and SHALL NOT interpret NOT-SATISFIED or INDETERMINATE as SATISFIED.	I, T
AERA-DAF-108	For each decision type and consequence class, policy SHALL identify hard gates, compensable dimensions, prohibited trade-offs, approving authorities, and the conditions and validity limits for any permitted trade-off.	I, A
AERA-DAF-109	Each permitted trade-off among assurance dimensions SHALL identify the accepted degradation or loss, compensating controls and their evidence, applicable decision types and consequence classes, accountable approving authority, validity, and reassessment triggers.	I, A
AERA-DAF-110	The DAF SHALL support the dispositions PERMIT, PERMIT-WITH-CONSTRAINTS, DEFER, ESCALATE, SIMULATE, QUARANTINE, and DENY with the semantics defined in Section 24; an implementation MAY use different external labels only when it documents a one-to-one semantic mapping.	I, T
AERA-DAF-111	A PERMIT-WITH-CONSTRAINTS disposition SHALL bind and enforce the narrowed proposal; it SHALL NOT authorize the original unconstrained proposal.	I, T
AERA-DAF-112	DEFER, ESCALATE, SIMULATE, QUARANTINE, and DENY SHALL NOT be accepted as authorization for an operational effect.	I, T
AERA-DAF-113	A permit SHALL state a bounded validity interval, bounded use count, or both.	I, T
AERA-DAF-114	The DAF SHALL record all applicable non-permit reasons even when it returns one primary disposition.	I, T
AERA-DAF-115	Unresolved policy conflict, ambiguous authority scope, or incomplete mandatory context SHALL NOT produce a permit.	A, T
AERA-DAF-116	State transitions SHALL be attributable, ordered, integrity-protected, and associated with stated time quality.	I, T
AERA-DAF-117	Re-evaluation after deferment, escalation, or simulation SHALL preserve the prior snapshot and create a new evaluation snapshot.	I, T
AERA-DAF-118	A decision transaction SHALL NOT enter verified closure until the enforcement receipt and required effect observations have been reconciled with the decision contract, or residual-risk closure has been explicitly approved.	I, T, D
AERA-ENF-101	Each material effect SHALL be mediated by an enforcement point that is independently controllable from the proposer to the degree required by consequence.	I, A, T
AERA-ENF-102	The enforcement point SHALL validate the authorization artifact, actor, target, canonical proposal digest, constraints, validity, use limit, and revocation state before execution.	I, T

ID	Normative requirement	Primary verification
AERA-ENF-103	The enforcement point SHALL reject proposal substitution, target substitution, actor substitution, replay, expired authorization, exhausted use, and use outside the named enforcement boundary.	T
AERA-ENF-104	Revocation after dispatch SHALL invoke the defined stop, containment, or recovery behavior and SHALL preserve evidence of any effect already initiated.	I, T, D
AERA-ENF-105	The enforcement path SHALL produce an integrity-protected receipt stating validation, accepted proposal, actual execution parameters, timing, result, and exceptions.	I, T
AERA-ENF-106	Required effect observation SHALL be independent of the proposer for CC-3 and CC-4 decisions unless the assurance case documents why independence is infeasible and identifies compensating controls.	I, A, T
AERA-ENF-107	The audit record SHALL reconstruct the causal chain from intent and delegation through proposal, evidence, vector, policy, disposition, enforcement, observed effect, override, exception, and recovery.	I, T, D
AERA-EVID-001	Decision evidence SHALL carry or resolve to source identity, provenance, observed/effective time, acquisition/transformation history, integrity, freshness, uncertainty/quality, relevant configuration/calibration, and restrictions.	I, T
AERA-EVID-002	High-consequence disposition SHALL use evidence or enforcement sufficiently independent of the proposer; independence SHALL address shared source, time, network, software, administration, and supply-chain dependencies.	A, T
AERA-EVID-003	Material contradictions SHALL be retained, surfaced to policy, and resolved or dispositioned; fusion SHALL NOT suppress contradiction to create unsupported confidence.	I, T
AERA-EVID-004	Transformed evidence SHALL preserve source-level traceability and declare transformations that prevent independent re-evaluation.	I, T
AERA-EVID-011	Data used to train, adapt, evaluate, or operate a consequence-material autonomous capability SHALL have documented provenance, quality, representativeness, limitations, permissible use, and change control.	I, A, T
AERA-EVID-101	Decision evidence SHALL identify the supported claim, source, provenance, transformations, integrity state, uncertainty, observation time, valid interval, and limitations.	I, T
AERA-EVID-102	The DAF SHALL evaluate evidence freshness and time/PNT quality against the requirements of the specific decision contract.	I, A, T
AERA-EVID-103	Evidence status SHALL distinguish source capability present, source monitored, evidence accepted, and evidence used in the computed assurance verdict.	I, T
AERA-EVID-104	Required source independence SHALL be defined against the relevant common-mode and administrative-control failure model; source count alone SHALL NOT establish independence.	I, A, T
AERA-EVID-105	Contradictory, withdrawn, corrected, stale, or compromised evidence SHALL trigger the policy-defined re-evaluation, constraint, deferment, escalation, quarantine, denial, or recovery behavior.	I, T
AERA-EVID-106	Each material decision class SHALL define stop conditions, degraded behavior, and a validated recovery envelope, and the DAF SHALL evaluate current recovery readiness before permit.	I, A, T, D
AERA-EVID-112	A decision-relevant state claim SHALL identify its typed state model and SHALL NOT treat ACCEPTED_BY_SERVICE, USED_IN_SOURCE_SOLUTION, USED_IN_AUTHORIZATION, and VALIDATED_EFFECTIVE as equivalent states.	I, A, T
AERA-GOV-001	Each autonomous decision class SHALL have a named human or governing body accountable for mission legitimacy, delegated authority, residual risk, suspension, and reassessment.	I, D
AERA-GOV-002	Governance SHALL review performance, overrides, near misses, adverse effects, anomalies, deviations, recovery, and material authority/policy/threat/consequence/obligation changes at defined cadence and triggers.	I, A
AERA-GOV-003	Applicable privacy, civil-rights, equity, safety, transparency, recordkeeping, redress, and data-governance obligations SHALL be identified for each decision class and translated into enforceable intent, policy, evidence, and oversight requirements.	I, A
AERA-GOV-004	A consequence-material decision class affecting people SHALL define notice, contestability, human review, and redress mechanisms, or SHALL document the legal and mission basis for inapplicability.	I, A, D
AERA-GOVR-001	Every normative change SHALL identify impact on definitions, requirements, interfaces, schemas, threats, crosswalks and conformance.	I, A, T and/or D as consequence requires
AERA-GOVR-002	Sector profiles SHALL preserve or strengthen the AERA core control floor.	I, A, T and/or D as consequence requires
AERA-GOVR-003	A conformance claim SHALL remain bound to the AERA edition assessed.	I, A, T and/or D as consequence requires
AERA-GOVR-004	External mappings SHALL be versioned, directional and reviewed on source or AERA revision.	I, A, T and/or D as consequence requires
AERA-GOVR-005	An urgent advisory SHALL identify scope, evidence, required action and expiration or incorporation path.	I, A, T and/or D as consequence requires
AERA-GOVR-006	The controlled publication process SHALL maintain a public, versioned change record that links each material comment or proposal to its disposition, rationale, affected artifacts, and effective edition.	I, A
AERA-GOVR-007	The publication process SHALL identify editor, contributor, reviewer, pilot, and assessor roles and SHALL disclose relevant competence, organizational affiliation, sponsorship, and conflicts of interest.	I

ID	Normative requirement	Primary verification
AERA-GOVR-008	A proposed normative revision SHALL receive a defined multistakeholder review period, preserve substantive dissent, and publish disposition before it is represented as a consensus release.	I, A
AERA-GOVR-009	Human-readable and machine-readable normative artifacts SHALL be generated from or reconciled to one authoritative controlled source, carry the same edition identifier, and pass semantic and integrity checks before release.	I, A, T
AERA-GOVR-010	A maturity, conformance, interoperability, or effectiveness claim SHALL state the assessed population, assessor independence, evidence period, known limitations, and whether multi-organization or inter-rater validation has been completed.	I, A
AERA-GOVR-011	Each public edition SHALL state its copyright, license, contribution terms, permitted uses, and intellectual-property process before external redistribution or standards submission.	I
AERA-GOVR-012	A released artifact SHALL be immutable; any byte-changing replacement SHALL receive a new build identifier and manifest, and any normative change SHALL receive a new specification version.	I, T
AERA-HUM-101	Every decision class SHALL designate an accountable human authority and the applicable human-control mode.	I, A, T and/or D as consequence requires
AERA-HUM-102	CC-3 and CC-4 operation SHALL use positive oversight unless a governance-approved preauthorized envelope is validated for the decision tempo.	I, A, T and/or D as consequence requires
AERA-HUM-103	A human approval interface SHALL present decisive evidence, uncertainty, alternatives, expected effects and recovery options.	I, A, T and/or D as consequence requires
AERA-HUM-104	Emergency authority SHALL remain purpose-limited, time-bounded, attributable and unable to waive constraints outside the issuer authority.	I, A, T and/or D as consequence requires
AERA-HUM-105	For a CC-4 decision, the accountable authority SHALL approve the decision-class envelope and every material change; each operational transaction SHALL require transaction-specific concurrence unless a tested preauthorized envelope defines exact bounds, independent deterministic interlocks, continuous observation, revocation, and expiry.	I, A, T, D
AERA-IF-001	Decision-relevant interfaces SHALL identify object schema and version, producer identity, integrity state and temporal validity.	I, A, T and/or D as consequence requires
AERA-IF-002	A material change to a bound object SHALL create a new version and trigger policy-defined reassessment.	I, A, T and/or D as consequence requires
AERA-IF-003	A consumer SHALL record whether an object was rejected, accepted, and used in the computed decision.	I, A, T and/or D as consequence requires
AERA-IF-004	A decision trace SHALL preserve causal links across all proposal-to-effect objects.	I, A, T and/or D as consequence requires
AERA-IF-005	Interface replay SHALL be rejected outside the use, sequence and temporal semantics of the bound transaction.	I, A, T and/or D as consequence requires
AERA-IF-006	Cross-domain disclosure constraints SHALL NOT remove the evidence necessary to validate authority and disposition.	I, A, T and/or D as consequence requires
AERA-IF-007	Corrections and withdrawals SHALL supersede rather than erase decision evidence.	I, A, T and/or D as consequence requires
AERA-IF-008	Authorization and enforcement objects SHALL use the same canonical action and target representation.	I, A, T and/or D as consequence requires
AERA-IF-009	Interface failures SHALL invoke a defined non-permit or degraded behavior.	I, A, T and/or D as consequence requires
AERA-IF-010	Interoperability tests SHALL include version mismatch, stale object, replay and material-substitution cases.	I, A, T and/or D as consequence requires
AERA-INT-001	Intent authorizing an effect SHALL be versioned, attributable, integrity-protected, temporally valid, and machine-testable for objectives, constraints, success, prohibitions, and stops.	I, T
AERA-INT-002	The implementation SHALL detect material objective/constraint/authority/policy conflicts and resolve by declared precedence or escalate; it SHALL NOT silently optimize unresolved conflict.	A, T
AERA-LRN-001	Behavior-changing model, prompt, memory, retrieval, policy, tool, data, orchestration, or equivalent artifacts SHALL be versioned, impact-analyzed, evaluated, approved, observable, and reversible before production authority.	I, T
AERA-LRN-002	Learning/planning/adaptation SHALL NOT increase authority, consequence, targets, tools, data rights, validity, or subdelegation except through separate attributable governance and authorization.	A, T
AERA-MAT-001	A maturity claim SHALL identify its decision population, consequence scope, dependencies, exclusions and assessment date.	I, A, T and/or D as consequence requires
AERA-MAT-002	Assessors SHALL score demonstrated operating behavior rather than planned or purchased capability.	I, A, T and/or D as consequence requires
AERA-MAT-003	The reported level SHALL NOT exceed the lowest evidenced level among consequence-material dimensions.	I, A, T and/or D as consequence requires
AERA-MAT-004	Level 3 SHALL require tested-effective evidence for every consequence-material control.	I, A, T and/or D as consequence requires
AERA-MAT-005	Levels 4 and 5 SHALL require sustained-effective evidence for the operating behavior claimed.	I, A, T and/or D as consequence requires

ID	Normative requirement	Primary verification
AERA-MAT-006	Adverse results, workarounds, overrides and unresolved anomalies SHALL remain in the evidence package.	I, A, T and/or D as consequence requires
AERA-MAT-007	Promotion thresholds SHALL be declared before formal assessment.	I, A, T and/or D as consequence requires
AERA-MAT-008	Material change SHALL trigger impact analysis and reassessment of affected dimension scores.	I, A, T and/or D as consequence requires
AERA-MAT-009	A SUSTAINED EFFECTIVE evidence claim SHALL identify the minimum observation interval, operating exposure, event and consequence coverage, change history, adverse results, and rationale for representativeness before assessment begins.	I, A
AERA-PNT-001	Each decision-relevant time or position claim SHALL identify its contributing sources, distribution or derivation path, reference time scale or coordinate frame as applicable, observation time, uncertainty, integrity state, required traceability, valid interval, and degraded-state behavior.	I, A, T
AERA-PNT-002	PNT reporting and decision logic SHALL distinguish PRESENT, MONITORED, ACCEPTED, and USED IN SOLUTION for each source and SHALL NOT treat those states as interchangeable.	I, T
AERA-PNT-003	Each decision contract that depends on time, order, freshness, synchronization, or position SHALL define and enforce bounds for evidence age, event order, time or position uncertainty, authority and policy expiration tolerance, holdover, and behavior when any bound is exceeded.	I, A, T
AERA-PNT-004	High-consequence PNT-dependent classes SHALL analyze/test common-mode failures and define acceptable alternate, holdover, local, or no-service state.	A, T
AERA-PR-001	For each autonomous decision class, the organization SHALL designate an accountable human or governing body that retains responsibility for mission legitimacy, delegated authority, residual-risk acceptance, suspension, and reassessment.	I, D
AERA-PR-002	An autonomous actor SHALL exercise only decision rights contained in a current, attributable, revocable delegation whose scope includes the actor, action, target, purpose, context, consequence ceiling, and validity interval.	I, T
AERA-PR-003	A model output SHALL NOT authorize a material effect unless the applicable authority, evidence, policy, consequence, temporal-validity, and recovery predicates are satisfied and an enforcement point accepts a current authorization artifact bound to the transaction.	A, T
AERA-PR-004	If an optimization objective conflicts with current operational intent, law, policy, or an applicable prohibition, the implementation SHALL apply the declared precedence rules; unresolved conflict SHALL produce a non-permit disposition.	I, T
AERA-PR-005	Each AERA layer and decision-relevant interface SHALL identify the identity, authorization, evidence-integrity, communications-security, resilience, logging, and recovery properties that it provides, requires, and verifies; allocation of a property to one layer SHALL NOT imply that another layer inherits it without validation.	I, A
AERA-PR-006	The DAF SHALL re-evaluate affected assurance claims before further effect whenever a policy-defined material change occurs in evidence, context, threat state, actor or system state, temporal state, or recovery readiness.	I, T
AERA-PR-007	Before authorizing a proposal that can exceed a validated recovery envelope, the accountable authority SHALL either prohibit the proposal or approve a consequence-specific control set that includes narrower enforceable bounds, additional independent prevention or inhibition, effect observation, and explicit residual-risk acceptance.	I, A, T
AERA-PR-008	Before a behavior-changing model, prompt, memory, retrieval source, policy, data set, tool, dependency, threshold, or orchestration artifact is used under production authority, the organization SHALL version and provenance-bind the artifact, analyze its impact, test and approve it, make its use observable, and provide a tested rollback or safe-replacement path.	I, T, D
AERA-PR-009	Each decision contract SHALL specify and enforce maximum evidence age, event-order and synchronization tolerances, time or position uncertainty bounds as applicable, and the disposition to use when a temporal bound is exceeded.	I, A, T
AERA-PR-010	The minimum control floor for a decision class SHALL be selected from its worst credible consequence and SHALL NOT be reduced on the basis of model type, implementation brand, vendor assertion, or claimed technical sophistication.	I, A
AERA-PR-011	For each CC-3 or CC-4 decision class, evidence or controls relied upon as independent SHALL have documented source, method, controller, administration, provider, software, network, power, supply-chain, and time-reference failure domains; correlated items SHALL NOT be counted as independent without an approved rationale and compensating controls.	I, A, T
AERA-PR-012	For every material decision transaction, the implementation SHALL retain a tamper-evident causal record that identifies who or what proposed, authorized, enforced, observed, overrode, or recovered the action; the evidence and policy used; the authorization and validity; and the observed result.	I, T, D
AERA-Q-001	A quantum-enabled decision service SHALL identify capability class, role/boundary, readiness, classical baseline, measured mission advantage, error/uncertainty, security assumptions, cost/dependencies, and exit criteria.	I, A, T
AERA-Q-002	A quantum-enabled service SHALL retain authenticated classical control, monitoring, and audit and provide validated classical fallback or explicit no-service state before material-effect use.	A, T
AERA-REC-001	Each material class SHALL define states, resources, authority, time bounds, irreversibility, and actions for arrest, containment, reversal, compensation, or manual transfer.	I, A
AERA-REC-002	Recovery, revocation, manual takeover, and state reconciliation SHALL be tested before production and after material change under credible fault/adversary conditions.	T, D

ID	Normative requirement	Primary verification
AERA-REC-003	Command issue/acknowledgment SHALL NOT prove effect; obtain consequence-appropriate effect evidence or mark unverified and inhibit dependent autonomous actions.	A, T
AERA-THEORY-001	Every autonomous action SHALL resolve to a valid authority delegation.	I, A, T and/or D as consequence requires
AERA-THEORY-002	Decision evaluation SHALL consider material cumulative effects when actions compose.	I, A, T and/or D as consequence requires
AERA-THEORY-003	Each decision class SHALL have an accountable human owner.	I, A, T and/or D as consequence requires
AERA-THR-001	When an authenticated evidence source can remain credential-valid after compromise, the implementation SHALL apply consequence-appropriate challenge, corroboration, or plausibility analysis before relying on that source for a material effect.	A, T
AERA-THR-002	Each cross-domain or delegated service hop SHALL preserve and validate the original initiator, delegation chain, purpose, consequence context, and requested effect before exercising target-domain authority.	I, A, T
AERA-THR-003	The Decision Assurance Fabric SHALL define capacity objectives and saturation thresholds, detect bypass and denial-of-assurance conditions, and invoke consequence-specific throttling or non-permit behavior before overload can become implicit authorization.	I, A, T
AERA-THR-004	Distributed enforcement points SHALL use monotonic policy and state updates, acknowledgement, deduplication, idempotent effect semantics, and distributed use accounting where duplicate or reordered execution could create a material effect.	I, A, T
AERA-THR-005	Audit evidence for high-consequence decision classes SHALL use independent replication or an equivalent survivability mechanism and SHALL detect omission, deletion, reordering, retiming, or incomplete causal chains.	I, A, T
AERA-THR-006	Human-oversight mechanisms SHALL define workload limits, consequence-prioritized queues, review-time and usability criteria, saturation detection, safe throttling, and timeout behavior that cannot convert absent review into permission.	I, A, T, D
AERA-VV-001	Before formal validation begins, the accountable validation authority SHALL approve the decision class, configurations, operating environments, acceptance criteria, measures, uncertainty treatment, thresholds, failure conditions, and required evidence.	I
AERA-VV-002	Validation of each material decision class SHALL include all applicable representative, boundary, adversarial, degraded, partitioned, common-mode, contradictory or stale evidence, identity, authority, policy, PNT, communications, tool or model, enforcement-bypass, recovery, and rollback conditions.	I, T
AERA-VV-003	Each CC-3 or CC-4 decision class SHALL maintain a configuration-bound assurance case that states the top claim, scope and context, assumptions, argument, subclaims, evidence, rebuttals, limitations, residual risk, validity, monitoring, and reassessment triggers.	I, A
AERA-VV-004	The accountable authority SHALL approve the assurance case for each CC-3 or CC-4 decision class before production authority is granted and after any material change that invalidates a claim or item of evidence.	I, A
AERA-VV-005	Each formal test SHALL define a decision oracle, expected result, tolerance, treatment of indeterminate outcomes, and defect-disposition rule before execution.	I, A
AERA-VV-006	A validation campaign SHALL predeclare coverage, sampling basis, sample size or stopping rule, statistical or epistemic uncertainty treatment, environment and configuration, and limits on inference to the claimed decision population.	I, A, T
AERA-VV-007	Each active normative requirement SHALL reference at least one profile- and consequence-specific verification procedure; a requirement lacking an approved procedure or sufficient current evidence SHALL be assessed INDETERMINATE rather than SATISFIED.	I, A
AERA-XW-001	Every external mapping SHALL identify the official source, publication status, version or release, access date, and mapping direction.	Crosswalk metadata; source register; sampled URL and version verification.
AERA-XW-002	An AERA conformance claim SHALL NOT be inferred from external certification, authorization, maturity, or framework alignment, and an AERA assessment SHALL NOT be represented as satisfying an external obligation without an independent assessment against that source.	Claim-language review; assessment scope; communications approval.
AERA-XW-003	External evidence SHALL be reused only when the asset, decision class, control implementation, operating context, assessment period, evidence integrity, and assurance level are shown to match the AERA claim.	Evidence-reuse determination; scope comparison; assessor approval.
AERA-XW-004	Mappings SHALL be reviewed when AERA changes, an external source is revised or superseded, applicable law or policy changes, or operational evidence reveals a semantic gap.	Change-monitoring record; mapping review log; retired mapping history.

Appendix B - Decision Contract Reference Schema

The schema below is a proposed informative logical representation, not an interoperability specification or mandated serialization. An implementation may use policy languages, signed tokens, structured events or safety cases. Normative semantic obligations remain in Sections 23 and 24 and the identified requirements. Agent identity/authorization and Zero Trust sources provide related architectural context [6], [11], [12].

Table 36. Field / Proposed informative content

Field	Proposed informative content	Assurance purpose
decision_id	Globally unique identifier	Correlates proposal, authorization, action and effect
actor	Workload identity, owner, role, integrity state	Establishes accountable execution subject
authority	Grantor, delegation ID, scope, expiry	Proves legitimate decision rights
intent	Mission objective and version	Binds decision to current purpose
action	Canonical operation and parameters	Prevents substitution after authorization
target	Resource, environment or stakeholder boundary	Defines effect destination
evidence	References, provenance, freshness, uncertainty	Supports decision-relevant claims
consequence	Class, credible harms and reversibility	Selects control rigor
confidence	Calibrated claim vector and limitations	Represents decision support without false precision
policy	Evaluated policy and rule versions	Enables reproducibility
time_pnt	Time/position source, uncertainty and integrity	Preserves temporal and spatial validity
recovery	Stop, rollback, containment and takeover plan	Bounds failure
disposition	PERMIT, PERMIT-WITH-CONSTRAINTS, DEFER, ESCALATE, SIMULATE, QUARANTINE or DENY	Records authorization result
effect	Observed result and verification state	Closes the assurance loop

Proposed logical object set (informative)

Table 37. Proposed logical object set (informative)

Object	Proposed semantic content	Illustrative binding or lifecycle rule
AuthorityGrant	Issuer and subject; parent grants; actions, targets, purposes and effects; prohibitions; consequence ceiling; predicates; validity; subdelegation; revocation; disconnected operation.	Child grants attenuate effective parent authority; parent expiry or revocation invalidates descendants.
EvidenceClaim	Claim and context; value or verdict; uncertainty; source identity and competence; control/failure domain; origin and transformations; time/PNT quality; integrity; use restrictions; correction reference.	Correction or withdrawal supersedes the claim and triggers policy-defined reassessment.
AssuranceAssessment	Transaction and snapshot; policy versions; consequence class; dimension claims, verdicts, criteria, evidence, limits, hard-gate status and validity; evaluator identity.	A failed or indeterminate hard gate remains visible and is not offset by aggregation.
AuthorizationDisposition	Primary disposition; rationale; canonical digest; actor/target/authority/intent bindings; constraints; validity/use count; obligations; invalidation triggers; reservations; enforcement boundary.	Only PERMIT and PERMIT-WITH-CONSTRAINTS authorize an effect, and only inside the exact binding.
EnforcementReceipt	Artifact validation; enforcement identity; executed digest; actual target, parameters and resources; start/stop time quality; status and exceptions.	Receipt states what the boundary accepted or attempted; it is not by itself proof of achieved effect.
EffectObservation	Observer and independence group; observed state; expected comparison; uncertainty and coverage; time quality; provenance; recovery trigger.	Effect observation closes the transaction or moves it to exception/recovery under the applicable normative rules.

Proposed schema invariants (informative)

- Include object identifiers, schema versions, producer identities and integrity bindings for decision-relevant objects.
- Share the canonical proposal representation between evaluation and enforcement; treat semantic substitution as a new transaction.
- Give each permit an explicit time/use bound and name continuing obligations and invalidation triggers.
- Carry quality and uncertainty with temporal fields, not timestamps alone.
- Reconcile the enforcement receipt and applicable effect observations before treating a transaction as verified.

Appendix C - Glossary and Acronyms

Table 38. Appendix C - Glossary and Acronyms

Term	Meaning	Term	Meaning
AERA	Autonomous Enterprise Reference Architecture	OT/ICS	Operational Technology / Industrial Control Systems
AI RMF	Artificial Intelligence Risk Management Framework	PNT	Positioning, Navigation and Timing
Assurance claim	A falsifiable statement about a decision property, evaluated against a criterion with identified evidence.	PQC	Post-Quantum Cryptography
Assurance vector	The set of independently visible claim verdicts used to derive a disposition; not a universal scalar score.	QKD	Quantum Key Distribution
Authority grant	A current, attributable and revocable delegation of bounded decision rights.	Recovery envelope	Validated states, resources, authorities and time limits for arrest, containment, reversal, compensation or takeover.
Authorization artifact	Integrity-protected permit bound to the exact transaction and enforcement obligations.	SLH-DSA	Stateless Hash-Based Digital Signature Algorithm
Canonical proposal	Unambiguous action representation in which a material semantic change alters the binding identifier.	TEVV	Test, Evaluation, Verification and Validation
CC	AERA consequence class, CC-0 through CC-4.	Workload identity	Identity of a non-human execution subject bound to owner, role and consequence-material runtime state.
CSF	Cybersecurity Framework	ZTA	Zero Trust Architecture
DAF	Decision Assurance Fabric	Acceptable risk	Residual risk explicitly accepted by an authority empowered for the decision class, scope, duration and consequence; not the absence of known defects.
Decision class	A governed population of actions sharing authority, intent, effect, consequence and assurance rules.	Achieved effect	Externally consequential state change supported by the required effect observations and reconciled with the transaction.
Decision contract	Machine-verifiable object binding proposal to authority, intent, evidence, consequence, time, oversight and recovery.	Attempted effect	Effect request accepted or initiated at an enforcement boundary; not proof that the intended external state was achieved.
Decision transaction	Lifecycle record from proposal through disposition, execution, observed effect and closure or recovery.	Claimant	Legal or organizational entity that makes and remains accountable for a formal AERA claim.
Degraded mode	Pre-engineered operating behavior used when a dependency or assurance condition is outside nominal bounds.	Conformance subject	Architecture, DAF service, decision class, mission thread, domain or sector-profile implementation identified by a formal claim.
Effect	Umbrella term for an attempted or achieved externally consequential change, commitment, command, disclosure, allocation or release; normative distinctions between attempted and achieved effect are defined in Section 3 and applicable requirements.	Consequence-material	Capable of changing consequence class, authorization disposition, control floor, recovery obligation or residual-risk acceptance.
Effect observation	Evidence describing the resulting state, preferably independent of the proposer and command path.	Credible consequence	Effect supported by scenario, hazard, threat, dependency or operational evidence and not dismissed solely because probability is uncertain or low.
Effect receipt	Integrity-protected record of what an enforcement boundary accepted, rejected, executed or failed.	Current	Within the decision-specific validity, freshness and context bounds established before assessment.
Enforcement point	Control boundary that validates authorization and mediates a material effect.	I1	Self-assessment performed or controlled by the function responsible for the assessed subject; no independence claim.
GNSS	Global Navigation Satellite System	I2	Independent internal assessment by a competent function separated from design and operation and without direct delivery accountability.
Hard gate	An assurance predicate that policy does not permit another dimension to compensate for.	I3	Independent external assessment by a competent entity operationally and contractually independent of the subject, with sponsor and conflicts disclosed.
HITL	Human in the loop	Independent evidence	Evidence whose relevant source, control, data, model, provider, clock, network, power and administrative failure domains satisfy the declared independence model.
Intent package	Versioned mission objective, priorities, constraints, measures, prohibitions and validity.	Material effect	Attempted or achieved externally consequential change whose credible consequence meets the decision-class materiality rule.
Material context change	Change capable of altering authority, evidence, consequence, policy, enforcement, time or recovery outcome.	Positive oversight	Human-control mode requiring timely, attributable action by an authorized reviewer rather than only post-effect notice.
ML-DSA	Module-Lattice-Based Digital Signature Algorithm	Semantic equivalence	Demonstrated preservation of required meaning, constraints, lifecycle behavior, failure semantics and observable evidence across representations or implementations.
ML-KEM	Module-Lattice-Based Key-Encapsulation Mechanism	Sufficient evidence	Evidence satisfying the predeclared claim criteria, coverage, quality, independence and uncertainty requirements for the consequence class.
Operational envelope	Validated boundary of context, behavior, performance and consequence in which authority applies.	Validated	Demonstrated against predeclared criteria in a configuration and environment representative of the stated claim; not merely reviewed or tested once.

Interpretive distinctions

Table 39. Interpretive distinctions

Distinction	AERA interpretation
Automation vs. autonomy	Automation executes predefined transitions; autonomy selects or sequences actions within a delegated envelope.
Access vs. authority	Access permits use of a resource; authority permits a class of decisions or effects. Access is necessary but may be insufficient.
Output vs. decision	A model output becomes a decision only when adopted into a governed action commitment.
Confidence vs. correctness	Confidence expresses support for a claim under assumptions; it does not prove the action is correct.
Explainability vs. assurance	An explanation may support review but does not establish identity, evidence integrity, authorization, safety or recovery.
Resilience vs. redundancy	Redundancy adds alternatives; resilience requires sufficiently independent alternatives and acceptable degraded operation.

Appendix D - Assurance Case Starter

This appendix provides a minimum structure for an AERA assurance case. It is intentionally claim-centered: a control list without an explicit argument does not establish that the decision class is acceptably governed. Assessment, systems-security, lifecycle and V&V sources provide related practices [14], [18], [19], [42], [44].

Table 40. Element / Required statement

Element	Required statement	Review question
Top claim	The bounded decision class is acceptably governed for the stated operational context.	Is the decision population and consequence boundary explicit?
Context	Mission, environment, stakeholders, authority, dependencies and exclusions.	Could a reviewer determine where the claim does not apply?
Strategy	Argument connecting authority, evidence, cyber integrity, timing, performance and recovery.	Does the argument address correlated and adversarial failure?
Subclaims	Competence, evidence adequacy, policy correctness, enforcement, temporal validity and recoverability.	Are subclaims independently falsifiable?
Evidence	Tests, analysis, operational data, calibration, exercises and configuration records.	Is evidence current, representative and traceable?
Rebuttals	Known counterarguments, anomalies, limitations and residual risk.	Were contrary results preserved rather than averaged away?
Decision	Approval, constraints, expiration, monitoring and reassessment trigger.	Who accepted residual risk and when must the claim be revisited?

Illustrative claim decomposition

- C1 - Authority: the actor can exercise only the decision rights explicitly delegated for this mission and time.
- C2 - Evidence: required state claims are sufficiently fresh, independent, provenance-rich and uncertainty-bounded.
- C3 - Integrity: the actor, policy, enforcement path and evidence chain are protected against the defined threat model.
- C4 - Performance: decision quality and latency meet the consequence-specific acceptance criteria across nominal and boundary conditions.
- C5 - Recovery: incorrect or compromised behavior can be arrested, contained, reversed or compensated within the validated envelope.
- C6 - Lifecycle: material change, drift and expired evidence trigger reassessment before authority expands.

Appendix E - Executive Adoption Checklist

Leadership should not authorize broader autonomous effects until the following questions have evidence-backed answers. “Yes” means demonstrated in the operationally representative decision path, not merely planned or technically possible.

Table 41. Domain / Evidence-backed decision question

Domain	Evidence-backed decision question	Status
Mission and ownership	Is the decision class tied to a current mission objective and named accountable owner?	☐ Ready ☐ Gap ☐ N/A
Authority	Is delegation explicit, bounded, revocable and protected against unauthorized subdelegation?	☐ Ready ☐ Gap ☐ N/A
Consequence	Are credible direct, secondary, cumulative and irreversible effects classified?	☐ Ready ☐ Gap ☐ N/A
Evidence	Are provenance, freshness, uncertainty, independence and contradiction handling defined?	☐ Ready ☐ Gap ☐ N/A
Identity	Are human, workload, device, service, model and data identities established at the required strength?	☐ Ready ☐ Gap ☐ N/A
Authorization	Is the decision disposition independently controlled and bound to the exact action and context?	☐ Ready ☐ Gap ☐ N/A
Timing/PNT	Are temporal and spatial integrity states known and consumed by the decision logic?	☐ Ready ☐ Gap ☐ N/A
Cybersecurity	Has the complete proposal-to-effect path been threat modeled and adversarially tested?	☐ Ready ☐ Gap ☐ N/A
Human control	Are approval, override, suspension and manual takeover roles operationally usable?	☐ Ready ☐ Gap ☐ N/A
Recovery	Has degraded operation, containment and recovery been exercised within acceptance limits?	☐ Ready ☐ Gap ☐ N/A
Learning	Can behavior-changing updates be detected, evaluated, reversed and prevented from expanding authority?	☐ Ready ☐ Gap ☐ N/A
Assurance	Does an approved assurance case state limitations, residual risk and reassessment triggers?	☐ Ready ☐ Gap ☐ N/A

GO / NO-GO RULE: If any gap affects authority validity, independent enforcement, safety interlock, temporal integrity or recovery for a high-consequence decision, production authority should not expand until the gap is resolved or a named accountable authority approves a time-bounded deviation with compensating controls.

Appendix F - Threat-Control-Test Traceability

This informative traceability appendix connects every DAT threat to canonical controls, a representative negative test and expected evidence. The companion

AERA_1.2_WD_Threat_Control_Test_Matrix.xlsx preserves the full prevention, detection, containment and recovery analysis. This appendix does not create additional requirements. A row is not closed merely because a control identifier is present; AERA-DA-035 remains controlling. AERA-THR-001 through AERA-THR-006 close the prior partial coverage of DAT-10, DAT-16, DAT-20, DAT-21, DAT-22 and DAT-25.

Table 42. Threat / Primary AERA controls

Threat	Primary AERA controls	Representative negative test	Expected evidence
DAT-01 Authority forgery	AERA-DA-001, AERA-DA-002, AERA-DA-003, AERA-DA-006, AERA-AUTH-101, AERA-AUTH-107, AERA-DA-033	Submit a well-formed, correctly signed grant whose signer is authenticated but lacks authority to grant the requested decision class; attempt the effect through both the normal DAF path and the effect interface. Expected result: no permit and no effect.	Approved trust-anchor and accountable-authority inventory; delegation schema; chain-validation trace; role-separation matrix; negative grant cases; enforcement denial receipt; revocation or quarantine record; effect-observation record showing no unauthorized effect.
DAT-02 Delegation-chain corruption	AERA-DA-002, AERA-DA-003, AERA-DA-005, AERA-AUTH-105, AERA-AUTH-107, AERA-AUTH-108, AERA-DA-031	Remove an intermediate parent, reorder two parents, and substitute a valid grant rooted in the wrong authority domain. Expected result: all variants are rejected before authorization and are distinguishable in the audit record.	Trust-anchor inventory; parent-reference and chain-validation logs; malformed-chain corpus and results; precedence tests; descendant invalidation evidence; causal reconstruction showing the exact fault.
DAT-03 Authority laundering / covert subdelegation	AERA-DA-002, AERA-DA-003, AERA-DA-004, AERA-DA-007, AERA-DA-022, AERA-DA-028, AERA-DA-031	Give Actor A a non-subdelegable grant, then have A invoke Actor B or a broker service to perform the effect using A's tool access. Expected result: target-domain enforcement rejects the request and records both the initiator and attempted deputy.	Parent and child grant comparison; subdelegation policy; workload identity and tool-credential records; transitive audit trace; denied effect receipt; credential revocation record; recovery transaction under a valid grant.
DAT-04 Stale-authority replay	AERA-DA-002, AERA-DA-005, AERA-DA-017, AERA-DA-024, AERA-DA-026, AERA-DA-033, AERA-AUTH-106, AERA-ENF-103	Authorize once, revoke the parent grant, isolate an enforcement cache until after the maximum propagation time, then replay the authorization. Expected result: rejection in connected and partitioned modes; no new effect.	Revocation-latency measurement; cache and partition test results; token validity/use records; enforcement rejection receipt; descendant invalidation trace; post-partition reconciliation evidence.
DAT-05 Workload identity substitution	AERA-DA-005, AERA-DA-007, AERA-DA-012, AERA-DA-024, AERA-DA-033, AERA-DA-034, AERA-ENF-103	Obtain a permit for workload version N, then execute the same proposal from version N+1 or a different model using the same logical actor identifier. Expected result: enforcement rejects actor substitution and the prior permit is invalidated.	Workload identity registry; version/configuration baseline; attestation or equivalent runtime-integrity evidence; actor-binding test; enforcement rejection receipt; monitoring alert; rollback and reauthorization trace.
DAT-06 Intent substitution or drift	AERA-DA-001, AERA-DA-008, AERA-DA-009, AERA-DA-026, AERA-DA-031, AERA-DA-034	Issue a permit under intent version 7, then supersede a prohibited outcome or priority without accountable-owner approval while retaining the prior intent identifier. Expected result: integrity/version failure, permit invalidation, and no effect.	Signed intent objects; intent change history; precedence and conflict tests; intent-to-effect trace; context-change alert; invalidation latency; rollback and new-transaction evidence.
DAT-07 Policy or configuration rollback	AERA-DA-008, AERA-DA-009, AERA-DA-010, AERA-DA-026, AERA-DA-031, AERA-DA-034	Replace the current restrictive policy bundle with an authentic but obsolete permissive version and replay a previously valid evaluator image. Expected result: rollback rejection and no permit.	Approved signed policy inventory; configuration baseline; disposition record with policy/evaluator versions; rollback and downgrade test results; alert and quarantine record; restoration and reassessment trace.
DAT-08 Consequence-class downgrade	AERA-DA-001, AERA-DA-006, AERA-DA-009, AERA-DA-011, AERA-DA-018, AERA-DA-027, AERA-CONS-102, AERA-CONS-103, AERA-CONS-105	Present the same physical or digital effect as CC-1 after it was approved as CC-3 by splitting it into smaller requests and averaging severity with reversible attributes. Expected result: classification remains at the controlling class and high-consequence controls remain active.	Approved classification rubric; accountable owner and role matrix; inter-rater and boundary-case results; cumulative-effect analysis; downgrade approval record if applicable; policy trace showing the controlling class.
DAT-09 Proposal mutation / TOCTOU substitution	AERA-DA-012, AERA-DA-017, AERA-DA-024, AERA-DA-026, AERA-ENF-102, AERA-ENF-103, AERA-ENF-105, AERA-IF-002	After authorization, change one consequential parameter or target while retaining the original token and transaction ID; repeat during a race between policy evaluation and dispatch. Expected result: both mutations are rejected before effect.	Canonicalization specification; digest and version records; mutation, race, target-substitution, and replay test results; enforcement receipt; effect observation; reauthorization trace.
DAT-10 Evidence fabrication or source impersonation	AERA-DA-013, AERA-DA-015, AERA-DA-016, AERA-DA-033, AERA-EVID-101, AERA-EVID-105, AERA-THR-001	Inject a fabricated observation using a compromised but otherwise valid source credential, then provide a conflicting independent observation. Expected result: the system does not treat authentication as truth, exposes the contradiction, and follows the configured non-permit path.	Source-identity and integrity records; evidence manifest; forged-source and contradiction test results; anomaly or plausibility alert; exclusion/quarantine policy trace; dependent-decision invalidation and reassessment evidence. AERA 1.1 RC1 gap-closure control retained in 1.2 WD: AERA-THR-001.
DAT-11 Provenance destruction or rebinding	AERA-DA-012, AERA-DA-013, AERA-DA-015, AERA-DA-030, AERA-DA-031, AERA-IF-002, AERA-IF-004, AERA-IF-007	Strip transformation metadata from a derived result and attach it to a different transaction with a compatible schema. Expected result: binding or lineage validation fails and no authorization relies on the artifact.	Evidence manifest and schema; raw/derived relationship records; transaction binding; lineage reconstruction; rebinding and missing-link tests; correction/withdrawal record; reassessment trace.
DAT-12 Independence illusion / correlated evidence	AERA-DA-014, AERA-DA-018, AERA-DA-021, AERA-DA-029, AERA-EVID-104	Present two application feeds as independent although both inherit one GNSS receiver, clock distribution path, and provider control plane; fault the common source. Expected result: the quorum loses independence and nominal authority is not preserved.	Failure-domain model; dependency graph; provenance and independence metadata; common-mode fault-injection results; quorum policy trace; degraded-mode and alternate-source evidence.
DAT-13 Confidence or calibration manipulation	AERA-DA-016, AERA-DA-018, AERA-DA-019, AERA-DA-023, AERA-DA-033	Inflate a scalar confidence score so it exceeds the permit threshold while authority validity or evidence adequacy is NOT-SATISFIED; separately alter calibration thresholds without approval. Expected result: no permit and a visible failed claim.	Assurance-vector schema; policy decision table; calibration report and population; dimension-failure and threshold-tampering tests; drift alerts; rollback/recalibration record.

Appendix F - Threat-Control-Test Traceability (continued)

Table 43. Appendix F - Threat-Control-Test Traceability (continued)

Threat	Primary AERA controls	Representative negative test	Expected evidence
DAT-14 Temporal or causal manipulation	AERA-DA-017, AERA-DA-020, AERA-DA-021, AERA-DA-026, AERA-DA-033, AERA-PNT-001, AERA-PNT-002, AERA-PNT-003	Delay or rewrite one evidence timestamp so stale data appears fresh while a common reference is spoofed; continue beyond characterized holdover. Expected result: the source is not accepted or used nominally and the transaction cannot permit outside temporal limits.	PNT evidence object; present/monitored/accepted/used instrumentation; source-selection and sequence logs; spoofing, delay, disagreement, and holdover tests; uncertainty characterization; degraded-mode and reauthorization trace.
DAT-15 Tool-mediated privilege escalation	AERA-DA-007, AERA-DA-012, AERA-DA-022, AERA-DA-024, AERA-DA-028, AERA-CY-001, AERA-CY-002, AERA-ENF-102, AERA-ENF-103	Use a valid tool credential from the authorized actor against a different target, API audience, action, or time window, including direct invocation outside the transaction. Expected result: all out-of-scope attempts are denied.	Workload identity and entitlement records; credential issuance and use logs; audience/target/action/time substitution tests; bypass test; enforcement denial receipt; revocation and reissuance trace.
DAT-16 Cross-domain confused deputy	AERA-DA-004, AERA-DA-007, AERA-DA-022, AERA-DA-024, AERA-DA-028, AERA-DA-031, AERA-ENF-102, AERA-ENF-103, AERA-THR-002	Have an authorized intermediary receive an unauthorized request and invoke the target using the intermediary's own standing credential without forwarding the original initiator/delegation. Expected result: target-domain enforcement rejects the request.	Multi-hop identity/delegation trace; target-domain policy test; service credential scope; ambient-authority negative case; enforcement receipt; transitive audit reconstruction. AERA 1.1 RC1 gap-closure control retained in 1.2 WD: AERA-THR-002.
DAT-17 Cumulative-effect / action-decomposition attack	AERA-DA-011, AERA-DA-027, AERA-DA-033, AERA-CONS-104, AERA-CONS-106, AERA-THEORY-002	Have two actors submit individually permissible operations that together exceed a shared safety, financial, cyber, or physical limit; vary order and concurrency. Expected result: aggregate reservation prevents the prohibited state.	Consequence and dependency model; shared-limit/reservation design; multi-actor sequence and concurrency tests; budget telemetry; denial/stop trace; recovery or compensation evidence.
DAT-18 Multi-agent collusion / consensus capture	AERA-DA-014, AERA-DA-027, AERA-DA-029, AERA-EVID-104	Populate a three-agent quorum with agents that have separate identities but share one controller, model lineage, data source, and operator; coordinate a false vote. Expected result: the votes do not satisfy the independence requirement.	Quorum policy; controller/model/data/operator provenance; failure-domain analysis; collusion and common-controller tests; disposition trace; independent-review reconstitution evidence.
DAT-19 External compute / oracle substitution	AERA-DA-013, AERA-DA-015, AERA-DA-017, AERA-DA-030, AERA-DA-033, AERA-DA-034, AERA-DA-036	Return a validly signed but stale or cross-request result from an external AI, optimizer, or quantum service, then make the provider unavailable. Expected result: misbound result rejection and controlled fallback without authority expansion.	Service contract and provider identity; request/result binding records; version and uncertainty metadata; stale/misbound/plausibility tests; provider-failure exercise; fallback characterization; independent enforcement receipt.
DAT-20 Assurance bypass / denial / fail-open	AERA-DA-006, AERA-DA-009, AERA-DA-018, AERA-DA-022, AERA-DA-023, AERA-DA-025, AERA-DA-033, AERA-CY-002, AERA-IF-009, AERA-THR-003	Saturate or partition the DAF, force a timeout, and attempt direct execution plus a default-allow response. Expected result: no silent permit, deterministic degraded behavior, and no unmediated effect.	Trust-boundary and bypass analysis; capacity, overload, timeout, and partition tests; fail-safe/degraded-mode evidence; effect-interface denial receipt; continuity and reconciliation exercise. AERA 1.1 RC1 gap-closure control retained in 1.2 WD: AERA-THR-003.
DAT-21 Enforcement desynchronization	AERA-DA-005, AERA-DA-010, AERA-DA-012, AERA-DA-017, AERA-DA-024, AERA-DA-025, AERA-DA-026, AERA-DA-033, AERA-ENF-102, AERA-ENF-105, AERA-IF-005, AERA-IF-010, AERA-THR-004	Delay policy/revocation replication to one enforcement point, then race duplicate or reordered requests across two points. Expected result: stale and duplicate attempts cannot create a second or unauthorized effect.	Policy/context version records; replication-lag and race tests; authorization use counter; enforcement acknowledgements/receipts; replay rejection; mismatch alert; reconciliation trace. AERA 1.1 RC1 gap-closure control retained in 1.2 WD: AERA-THR-004.
DAT-22 Audit-chain corruption	AERA-DA-020, AERA-DA-031, AERA-DA-035, AERA-DA-036, AERA-DAF-116, AERA-ENF-105, AERA-ENF-107, AERA-IF-004, AERA-IF-007, AERA-THR-005	Delete one disposition, reorder two state transitions, rewrite a timestamp, and forge an enforcement receipt while leaving local logs internally consistent. Expected result: integrity or completeness checks identify each break and reconstruction cannot falsely close the transaction.	Cryptographic/log-integrity records; time-quality metadata; state-transition and enforcement receipts; deletion/reorder/retime/forgery tests; causal reconstruction; correction record; residual-risk disposition. AERA 1.1 RC1 gap-closure control retained in 1.2 WD: AERA-THR-005.
DAT-23 Recovery inhibition / disengagement suppression	AERA-DA-006, AERA-DA-022, AERA-DA-023, AERA-DA-032, AERA-DA-033, AERA-EVID-106, AERA-ENF-104, AERA-HUM-104	Compromise the actor and primary control plane, then have the actor refuse stop, revoke, rollback, and quarantine commands. Expected result: an independent recovery path arrests the effect within the validated envelope.	Recovery envelope and authority assignment; recovery-path independence analysis; protected override records; compromised-actor and control-plane exercise; time-to-safe-state measurement; effect verification and residual-exposure record.
DAT-24 Learning-loop poisoning / unauthorized behavior change	AERA-DA-008, AERA-DA-010, AERA-DA-013, AERA-DA-019, AERA-DA-026, AERA-DA-031, AERA-DA-033, AERA-DA-034, AERA-MAT-008	Poison feedback or memory to introduce a prohibited behavior without changing the nominal model identifier; separately alter a threshold through an unapproved online-learning path. Expected result: change is detected, authority does not expand, and rollback succeeds.	Configuration and data provenance baseline; predeclared acceptance tests; approval and staged-release record; drift/calibration/integrity alerts; unauthorized-change test; rollback evidence; material-change reassessment.
DAT-25 Oversight saturation / automation-bias exploitation	AERA-DA-001, AERA-DA-006, AERA-DA-011, AERA-DA-016, AERA-DA-023, AERA-DA-029, AERA-DA-032, AERA-DA-035, AERA-HUM-102, AERA-HUM-103, AERA-THR-006	Flood reviewers with mixed low- and high-consequence requests, compressed deadlines, repetitive alerts, and misleading but plausible explanations; test whether approval accuracy or independence collapses or timeout becomes permission.	Human-factors test plan; consequence-prioritized queue and throttle evidence; workload/latency/error telemetry; explanation-quality criteria; approval and second-person-review records; adverse-result analysis; reassessment and recovery record. AERA 1.1 RC1 gap-closure control retained in 1.2 WD: AERA-THR-006.

References

REFERENCE STATUS: External mappings were checked against publisher-controlled sources on 7 August 2026. AI RMF 1.0 remains the final mapped baseline while revision activity is reported [1], [2]. NIST IR 8596, IR 8323 Rev. 2, IR 8547 and Privacy Framework 1.1 are drafts [8], [23], [30], [61]. The NIST agent initiative and NCCoE concept paper are ongoing work, not standards or endorsements [5], [6]. FIPS 203-205 specify algorithms; implementation, protocol, migration and agility claims require separate evidence [25]-[31].

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